

ECE 513: VLSI Design Flow

# Intelligent Toll Booth Managment System: RTL to GDS Flow

SUBMITTED BY

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# Chapter 1

## INTRODUCTION

The Intelligent Toll Booth Management System operates based on a finite state machine (FSM), which allows it to process vehicles efficiently while responding to various operational conditions such as vehicle detection, account balance verification, transaction processing, and barrier control. Each state in the FSM represents a different mode of operation, and the system transitions between these states according to the inputs it receives from its sensors and internal status signals. The toll gate has multiple input sources which continuously provide data to the system. The primary inputs include the clock signal for synchronization, a reset signal for system initialization, vehicle entry and exit sensors that detect vehicle presence, and an 8-bit vehicle identification input that provides the unique identifier for each vehicle approaching the toll booth.

The IDLE state is the system's default operational state, where it remains ready to process vehicles while keeping the barrier closed for security. The system transitions from this state to the READ\_ID state when a vehicle is detected by the entry sensor. In the READ\_ID state, the system captures the vehicle's identification code from the input port. This allows the system to identify the specific account associated with the approaching vehicle. Once the vehicle ID has been successfully captured and validated, the system transitions to the CHECK\_BALANCE state, where it verifies whether the account has sufficient funds to cover the toll fee.

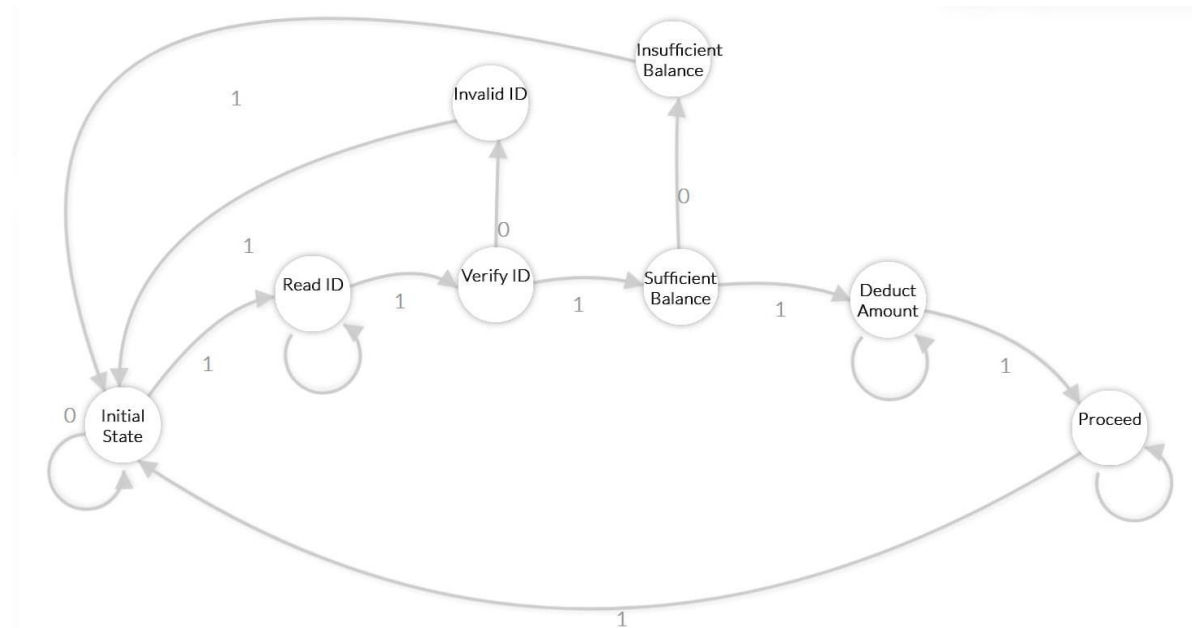
In the CHECK\_BALANCE state, the system continuously monitors the account balance against the predefined toll amount. If the account has sufficient funds ( $\text{balance} \geq 10$  units), the system transitions to the CHARGE\_ACCOUNT state, where it processes the transaction by deducting the toll amount from the vehicle's account. During this charging process, the system provides visual feedback through a blue LED indication and displays "CHARGE" on the output display. Once the transaction is successfully completed, the system transitions to the OPEN\_GATE state, where it commands the barrier to open and displays "GO" with a green LED indication, allowing the vehicle to pass through.

In the OPEN\_GATE state, the system continues to monitor the exit sensor for vehicle departure. Once the vehicle has completely passed through and triggers the exit sensor, the system transitions to the CLOSE\_GATE state. This is a critical safety state where the system ensures the barrier returns to its closed position. The barrier control mechanism uses a position counter that tracks the barrier's movement from fully closed (position 0) to fully open (position 7), providing precise feedback through `gate_is_open` and `gate_is_closed` status signals. Once the barrier has fully closed, the system returns to the IDLE state to await the next vehicle.

However, if during the CHECK\_BALANCE state the system detects insufficient funds in the vehicle's account, it immediately transitions to the REJECT\_TRANSACTION state. In this state, the system displays "FAIL" on the output display with a red LED indication, providing clear feedback to the driver about the rejected transaction. After showing this failure message for one clock cycle, the system automatically returns to the IDLE state, ready to process the next vehicle or allow for manual intervention.

Throughout this entire process, the system constantly monitors for reset conditions. If a reset signal is detected in any state, the system immediately transitions back to the IDLE state for

safety and reinitialization, halting all ongoing operations and resetting all internal registers and memory states. The reset signal can return the system to the IDLE state from any operational state, ensuring reliable recovery from any unexpected conditions or system errors.



## Chapter 2



# RTL CODE AND TESTBENCH

The RTL codes for the different modules are given below. The first code shown is for the module

```
toll_gate:
`timescale 1ns / 1ps

// DESCRIPTION: This is the main FSM that orchestrates the entire
// toll booth operation by interacting with submodules.

module toll_gate (

    // Port List

    clk,

    reset,

    sensor_vehicle_enter,

    sensor_vehicle_exit,

    vehicle_id_in,

    barrier_open_cmd,

    barrier_close_cmd,

    led_status,

    display_out

);

// Port Declarations

input clk;

input reset;
```

```
input sensor_vehicle_enter;

input sensor_vehicle_exit;

input [7:0] vehicle_id_in;


// Outputs to the Environment

output barrier_open_cmd;

output barrier_close_cmd;

output reg [1:0] led_status;

output reg [7:0] display_out;


// FSM State Parameters (Expanded for robust operation)

localparam IDLE          = 3'b000;

localparam READ_ID       = 3'b001;

localparam CHECK_BALANCE = 3'b010;

localparam CHARGE_ACCOUNT = 3'b011;

localparam REJECT_TRANSACTION = 3'b100;

localparam OPEN_GATE     = 3'b101;

localparam CLOSE_GATE    = 3'b110;


// FSM State Registers

reg [2:0] state, next_state;


// Internal control signals from FSM to submodules (must be 'reg')

reg read_id_en;
```

```
reg balance_check_req;

reg charge_req;

reg open_gate_en;

reg close_gate_en;


// Internal status signals from submodules to FSM (must be 'wire')

wire id_valid;

wire [3:0] id_to_check;

wire sufficient_balance;

wire charge_done;

wire gate_is_open;

wire gate_is_closed;


// Signals driven by a submodule's output port MUST be wires.

wire barrier_open_cmd;

wire barrier_close_cmd;


// I. STATE REGISTER (Sequential Logic with SYNCHRONOUS RESET)

always @(posedge clk) begin

    if (reset)

        state <= IDLE;
```

```

        else

            state <= next_state;

    end

// II. NEXT STATE & OUTPUT LOGIC (Single Combinational Block)

always @(*) begin

    // Default assignments for all control signals

    next_state          = state;

    read_id_en          = 1'b0;

    balance_check_req   = 1'b0;

    charge_req          = 1'b0;

    open_gate_en        = 1'b0;

    close_gate_en       = 1'b0;

    led_status          = 2'b01; // Default to RED LED

    display_out         = "----";

    //FSM Logic

    case (state)

        IDLE: begin

            display_out = "IDLE";

            close_gate_en = 1'b1; // Ensure gate is closed

```

```
        if (sensor_vehicle_enter) begin

            next_state = READ_ID;

        end

    end

    READ_ID: begin

        display_out = "READ";

        read_id_en = 1'b1; // Enable ID reading

        if (id_valid) begin

            next_state = CHECK_BALANCE;

        end

    end

    CHECK_BALANCE: begin

        display_out = "CHECK";

        balance_check_req = 1'b1;

        if (sufficient_balance) begin

            next_state = CHARGE_ACCOUNT;

        end else begin

            next_state = REJECT_TRANSACTION;

        end

    end

end
```

```
valid

CHARGE_ACCOUNT: begin

    led_status = 2'b11; // BLUE (Processing)

    display_out = "CHARGE";

    // Keep balance check active so 'sufficient_balance' remains

    balance_check_req = 1'b1;

    charge_req = 1'b1;

    if (charge_done) begin

        next_state = OPEN_GATE;

    end

end

REJECT_TRANSACTION: begin

    display_out = "FAIL";

    // After showing fail message for one cycle, return to idle

    next_state = IDLE;

end

OPEN_GATE: begin

    led_status = 2'b10; // GREEN LED

    display_out = "GO";

    open_gate_en = 1'b1; // Command the barrier to open
```

```

        if (sensor_vehicle_exit) begin

            next_state = CLOSE_GATE;

        end

    end

CLOSE_GATE: begin

    display_out    = "CLOSE";

    close_gate_en = 1'b1; // Command the barrier to close

    if (gate_is_closed) begin

        next_state = IDLE;

    end

end

default: begin

    next_state = IDLE;

end

endcase

end

// III. MODULE INSTANTIATIONS

```

```
vehicle_interface u_vehicle_interface (  
  
    .clk(clk),  
  
    .reset(reset),  
  
    .fsm_read_en(read_id_en),  
  
    .vehicle_id_in(vehicle_id_in),  
  
    .id_to_check(id_to_check),  
  
    .id_valid(id_valid)  
);
```

```
account_balance u_account_balance (  
  
    .clk(clk),  
  
    .reset(reset),  
  
    .id_to_check(id_to_check),  
  
    .fsm_check_req(balance_check_req),  
  
    .fsm_charge_req(charge_req),  
  
    .sufficient_balance(sufficient_balance),  
  
    .charge_done(charge_done)  
);
```

```
barrier_ctrl u_barrier_ctrl (  
  
    .clk(clk),  
  
    .reset(reset),
```



```

        .fsm_open_en(open_gate_en),

        .fsm_close_en(close_gate_en),

        .barrier_open_cmd(barrier_open_cmd),

        .barrier_close_cmd(barrier_close_cmd),

        .gate_is_open(gate_is_open),

        .gate_is_closed(gate_is_closed)

    );

endmodule


// MODULE: vehicle_interface

// DESCRIPTION: Captures the vehicle ID when enabled by the FSM.

module vehicle_interface (

    clk, reset, fsm_read_en, vehicle_id_in, id_to_check, id_valid

);

    input clk;

    input reset;

    input fsm_read_en;

    input [7:0] vehicle_id_in;

    output reg [3:0] id_to_check;

    output reg id_valid;


    // MODIFIED: Converted to synchronous reset.

```

```
always @(posedge clk) begin

    if (reset) begin

        id_to_check <= 4'h00;

        id_valid <= 1'b0;

    end else begin

        if (fsm_read_en) begin

            id_to_check <= vehicle_id_in;

            id_valid <= 1'b1;

        end else begin

            id_valid <= 1'b0;

        end

    end

end

endmodule
```

```
// MODULE: account_balance
```

```
// DESCRIPTION: Manages a simple memory-based account database.
```

```
module account_balance (

    clk, reset, id_to_check, fsm_check_req, fsm_charge_req,

    sufficient_balance, charge_done
```

```

);

input clk;

input reset;

input [7:0] id_to_check;

input fsm_check_req;

input fsm_charge_req;

output reg sufficient_balance;

output reg charge_done;


localparam TOLL_AMOUNT = 16'd10;

reg [15:0] balance_db [0:15];

wire [3:0] db_addr = id_to_check[3:0];


// Use of a synthesizable reset to initialize the memory.

always @(posedge clk) begin

    if (reset) begin

        charge_done <= 1'b0;

        // Initialize memory on reset

        balance_db[4'h01] <= 16'd100; // Has funds

        balance_db[4'h02] <= 16'd5;   // No funds

        balance_db[4'h03] <= 16'd20;  // Has funds


    end else begin

```

```

        charge_done <= 1'b0; // Default to not done

        if (fsm_charge_req && sufficient_balance) begin

            balance_db[db_addr] <= balance_db[db_addr] - TOLL_AMOUNT;

            charge_done <= 1'b1;

        end

    end

end

// Combinational logic for checking balance remains the same.
always @(*) begin

    if (fsm_check_req) begin

        if (balance_db[db_addr] >= TOLL_AMOUNT)

            sufficient_balance = 1'b1;

        else

            sufficient_balance = 1'b0;

        end else begin

            sufficient_balance = 1'b0;

        end

    end

endmodule

// MODULE: barrier_ctrl

```

```
// DESCRIPTION: Simulates the physical barrier motor and position.
```

```
module barrier_ctrl (  
    clk, reset, fsm_open_en, fsm_close_en,  
    barrier_open_cmd, barrier_close_cmd, gate_is_open, gate_is_closed  
);  
  
    input clk;  
  
    input reset;  
  
    input fsm_open_en;  
  
    input fsm_close_en;  
  
    output reg barrier_open_cmd;  
  
    output reg barrier_close_cmd;  
  
    output gate_is_open;  
  
    output gate_is_closed;  
  
  
    reg [2:0] barrier_position_counter;  
  
    assign gate_is_open = (barrier_position_counter == 3'd7);  
  
    assign gate_is_closed = (barrier_position_counter == 3'd0);  
  
  
    always @(posedge clk) begin  
  
        if (reset) begin  
  
            barrier_open_cmd <= 1'b0;  
  
            barrier_close_cmd <= 1'b1;  

```

```

        barrier_position_counter <= 3'd0;

    end else begin

        if (fsm_open_en && !gate_is_open) begin

            barrier_open_cmd <= 1'b1;

            barrier_close_cmd <= 1'b0;

            barrier_position_counter <= barrier_position_counter + 1;

        end else if (fsm_close_en && !gate_is_closed) begin

            barrier_open_cmd <= 1'b0;

            barrier_close_cmd <= 1'b1;

            barrier_position_counter <= barrier_position_counter - 1;

        end else begin

            barrier_open_cmd <= 1'b0;

            barrier_close_cmd <= 1'b0;

            if (fsm_close_en && gate_is_closed)

                barrier_close_cmd <= 1'b1;

            else if (fsm_open_en && gate_is_open)

                barrier_open_cmd <= 1'b1;

        end

    end

end

endmodule

```

The following is the first testbench to confirm the functionality of the RTL code provided previously:

```
`timescale 1ns / 1ps

module tb_toll_gate_enhanced;

    reg clk;
    reg reset;
    reg sensor_vehicle_enter;
    reg sensor_vehicle_exit;
    reg [7:0] vehicle_id_in;

    wire barrier_open_cmd;
    wire barrier_close_cmd;
    wire [1:0] led_status;
    wire [7:0] display_out;

    toll_gate uut (
        .clk(clk),
        .reset(reset),
        .sensor_vehicle_enter(sensor_vehicle_enter),
        .sensor_vehicle_exit(sensor_vehicle_exit),
        .vehicle_id_in(vehicle_id_in),
        .barrier_open_cmd(barrier_open_cmd),
        .barrier_close_cmd(barrier_close_cmd),
        .led_status(led_status),
```

```

        .display_out(display_out)
    );

// Clock Generation
initial begin
    clk = 0;
    forever #5 clk = ~clk;
end

initial begin
    // Initialize signals
    reset = 1;
    sensor_vehicle_enter = 0;
    sensor_vehicle_exit = 0;
    vehicle_id_in = 8'h00;

    // Apply reset
    #20;
    reset = 0;
    #10;

    $monitor("Time=%0t | State=%b | Enter=%b Exit=%b ID=%h | Open=%b Close=%b  
LED=%b Display=%s",
        $time, uut.state, sensor_vehicle_enter, sensor_vehicle_exit, vehicle_id_in,
        barrier_open_cmd, barrier_close_cmd, led_status, display_out);

    // SCENARIO 1: Successful transaction with ID 01
    $display("\n=== SCENARIO 1: Successful Transaction (ID=01) ===");
    @(posedge clk);
    sensor_vehicle_enter = 1;
    vehicle_id_in = 8'h01;
    @(posedge clk);

```



```

@(posedge clk);
sensor_vehicle_enter = 0;

// Wait for FSM to go through READ_ID -> CHECK_BALANCE ->
CHARGE_ACCOUNT -> OPEN_GATE
#100;

// Trigger exit sensor while gate is opening
sensor_vehicle_exit = 1;
@(posedge clk);
@(posedge clk);
sensor_vehicle_exit = 0;

// Wait for gate to close
#150;

// SCENARIO 2: Insufficient funds (ID=02)
$display("\n=== SCENARIO 2: Insufficient Funds (ID=02) ===");
@(posedge clk);
sensor_vehicle_enter = 1;
vehicle_id_in = 8'h02;
@(posedge clk);
@(posedge clk);
sensor_vehicle_enter = 0;

// Wait for rejection
#50;

// SCENARIO 3: Valid ID with multiple checks (ID=03)
$display("\n=== SCENARIO 3: Valid Transaction (ID=03) ===");
@(posedge clk);
sensor_vehicle_enter = 1;

```

```
vehicle_id_in = 8'h03;
@(posedge clk);
@(posedge clk);
sensor_vehicle_enter = 0;

#100;
sensor_vehicle_exit = 1;
@(posedge clk);
@(posedge clk);
sensor_vehicle_exit = 0;
#150;

// SCENARIO 4: Test edge cases in vehicle_interface
$display("\n=== SCENARIO 4: Vehicle Interface Edge Cases ===");

// Test with different ID values to cover all bits
@(posedge clk);
sensor_vehicle_enter = 1;
vehicle_id_in = 8'hFF; // All 1s
@(posedge clk);
sensor_vehicle_enter = 0;
#50;

@(posedge clk);
sensor_vehicle_enter = 1;
vehicle_id_in = 8'h55; // Alternating bits
@(posedge clk);
sensor_vehicle_enter = 0;
#50;

@(posedge clk);
```

```

sensor_vehicle_enter = 1;
vehicle_id_in = 8'hAA; // Alternating bits
@(posedge clk);
sensor_vehicle_enter = 0;
#50;

// SCENARIO 5: Test barrier_ctrl edge cases
$display("\n=== SCENARIO 5: Barrier Control Edge Cases ===");

// Normal operation with ID=01
@(posedge clk);
sensor_vehicle_enter = 1;
vehicle_id_in = 8'h01;
@(posedge clk);
@(posedge clk);
sensor_vehicle_enter = 0;

// Wait until gate fully opens
wait(uut.u_barrier_ctrl.gate_is_open == 1'b1);
$display("Gate fully opened at time %0t", $time);

// Hold at open position
#30;

// Exit while gate is fully open
sensor_vehicle_exit = 1;
@(posedge clk);
@(posedge clk);
sensor_vehicle_exit = 0;

// Wait for gate to fully close

```

```
wait(uut.u_barrier_ctrl.gate_is_closed == 1'b1);  
$display("Gate fully closed at time %0t", $time);  
#30;
```

```
// SCENARIO 6: Test account_balance edge cases
```

```
$display("\n=== SCENARIO 6: Account Balance Edge Cases ===");
```

```
// Test uninitialized memory location
```

```
@(posedge clk);  
sensor_vehicle_enter = 1;  
vehicle_id_in = 8'h0F; // Different address  
@(posedge clk);  
@(posedge clk);  
sensor_vehicle_enter = 0;  
#50;
```

```
// SCENARIO 7: Reset during operation
```

```
$display("\n=== SCENARIO 7: Reset During Operation ===");
```

```
@(posedge clk);  
sensor_vehicle_enter = 1;  
vehicle_id_in = 8'h01;  
@(posedge clk);  
@(posedge clk);  
sensor_vehicle_enter = 0;  
#20;
```

```
// Apply reset while in middle of transaction
```

```
reset = 1;  
#20;  
reset = 0;  
#30;
```

```

// SCENARIO 8: Rapid successive transactions

$display("\n=== SCENARIO 8: Rapid Successive Transactions ===");

@(posedge clk);
sensor_vehicle_enter = 1;
vehicle_id_in = 8'h01;

@(posedge clk);
sensor_vehicle_enter = 0;

#100;

sensor_vehicle_exit = 1;

@(posedge clk);
sensor_vehicle_exit = 0;

#150;


// Immediately start another transaction

@(posedge clk);
sensor_vehicle_enter = 1;
vehicle_id_in = 8'h03;

@(posedge clk);
sensor_vehicle_enter = 0;

#100;

sensor_vehicle_exit = 1;

@(posedge clk);
sensor_vehicle_exit = 0;

#150;


// SCENARIO 9: Test all FSM states and transitions

$display("\n=== SCENARIO 9: Complete FSM State Coverage ===");


// Ensure default case is hit (shouldn't happen normally)

// Force state to invalid value using force/release

```

```
force uut.state = 3'b111;
@(posedge clk);
release uut.state;
@(posedge clk);
#30;

// SCENARIO 10: Extended gate operations
$display("\n=== SCENARIO 10: Extended Gate Operations ===");
@(posedge clk);
sensor_vehicle_enter = 1;
vehicle_id_in = 8'h01;
@(posedge clk);
sensor_vehicle_enter = 0;

// Gate opening start
#60;

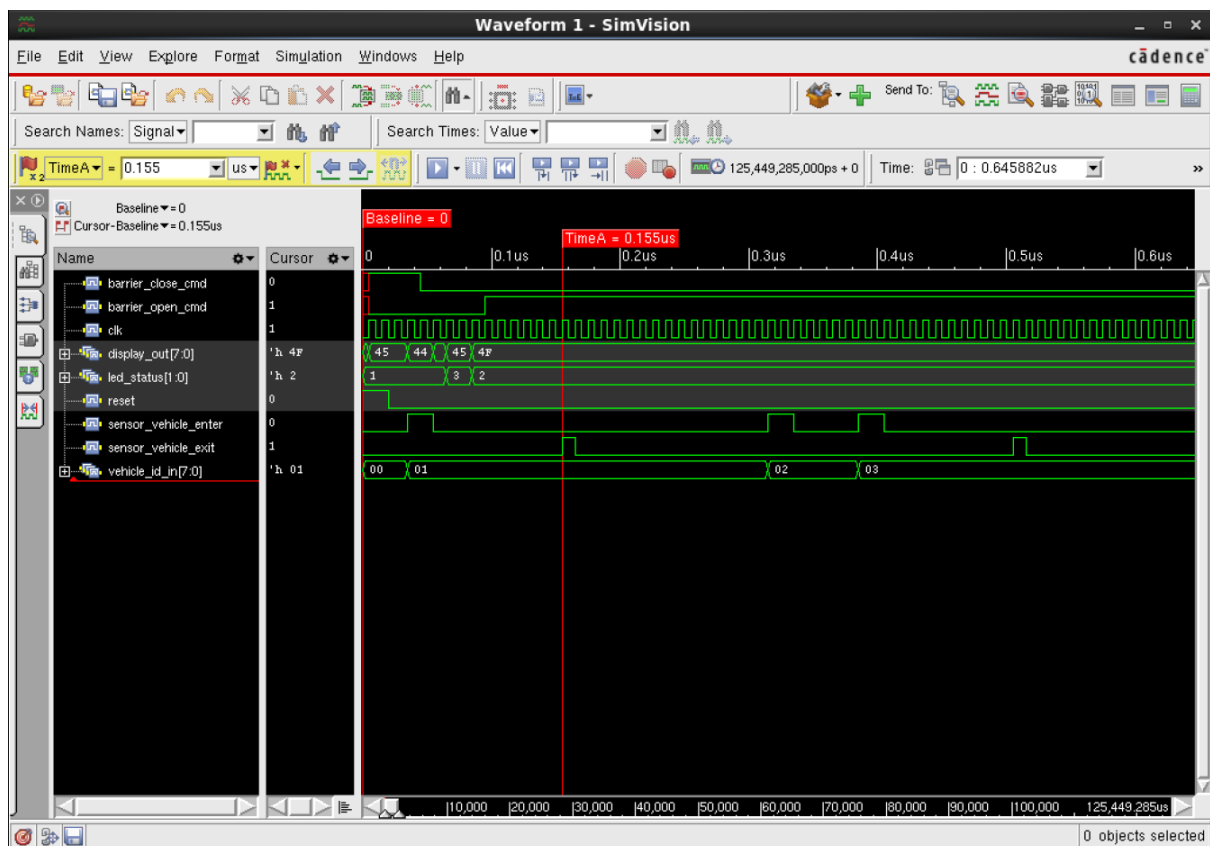
// Exit before gate fully opens
sensor_vehicle_exit = 1;
@(posedge clk);
@(posedge clk);
sensor_vehicle_exit = 0;

// Wait for complete close
#200;

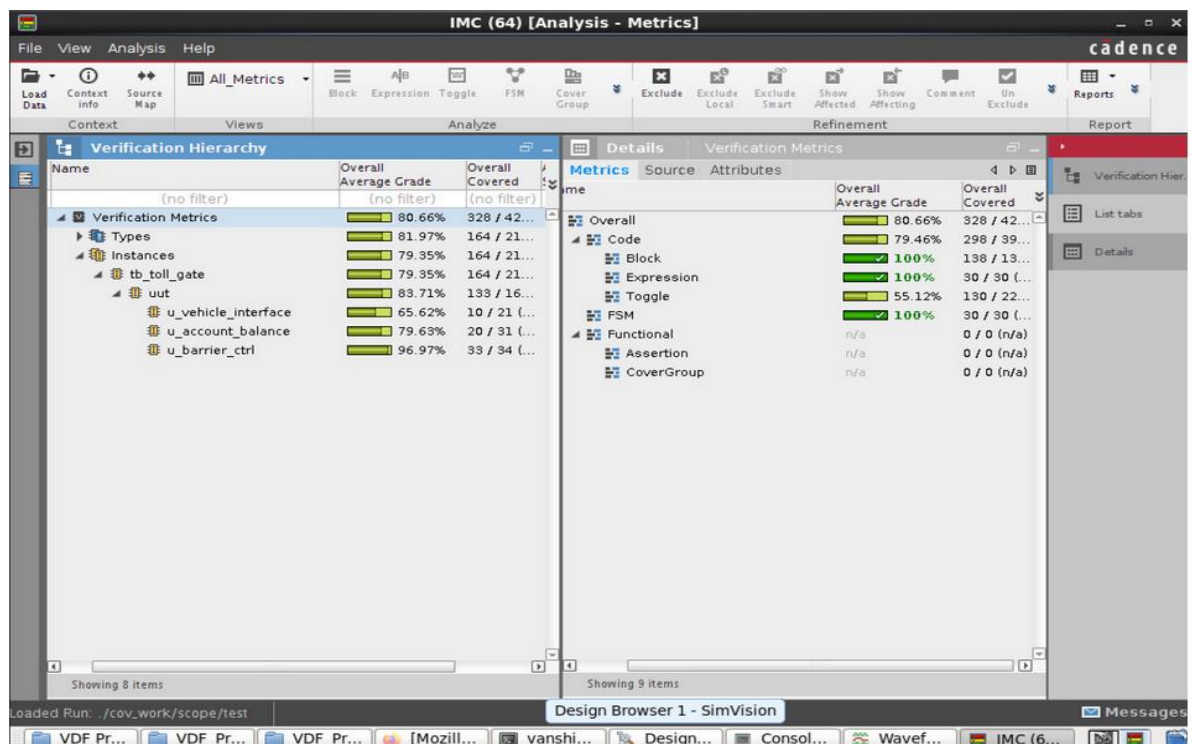
$display("\n=== All Test Scenarios Completed ===");
#100;
$finish;
end
```

```
// Coverage monitors  
initial begin  
    $dumpfile("tb_toll_gate.vcd");  
    $dumpvars(0, tb_toll_gate_enhanced);  
end  
  
endmodule
```

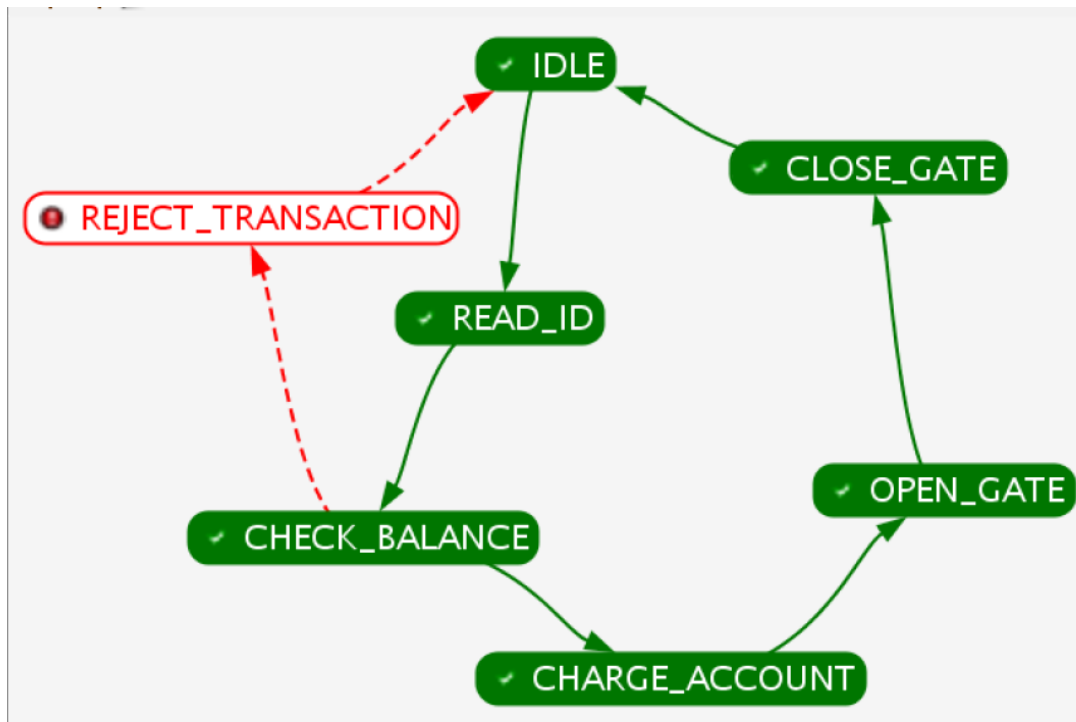
The simulation waveform for the testbench is shown below:



The coverage report for the testbench is given below







In the testbench, READ\_ID and VEHICLE\_ID\_IN signals are not thoroughly exercised during the operation of the Toll Management System, resulting in lowest module level coverage of U\_VEHICLE\_INTERFACE at 65.62%. Additionally, various balance scenarios of ACCOUNT\_DB module were not fully evaluated, thus, indicating the need for more test cases to implement all the presented scenarios. As a result, the FSM coverage is at 80.66%.

The second version of testbench code that validates the functionality of the RTL code used.

```
`timescale 1ns / 1ps
```

```
module tb_toll_gate;
```

```
    reg clk;
```

```
    reg reset;
```

```
    reg sensor_vehicle_enter;
```

```
    reg sensor_vehicle_exit;
```

```
    reg [7:0] vehicle_id_in;
```

```
    integer i;
```

```
    wire barrier_open_cmd;
```

```
    wire barrier_close_cmd;
```

```
    wire [1:0] led_status;
```

```
    wire [7:0] display_out;
```

```
    toll_gate uut (
```

```
        .clk(clk),
```

```
        .reset(reset),
```

```
        .sensor_vehicle_enter(sensor_vehicle_enter),
```

```
        .sensor_vehicle_exit(sensor_vehicle_exit),
```

```
        .vehicle_id_in(vehicle_id_in),
```

```
        .barrier_open_cmd(barrier_open_cmd),
```

```
        .barrier_close_cmd(barrier_close_cmd),
```

```
        .led_status(led_status),
```

```
        .display_out(display_out)
```

```
    );
```

```

// Clock Generation
initial begin
    clk = 0;
    forever #5 clk = ~clk;
end

initial begin
    // Initialize signals
    reset = 1;
    sensor_vehicle_enter = 0;
    sensor_vehicle_exit = 0;
    vehicle_id_in = 8'h00;

    // Apply reset
    #20;
    reset = 0;
    #10;

    $monitor("Time=%0t | State=%b | Enter=%b Exit=%b ID=%h | Open=%b  

    Close=%b LED=%b Display=%h",
        $time, uut.state, sensor_vehicle_enter, sensor_vehicle_exit,
        vehicle_id_in,
        barrier_open_cmd, barrier_close_cmd, led_status, display_out);

    // SCENARIO 1: Successful transaction with ID 01
    $display("\n=== SCENARIO 1: Successful Transaction (ID=01) ===");

```

```

    @(posedge clk);
    sensor_vehicle_enter = 1;
    vehicle_id_in = 8'h01;
    @(posedge clk);
    @(posedge clk);
    sensor_vehicle_enter = 0;

    // Wait for FSM to go through READ_ID -> CHECK_BALANCE ->
    CHARGE_ACCOUNT -> OPEN_GATE
    #100;

    // Trigger exit sensor while gate is opening
    sensor_vehicle_exit = 1;
    @(posedge clk);
    @(posedge clk);
    sensor_vehicle_exit = 0;

    // Wait for gate to close
    #150;

    // SCENARIO 2: Insufficient funds (ID=02)
    $display("\n=== SCENARIO 2: Insufficient Funds (ID=02) ===");
    @(posedge clk);
    sensor_vehicle_enter = 1;
    vehicle_id_in = 8'h02;
    @(posedge clk);
    @(posedge clk);

```

```
sensor_vehicle_enter = 0;
```

```
// Wait for rejection
```

```
#50;
```

```
// SCENARIO 3: Valid ID with multiple checks (ID=03)
```

```
$display("\n=== SCENARIO 3: Valid Transaction (ID=03) ===");
```

```
@(posedge clk);
```

```
sensor_vehicle_enter = 1;
```

```
vehicle_id_in = 8'h03;
```

```
@(posedge clk);
```

```
@(posedge clk);
```

```
sensor_vehicle_enter = 0;
```

```
#100;
```

```
sensor_vehicle_exit = 1;
```

```
@(posedge clk);
```

```
@(posedge clk);
```

```
sensor_vehicle_exit = 0;
```

```
#150;
```

```
// SCENARIO 4: Test edge cases in vehicle_interface
```

```
$display("\n=== SCENARIO 4: Vehicle Interface Edge Cases ===");
```

```
// Test with different ID values to cover all bits
```

```
@(posedge clk);
```

```
sensor_vehicle_enter = 1;  
vehicle_id_in = 8'hFF; // All 1s  
@(posedge clk);  
sensor_vehicle_enter = 0;  
#50;
```

```
@(posedge clk);  
sensor_vehicle_enter = 1;  
vehicle_id_in = 8'h55; // Alternating bits  
@(posedge clk);  
sensor_vehicle_enter = 0;  
#50;
```

```
@(posedge clk);  
sensor_vehicle_enter = 1;  
vehicle_id_in = 8'hAA; // Alternating bits  
@(posedge clk);  
sensor_vehicle_enter = 0;  
#50;
```

```
// SCENARIO 5: Test barrier_ctrl edge cases  
$display("\n=== SCENARIO 5: Barrier Control Edge Cases ===");
```

```
// Normal operation with ID=01  
@(posedge clk);  
sensor_vehicle_enter = 1;
```

```
vehicle_id_in = 8'h01;
@(posedge clk);
@(posedge clk);
sensor_vehicle_enter = 0;

// Wait until gate fully opens (approximate timing)
repeat(20) @(posedge clk);
$display("Gate fully opened at time %0t", $time);

// Hold at open position
#30;

// Exit while gate is fully open
sensor_vehicle_exit = 1;
@(posedge clk);
@(posedge clk);
sensor_vehicle_exit = 0;

// Wait for gate to fully close (approximate timing)
repeat(20) @(posedge clk);
$display("Gate fully closed at time %0t", $time);
#30;

// SCENARIO 6: Test account_balance edge cases
$display("\n=== SCENARIO 6: Account Balance Edge Cases ===");
```

```
// Test uninitialized memory location
@(posedge clk);
sensor_vehicle_enter = 1;
vehicle_id_in = 8'h0F; // Different address
@(posedge clk);
@(posedge clk);
sensor_vehicle_enter = 0;
#50;
```

```
// SCENARIO 7: Reset during operation
$display("\n=== SCENARIO 7: Reset During Operation ===");
@(posedge clk);
sensor_vehicle_enter = 1;
vehicle_id_in = 8'h01;
@(posedge clk);
@(posedge clk);
sensor_vehicle_enter = 0;
#20;
```

```
// Apply reset while in middle of transaction
reset = 1;
#20;
reset = 0;
#30;
```

```
// SCENARIO 8: Rapid successive transactions
```



```
$display("\n=== SCENARIO 8: Rapid Successive Transactions ===");  
  
@(posedge clk);  
sensor_vehicle_enter = 1;  
vehicle_id_in = 8'h01;  
  
@(posedge clk);  
sensor_vehicle_enter = 0;  
  
#100;  
  
sensor_vehicle_exit = 1;  
  
@(posedge clk);  
sensor_vehicle_exit = 0;  
  
#150;  
  
  
// Immediately start another transaction  
  
@(posedge clk);  
sensor_vehicle_enter = 1;  
vehicle_id_in = 8'h03;  
  
@(posedge clk);  
sensor_vehicle_enter = 0;  
  
#100;  
  
sensor_vehicle_exit = 1;  
  
@(posedge clk);  
sensor_vehicle_exit = 0;  
  
#150;  
  
  
// SCENARIO 9: Test all FSM states and transitions  
  
$display("\n=== SCENARIO 9: Complete FSM State Coverage ===");
```

```
// Ensure default case is hit (shouldn't happen normally)
// Force state to invalid value using force/release
force uut.state = 3'b111;
@(posedge clk);
release uut.state;
@(posedge clk);
#30;

// SCENARIO 10: Extended gate operations
$display("\n=== SCENARIO 10: Extended Gate Operations ===");
@(posedge clk);
sensor_vehicle_enter = 1;
vehicle_id_in = 8'h01;
@(posedge clk);
sensor_vehicle_enter = 0;

// gate starts opening
#60;

// Exit before gate fully opens
sensor_vehicle_exit = 1;
@(posedge clk);
@(posedge clk);
sensor_vehicle_exit = 0;
```

```
// Wait for complete close
#200;

// SCENARIO 11: Test all possible combinations of sensors
$display("\n=== SCENARIO 11: Sensor Combinations ===");

// Test all combinations of enter and exit sensors
@(posedge clk);
sensor_vehicle_enter = 0;
sensor_vehicle_exit = 0;

@(posedge clk);
#10;

@(posedge clk);
sensor_vehicle_enter = 1;
sensor_vehicle_exit = 0;

@(posedge clk);
#10;

@(posedge clk);
sensor_vehicle_enter = 0;
sensor_vehicle_exit = 1;

@(posedge clk);
#10;

@(posedge clk);
sensor_vehicle_enter = 1;
```

```
sensor_vehicle_exit = 1;
@(posedge clk);
#10;

// Test rapid toggling of sensors
i = 0;
while (i < 5) begin
    @(posedge clk);
    sensor_vehicle_enter = ~sensor_vehicle_enter;
    sensor_vehicle_exit = ~sensor_vehicle_exit;
    @(posedge clk);
    i = i + 1;
end

// SCENARIO 12: Test boundary conditions
$display("\n=== SCENARIO 12: Boundary Conditions ===");

// Test with maximum ID value
@(posedge clk);
sensor_vehicle_enter = 1;
vehicle_id_in = 8'hFF;
@(posedge clk);
sensor_vehicle_enter = 0;
#50;

// Test with minimum ID value (0)
```

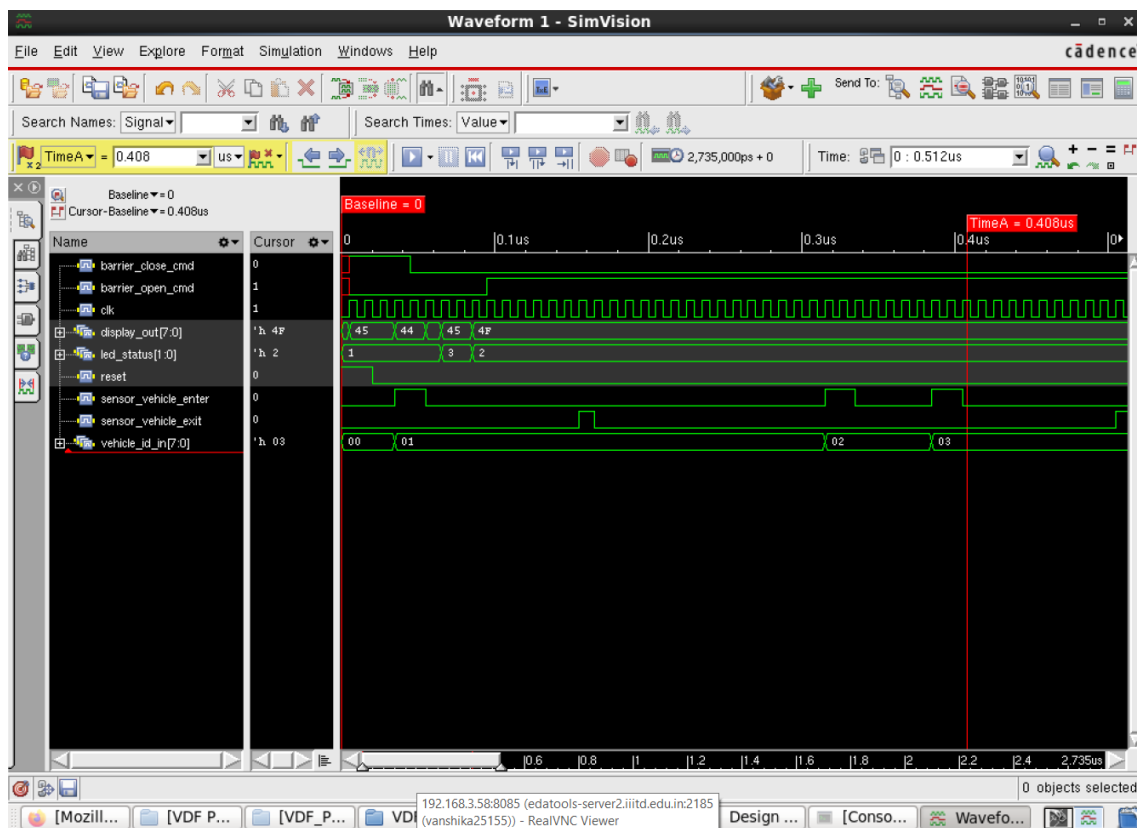
```
    @(posedge clk);
    sensor_vehicle_enter = 1;
    vehicle_id_in = 8'h00;
    @(posedge clk);
    sensor_vehicle_enter = 0;
    #50;

    $display("\n=== All Test Scenarios Completed ===");
    #100;
    $finish;
end

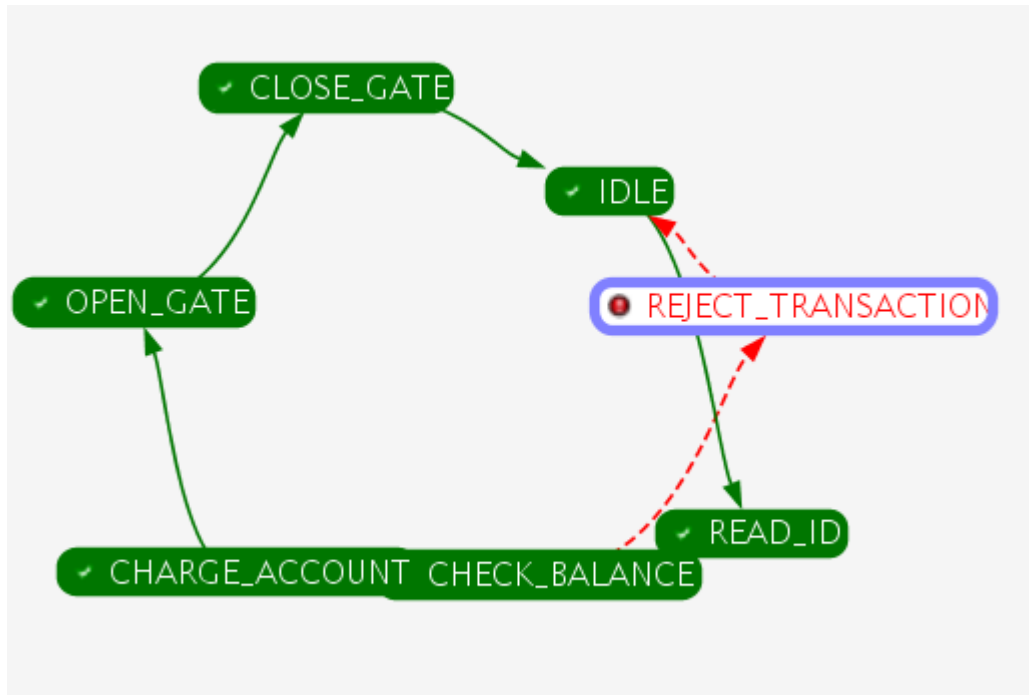
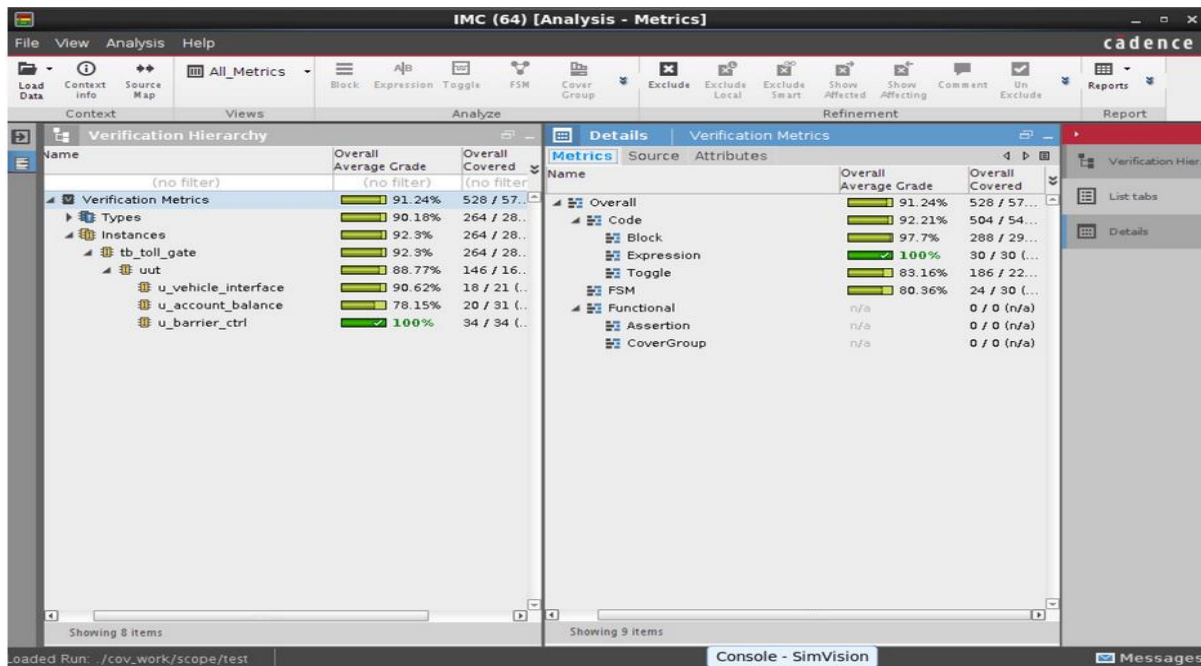
// Coverage monitors
initial begin
    $dumpfile("tb_toll_gate.vcd");
    $dumpvars(0, tb_toll_gate);
end

endmodule
```

The simulation waveform for the testbench is shown below:



The coverage report for the testbench is given below:



In the testbench, the state transitions from CHECK\_BALANCE to REJECT\_TRANSACTION and from REJECT\_TRANSACTION to IDLE are not performed. The RESET functionality has not been evaluated. Additionally, the vehicle\_id and account\_db registers are not fully utilized during the operation of the Toll Management System. As a result, the FSM coverage is at 91.24%.

The third version of testbench code that validates the functionality of the RTL code used.

```
`timescale 1ns / 1ps

module tb_toll_gate_coverage;

    reg clk;
    reg reset;
    reg sensor_vehicle_enter;
    reg sensor_vehicle_exit;
    reg [7:0] vehicle_id_in;

    wire barrier_open_cmd;
    wire barrier_close_cmd;
    wire [1:0] led_status;
    wire [7:0] display_out;

    integer i; // for loops

    // Instantiate the Device Under Test (DUT)
    toll_gate uut (
        .clk(clk),
        .reset(reset),
        .sensor_vehicle_enter(sensor_vehicle_enter),
        .sensor_vehicle_exit(sensor_vehicle_exit),
        .vehicle_id_in(vehicle_id_in),
        .barrier_open_cmd(barrier_open_cmd),
        .barrier_close_cmd(barrier_close_cmd),
```



```
.led_status(led_status),  
.display_out(display_out)  
);
```

```
// Clock Generation
```

```
initial begin
```

```
    clk = 0;
```

```
    forever #5 clk = ~clk;
```

```
end
```

```
// Main Test Sequence
```

```
initial begin
```

```
    // 1. Initialize and Reset
```

```
    $display("--- [START] Initial Reset ---");
```

```
    reset = 1;
```

```
    sensor_vehicle_enter = 0;
```

```
    sensor_vehicle_exit = 0;
```

```
    vehicle_id_in = 8'h00;
```

```
    #20;
```

```
    reset = 0;
```

```
    #10;
```

```
    // 2. Setup Monitoring
```

```
    $monitor("Time=%0t | State=%s | Enter=%b, Exit=%b, ID_in=%h |  
OpenCmd=%b, CloseCmd=%b, LED=%b, Display=%s",
```

```
        $time, uut.state, sensor_vehicle_enter, sensor_vehicle_exit,  
        vehicle_id_in,
```

```
        barrier_open_cmd, barrier_close_cmd, led_status, display_out);

// --- SCENARIO 1: Basic Functional Tests ---
$display("\n--- [START] Basic Functional Tests ---");
run_transaction(8'h01, 1); // Successful Transaction
run_transaction(8'h02, 0); // Failed Transaction (insufficient funds)

// --- SCENARIO 2: Mid-Operation Reset Test ---
$display("\n--- [START] Mid-Operation Reset Test ---");
sensor_vehicle_enter = 1;
vehicle_id_in = 8'h01;
#30; // Let it get past the IDLE state
sensor_vehicle_enter = 0;
$display("INFO: FSM is active. Asserting reset now.");
#20;
reset = 1; // Assert reset mid-operation
#20; // Hold reset
$display("INFO: Releasing reset. FSM should be in IDLE state.");
reset = 0;
#20;
$display("INFO: Verifying recovery by starting a new transaction.");
run_transaction(8'h03, 1); // Verify recovery

// --- SCENARIO 3: Back-to-Back Vehicle Test ---
$display("\n--- [START] Back-to-Back Vehicle Test ---");
run_transaction_and_interrupt(8'h01); // Vehicle 1 starts
```

```
// Vehicle 2 will be handled by the task.
```

```
// --- SCENARIO 4: Randomized Stress Test ---
```

```
$display("\n--- [START] Randomized Stress Test (50 iterations) ---");
```

```
for (i = 0; i < 50; i = i + 1) begin
```

```
    // Use random to get a random 8-bit vehicle ID.
```

```
    // The DUT will determine if it's a pass or fail.
```

```
    run_transaction($urandom, $urandom % 2);
```

```
end
```

```
// --- End Simulation ---
```

```
#100;
```

```
$display("\n--- Comprehensive Testbench Finished ---");
```

```
$finish;
```

```
end
```

```
// Task to run a standard transaction
```

```
task run_transaction;
```

```
    input [7:0] id;
```

```
    input is_successful_guess; // Testbench doesn't know balance, so this is a  
    guess for timing
```

```
begin
```

```
    $display("\n--- Testing Vehicle ID: %h ---", id);
```

```
    @(posedge clk);
```

```
    sensor_vehicle_enter = 1;
```

```
    vehicle_id_in = id;
```

```

    @(posedge clk); @(posedge clk);
    sensor_vehicle_enter = 0;

    // Wait for a period. If the gate opens, assume success.
    #80;
    if (barrier_open_cmd == 1'b1) begin
        $display("INFO: Gate is opening for ID %h. Assuming success.", id);
        #100; // Extra wait for gate to open fully
        sensor_vehicle_exit = 1;
        @(posedge clk); @(posedge clk);
        sensor_vehicle_exit = 0;
        #100; // Wait for gate to close
    end else begin
        $display("INFO: Gate did not open for ID %h. Assuming failure.", id);
        #40; // Wait for FSM to return to idle on failure
    end
    vehicle_id_in = 8'h00;
    #40; // Idle time
end
endtask

// Special task for the back-to-back test
task run_transaction_and_interrupt;
    input [7:0] id_v1;

begin

```

```

$display("--- Vehicle 1 (ID: %h) arriving... ---", id_v1);
@(posedge clk);
sensor_vehicle_enter = 1;
vehicle_id_in = id_v1;
@(posedge clk); @(posedge clk);
sensor_vehicle_enter = 0;

#180; // Wait for gate to open and vehicle to pass
sensor_vehicle_exit = 1;
@(posedge clk);
$display("INFO: Vehicle 1 is exiting. Vehicle 2 (ID: FF) arriving now.");
// As V1 exits, V2 arrives
sensor_vehicle_enter = 1;
vehicle_id_in = 8'hFF;
@(posedge clk);
sensor_vehicle_exit = 0; // V1 has cleared

// FSM should start closing the gate for V1, ignoring V2 for now
#80; // Wait for gate to close and FSM to return to IDLE

$display("INFO: FSM should now process Vehicle 2.");
@(posedge clk);
@(posedge clk);
sensor_vehicle_enter = 0;

// --- SCENARIO 6: Stuck Vehicle (Timeout Test) ---

```

```
$display("\n--- [START] Stuck Vehicle / Timeout Test ---");  
  
// Start a normal successful transaction  
run_transaction(8'h1A, 1);  
  
// Now, start another transaction but never trigger the exit sensor  
$display("\n--- Testing Vehicle ID: %h (will get stuck) ---", 8'h1B);  
@(posedge clk);  
sensor_vehicle_enter = 1;  
vehicle_id_in = 8'h1B;  
@(posedge clk); @(posedge clk);  
sensor_vehicle_enter = 0;  
  
// Wait for the gate to open  
#120;  
  
$display("INFO: Gate is OPEN, but vehicle will not exit. Waiting for a long  
time...");  
  
// In a real design, the FSM would be stuck here. We will wait and then reset.  
#1000; // Wait for a long duration  
  
$display("INFO: System is stuck as expected. Resetting to recover.");  
reset = 1;  
#20;  
reset = 0;  
#20;  
  
$display("INFO: System recovered. Running one final transaction.");
```

```
run_transaction(8'h1C, 1); // Verify system works after recovery
```

```
$display("\n--- [END] Stuck Vehicle / Timeout Test ---");
```

```
    // Run the rest of the transaction for Vehicle 2
```

```
    #180;
```

```
    sensor_vehicle_exit = 1;
```

```
    @(posedge clk); @(posedge clk);
```

```
    sensor_vehicle_exit = 0;
```

```
    #100;
```

```
    vehicle_id_in = 8'h00;
```

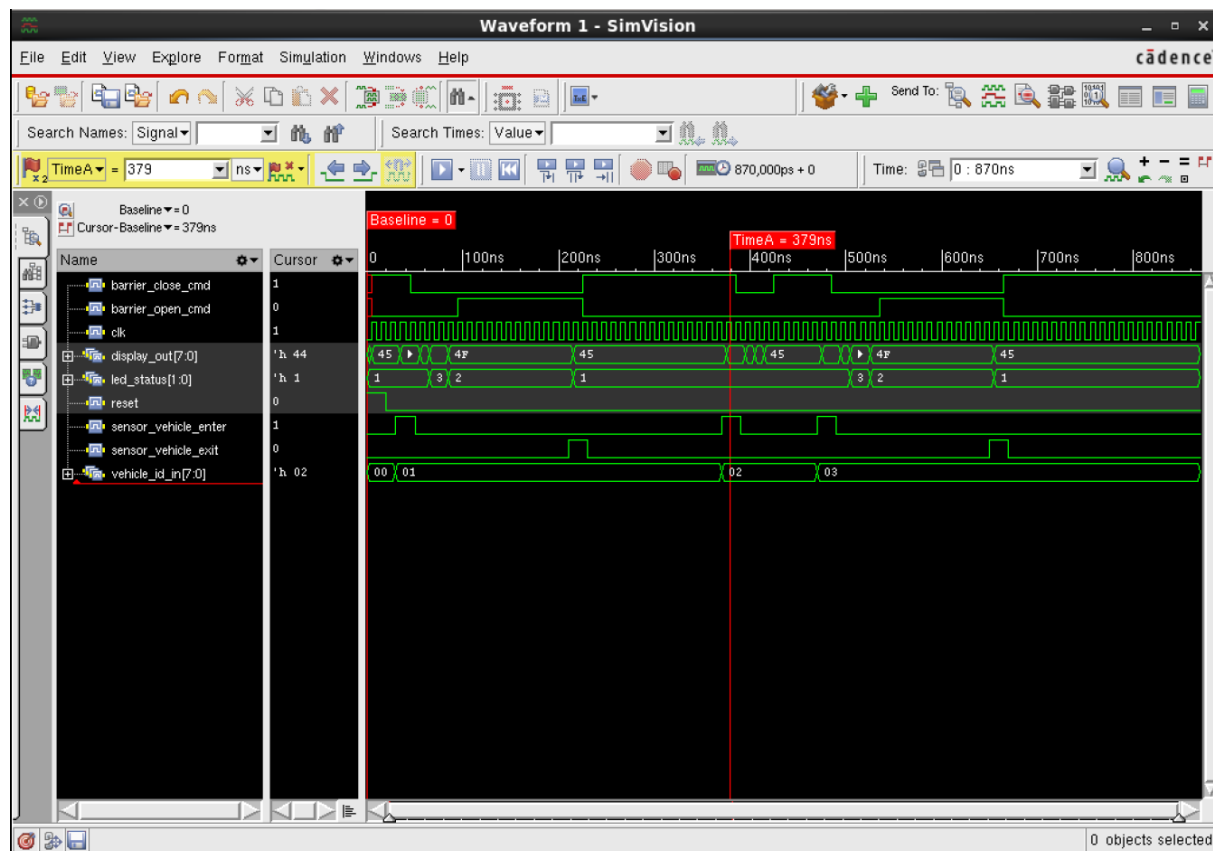
```
    #40;
```

```
end
```

```
endtask
```

```
endmodule
```

The simulation waveforms for the testbench are shown below:



The waveform presents the operational sequence of the intelligent toll booth management system, capturing key signals that demonstrate the system's response to multiple vehicle transactions. The clock signal (*clk*) establishes the timing foundation with a regular 10ns period, ensuring all state transitions occur synchronously on rising edges. The reset signal remains low throughout this segment, indicating the system is operating under normal conditions without initialization or recovery sequences.

The *sensor\_vehicle\_enter* signal pulses high every time a new vehicle is detected which indicates the triggering the FSM to begin processing a new vehicle. The *vehicle\_id\_in* bus, then displays different identification codes — first 01, then 02, and finally 03 — representing successive vehicles approaching the toll gate. For each vehicle, the system captures the ID, checks the account balance, and proceeds accordingly. When ID 01 is processed, the *display\_out* updates to show “IDLE”, then “READ”, “CHECK”, “CHARGE”, and finally “GO” as the transaction completes and the barrier opens. The *led\_status* signal transitions from



red (idle/reject) to blue (processing) and then green (go), providing clear visual feedback aligned with the system's state.

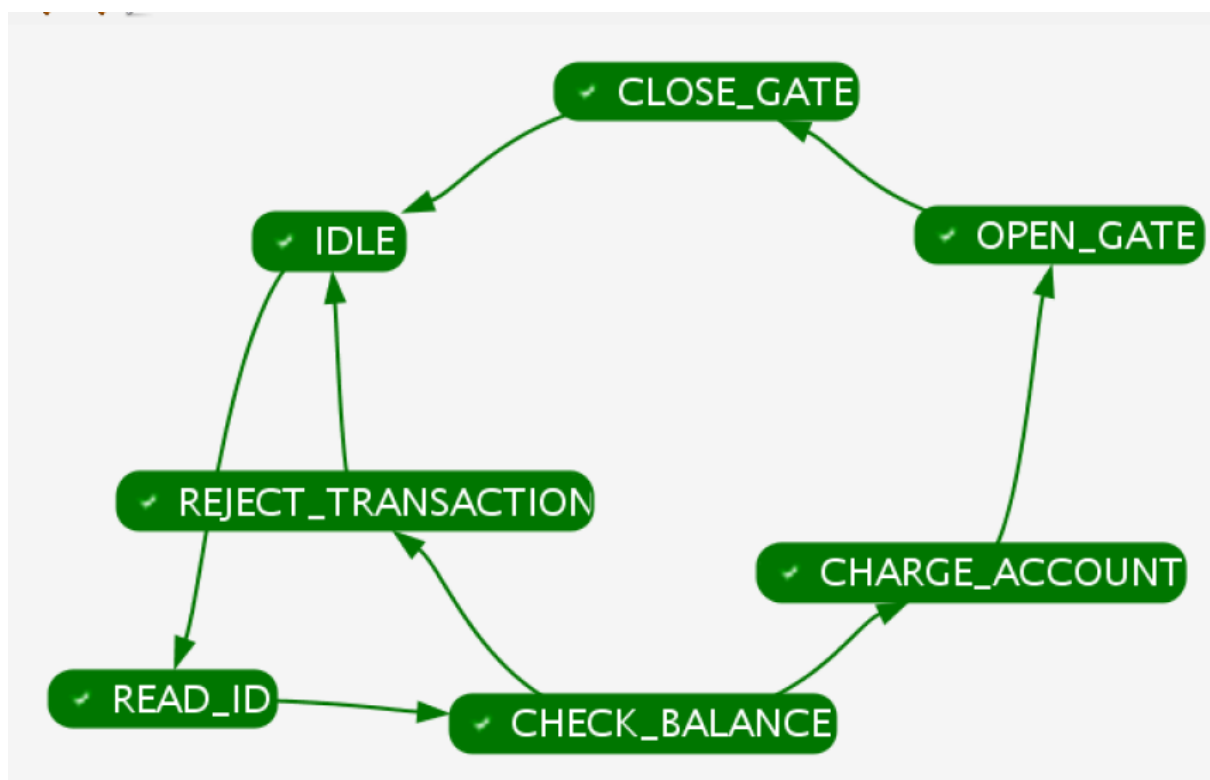
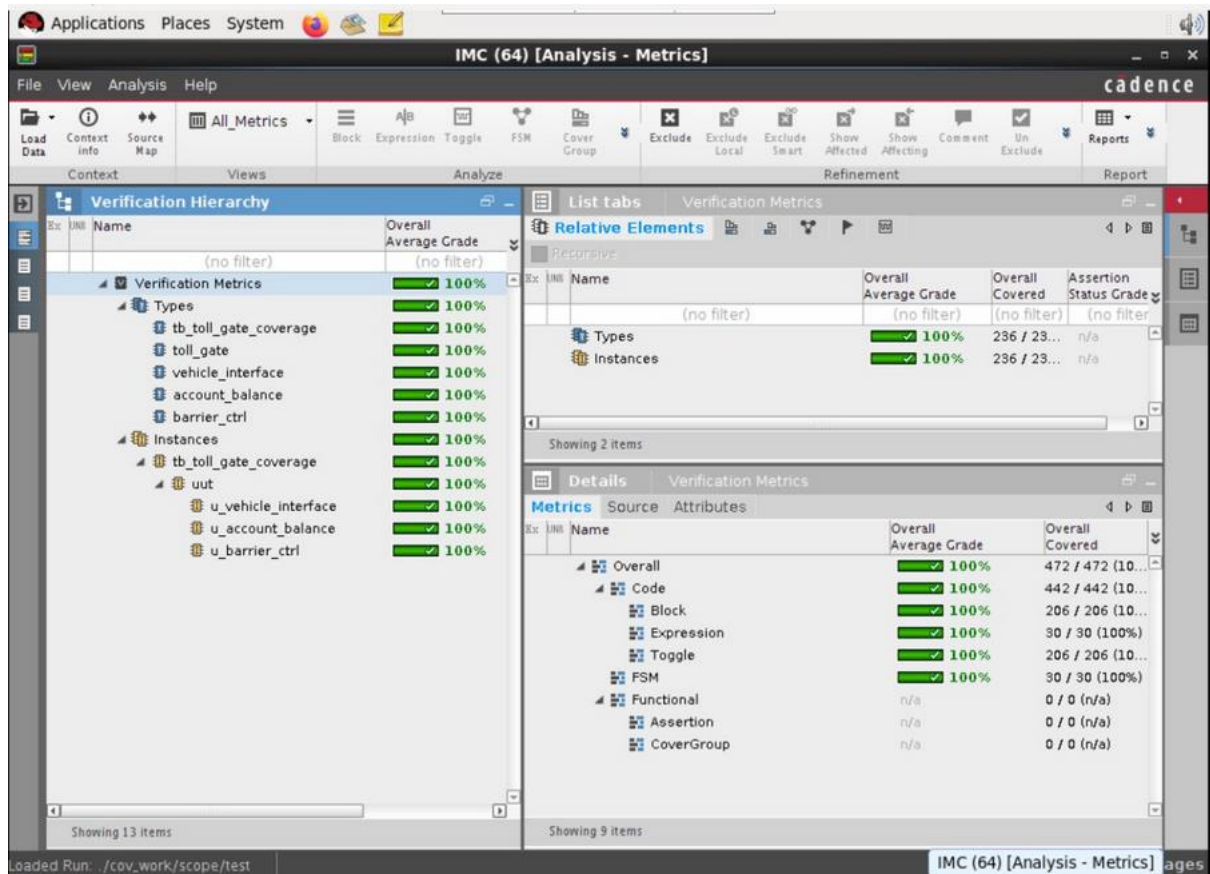
When vehicle ID 02 is detected, the system recognises it as an account with insufficient funds and thus, follows a different path. After reading the ID and checking the balance, it immediately transitions to *REJECT\_TRANSACTION*, displaying "FAIL" on the output and keeping the LED red. No barrier opening occurs in this case, and the system quickly returns to IDLE. This behavior is clearly visible in the waveform through the absence of *barrier\_open\_cmd* activation during this cycle.

For vehicle ID 03, the system repeats the successful transaction sequence: ID capture, balance verification, charging, and barrier opening. The *display\_out* cycles through its status messages accordingly, and the *led\_status* once again shifts to green during the GO phase. The *sensor\_vehicle\_exit* signal pulses briefly after each barrier opening, signaling that the vehicle has passed through and prompting the system to close the barrier via the *barrier\_close\_cmd* signal.

Throughout the simulation, the barrier control logic demonstrates realistic mechanical behavior, the *barrier\_open\_cmd* and *barrier\_close\_cmd* signals activate only when needed, and their duration reflects the multi-cycle nature of physical barrier movement. The *display\_out* signal provides continuous textual feedback, showing ASCII representations of the current state.

Therefore, the waveform confirms that the toll gate system reliably processes multiple vehicles in sequence, correctly handling both valid and invalid transactions while maintaining proper synchronization between input detection, internal processing, and output control. The consistent timing relationships and predictable state transitions validate the robustness of the design.

The coverage report for the testbench is given below:



The testbench was enhanced by incorporating numerous additional test cases to assess all the scenarios outlined in the RTL code. Consequently, complete coverage of 100% was attained for the specified FSM.

## Chapter 3

# LOGIC SYNTHESIS

The output produced by logic synthesis is a netlist, and we supply the tool with RTL code, a library file, and a constraints file as input. This netlist uses instances of cells from the library to represent the design structurally. Reports on timing, cell usage, and power consumption are also produced by the synthesis tool. Internal power, switching power, and leakage power are the three divisions of the power report. The biggest source of the netlist's overall power usage is the registers. The timing report indicates design slack by highlighting any timing violations based on the clock period and uncertainties listed in the constraints file. In the meantime, the cell report gives the overall area of the design and the number of cells that were instantiated in the netlist while respecting the constraints.

Essential information about frequently used cells, including drive strength, input/output pin information, cell delays, rise and fall transitions, power characteristics, and operating conditions (PVT), is contained in the library file. The logic synthesis of the RTL code discussed in the previous part is performed in this section under three distinct circumstances. They are for an intermediate constraint, tight timing constraints, and minimum area (with very loose timing constraints). Below is the tcl script that was used to execute the simulation. We just modify the input constraint file to synthesise appropriately for the other synthesis. Cadence has completed the synthesis in Genus.

```
read_lib /home/prerana25172/Desktop/cmos65/VDF_Project/synthesis/min_area/slow.lib
read_hdl /home/prerana25172/Desktop/cmos65/VDF_Project/synthesis/min_area/toll.v

elaborate

read_sdc cons.sdc
synthesize -to_mapped -effort medium

write_sdf -timescale ns -nonegchecks -recr split -edges check edge > syn_report_toll_minarea/delays_toll_minarea.sdf
check_timing_intent > syn_report_toll_minarea/timing_intent_toll_minarea.rep

write_hdl > syn_report_toll_minarea/netlist.v
write_sdc > syn_report_toll_minarea/dc_file_for_physical_design_toll_minarea.sdc
write_script > syn_report_toll_minarea/synthesis_script_sdc_toll_minarea.g

report_timing > syn_report_toll_minarea/synthesis_timing_report_toll_minarea.rep
report_power > syn_report_toll_minarea/synthesis_power_report_toll_minarea.rep
report_gates > syn_report_toll_minarea/synthesis_cell_report_toll_minarea.rep
report_area > syn_report_toll_minarea/synthesis_area_report_toll_minarea.rep

gui_show
```

## Synthesis for Minimum Area

Since the timing requirements for synthesising a design must be kept extremely loose, we greatly extend the clock's duration, which causes the design to be delayed. Additionally, by easing the timing restrictions, the synthesis tool can select slower, smaller cells rather than faster cells. Performance suffers as a result of the smaller area. It is possible to instruct the synthesis tool to use cells with lower driving strength, which are usually smaller in size. It's crucial to strike a balance between area reduction and timing objectives because this trade-off may result in further delays. Below is the constraint file for synthesising in the smallest possible space.

```
# Constraints File for Toll Management System Controller (Toll Module)

# Clock Definition
create_clock -name clk -period 12 [get_ports "clk"]

# Clock Latency
set_clock_latency 0.01 -source -late [get_clocks "clk"]
set_clock_latency 0.01 -source -early [get_clocks "clk"]

# Clock Transition
set_clock_transition -rise 0.6 [get_clocks "clk"]
set_clock_transition -fall 0.6 [get_clocks "clk"]

# Optional: Uncomment to add uncertainty
# set_clock_uncertainty -setup 0.01 [get_clocks "clk"]
# set_clock_uncertainty -hold 0.01 [get_clocks "clk"]

# Input Delays
set_input_delay -min 0.5 -clock [get_clocks "clk"] [all_inputs]
```

The area report after the synthesis using the above constraint file is shown below.

```
=====
Generated by:      Genus(TM) Synthesis Solution 19.13-s073_1
Generated on:      Oct 11 2025 06:21:35 pm
Module:           toll_gate
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

| Instance            | Module            | Cell Count | Cell Area | Net Area | Total Area | Wireload   |
|---------------------|-------------------|------------|-----------|----------|------------|------------|
| toll_gate           |                   | 642        | 7082.313  | 0.000    | 7082.313   | <none> (D) |
| u_account_balance   | account_balance   | 559        | 6591.842  | 0.000    | 6591.842   | <none> (D) |
| u_barrier_ctrl      | barrier_ctrl      | 31         | 181.656   | 0.000    | 181.656    | <none> (D) |
| u_vehicle_interface | vehicle_interface | 14         | 120.347   | 0.000    | 120.347    | <none> (D) |

(D) = wireload is default in technology library

The power report for the synthesized netlist is shown in the figure below.

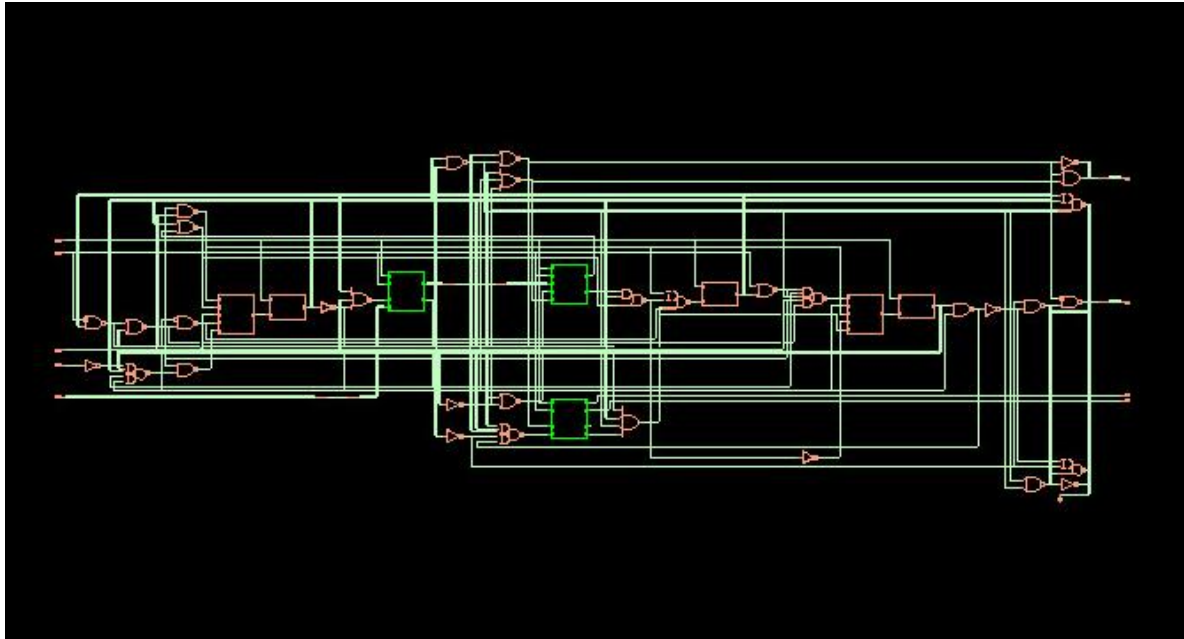
Instance: /toll\_gate  
Power Unit: W  
PDB Frames: /stim#0/frame#0

| Category   | Leakage     | Internal    | Switching   | Total       | Row%    |
|------------|-------------|-------------|-------------|-------------|---------|
| memory     | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| register   | 3.23906e-05 | 4.16738e-04 | 6.00593e-06 | 4.55135e-04 | 84.84%  |
| latch      | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| logic      | 9.12541e-06 | 2.56782e-05 | 1.74128e-05 | 5.22164e-05 | 9.73%   |
| bbox       | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| clock      | 0.00000e+00 | 0.00000e+00 | 2.91397e-05 | 2.91397e-05 | 5.43%   |
| pad        | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| pm         | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| Subtotal   | 4.15160e-05 | 4.42416e-04 | 5.25584e-05 | 5.36491e-04 | 100.00% |
| Percentage | 7.74%       | 82.46%      | 9.80%       | 100.00%     | 100.00% |

The timing report for the netlist is shown in the two figures below.

|                                                                                           |                                                |                                           |       |      |           |        |      |       |       |         |          |
|-------------------------------------------------------------------------------------------|------------------------------------------------|-------------------------------------------|-------|------|-----------|--------|------|-------|-------|---------|----------|
| Generated by:                                                                             |                                                | Genus(TM) Synthesis Solution 19.13-s073_1 |       |      |           |        |      |       |       |         |          |
| Generated on:                                                                             |                                                | Oct 11 2025 06:21:35 pm                   |       |      |           |        |      |       |       |         |          |
| Module:                                                                                   |                                                | toll_gate                                 |       |      |           |        |      |       |       |         |          |
| Operating conditions:                                                                     |                                                | slow (balanced_tree)                      |       |      |           |        |      |       |       |         |          |
| Wireload mode:                                                                            |                                                | enclosed                                  |       |      |           |        |      |       |       |         |          |
| Area mode:                                                                                |                                                | timing library                            |       |      |           |        |      |       |       |         |          |
| =====                                                                                     |                                                |                                           |       |      |           |        |      |       |       |         |          |
| Path 1: MET (6968 ps) Setup Check with Pin u_account_balance/balance_db_reg[7][15]/CK->SE |                                                |                                           |       |      |           |        |      |       |       |         |          |
| Group: clk                                                                                |                                                |                                           |       |      |           |        |      |       |       |         |          |
| Startpoint: (R) u_vehicle_interface/id_to_check_reg[3]/CK                                 |                                                |                                           |       |      |           |        |      |       |       |         |          |
| Clock: (R) clk                                                                            |                                                |                                           |       |      |           |        |      |       |       |         |          |
| Endpoint: (F) u_account_balance/balance_db_reg[7][15]/SE                                  |                                                |                                           |       |      |           |        |      |       |       |         |          |
| Clock: (R) clk                                                                            |                                                |                                           |       |      |           |        |      |       |       |         |          |
|                                                                                           | Capture                                        | Launch                                    |       |      |           |        |      |       |       |         |          |
| Clock Edge:+                                                                              | 12000                                          | 0                                         |       |      |           |        |      |       |       |         |          |
| Src Latency:+                                                                             | 10                                             | 10                                        |       |      |           |        |      |       |       |         |          |
| Net Latency:+                                                                             | 0 (I)                                          | 0 (I)                                     |       |      |           |        |      |       |       |         |          |
| Arrival:=                                                                                 | 12010                                          | 10                                        |       |      |           |        |      |       |       |         |          |
| Setup:-                                                                                   | 417                                            |                                           |       |      |           |        |      |       |       |         |          |
| Required Time:=                                                                           | 11593                                          |                                           |       |      |           |        |      |       |       |         |          |
| Launch Clock:-                                                                            | 10                                             |                                           |       |      |           |        |      |       |       |         |          |
| Data Path:-                                                                               | 4614                                           |                                           |       |      |           |        |      |       |       |         |          |
| Slack:=                                                                                   | 6968                                           |                                           |       |      |           |        |      |       |       |         |          |
| -----                                                                                     |                                                |                                           |       |      |           |        |      |       |       |         |          |
| #                                                                                         | Timing Point                                   | Flags                                     | Arc   | Edge | Cell      | Fanout | Load | Trans | Delay | Arrival | Instance |
| #                                                                                         |                                                |                                           |       |      |           | (fF)   | (ps) | (ps)  | (ps)  | (ps)    | Location |
| #                                                                                         | -----                                          |                                           |       |      |           |        |      |       |       |         |          |
|                                                                                           | u_vehicle_interface/id_to_check_reg[3]/CK      | -                                         | -     | R    | (arrival) | 254    | -    | 600   | -     | 10      | (-, -)   |
|                                                                                           | u_vehicle_interface/id_to_check_reg[3]/Q       | -                                         | CK->Q | R    | DFFQX1    | 12     | 31.3 | 271   | 528   | 538     | (-, -)   |
|                                                                                           | u_account_balance/g17076/Y                     | -                                         | A->Y  | F    | CLKINX1   | 21     | 76.8 | 542   | 443   | 980     | (-, -)   |
|                                                                                           | u_account_balance/g17074/Y                     | -                                         | A->Y  | R    | NOR2XL    | 15     | 30.1 | 897   | 753   | 1734    | (-, -)   |
|                                                                                           | u_account_balance/g17070/Y                     | -                                         | A->Y  | F    | CLKINX1   | 4      | 6.8  | 245   | 134   | 1867    | (-, -)   |
|                                                                                           | u_account_balance/g17044/Y                     | -                                         | B->Y  | R    | NOR2XL    | 9      | 22.7 | 677   | 541   | 2408    | (-, -)   |
|                                                                                           | u_account_balance/g16923/Y                     | -                                         | B1->Y | F    | A0I221X1  | 1      | 2.8  | 282   | 165   | 2573    | (-, -)   |
|                                                                                           | u_account_balance/g16910/Y                     | -                                         | B0->Y | R    | OAI21X1   | 1      | 2.8  | 163   | 93    | 2666    | (-, -)   |
|                                                                                           | u_account_balance/g16904/Y                     | -                                         | C0->Y | F    | A0I211X1  | 1      | 1.7  | 204   | 52    | 2718    | (-, -)   |
|                                                                                           | u_account_balance/g16896/Y                     | -                                         | B->Y  | R    | NAND2XL   | 2      | 3.5  | 116   | 112   | 2830    | (-, -)   |
|                                                                                           | u_account_balance/g16891/Y                     | -                                         | B->Y  | F    | NOR2XL    | 2      | 5.5  | 103   | 91    | 2921    | (-, -)   |
|                                                                                           | u_account_balance/g16890/Y                     | -                                         | A->Y  | R    | INVX1     | 1      | 1.7  | 43    | 48    | 2969    | (-, -)   |
|                                                                                           | u_account_balance/g16888/Y                     | -                                         | C->Y  | R    | OR4X1     | 5      | 8.7  | 94    | 139   | 3108    | (-, -)   |
|                                                                                           | u_account_balance/g16887/Y                     | -                                         | AN->Y | R    | NAND3BX1  | 1      | 2.8  | 60    | 117   | 3225    | (-, -)   |
|                                                                                           | u_account_balance/g16886/Y                     | -                                         | A2->Y | F    | OAI31X1   | 3      | 7.3  | 150   | 114   | 3338    | (-, -)   |
|                                                                                           | u_account_balance/g16885/Y                     | -                                         | B->Y  | R    | NOR2BX1   | 3      | 5.1  | 121   | 125   | 3463    | (-, -)   |
|                                                                                           | u_account_balance/g3679/Y                      | -                                         | AN->Y | R    | NOR2BX1   | 2      | 3.4  | 118   | 146   | 3609    | (-, -)   |
|                                                                                           | u_account_balance/g3668/Y                      | -                                         | B->Y  | R    | AND2X1    | 4      | 10.1 | 115   | 198   | 3806    | (-, -)   |
|                                                                                           | u_account_balance/g3644/Y                      | -                                         | C->Y  | F    | NAND3X1   | 15     | 57.0 | 1153  | 818   | 4624    | (-, -)   |
|                                                                                           | u_account_balance/balance_db_reg[7][15]/SE <<< | -                                         | -     | F    | SDDFQX1   | 15     | -    | -     | 0     | 4624    | (-, -)   |
| #                                                                                         | -----                                          |                                           |       |      |           |        |      |       |       |         |          |

The overall synthesized netlist schematic is shown below.



## Synthesis for Minimum Timing

To have the best timing performance, the tool should choose cells with the maximum area. Larger cells are usually able to handle more current due to their higher driving strength, which reduces delay. In order to attain a tiny negative slack and make sure the design satisfies the timing requirements as precisely as feasible, we describe a clock period and include clock uncertainties.

```
read_lib /home/prerana25172/Desktop/cmos65/VDF_Project/synthesis/min_time/slow.lib
read_hdl /home/prerana25172/Desktop/cmos65/VDF_Project/synthesis/min_time/toll.v

elaborate

read_sdc cons.sdc
synthesize -to_mapped -effort medium

write_sdf -timescale ns -nonegchecks -recr split -edges check edge > syn_report_toll_minarea/delays_toll_minarea.sdf
check_timing_intent > syn_report_toll_minarea/timing_intent_toll_minarea.rep

write_hdl > syn_report_toll_minarea/netlist.v
write_sdc > syn_report_toll_minarea/dc_file_for_physical_design_toll_minarea.sdc
write_script > syn_report_toll_minarea/synthesis_script_sdc_toll_minarea.g

report_timing > syn_report_toll_minarea/synthesis_timing_report_toll_minarea.rep
report_power > syn_report_toll_minarea/synthesis_power_report_toll_minarea.rep
report_gates > syn_report_toll_minarea/synthesis_cell_report_toll_minarea.rep
report_area > syn_report_toll_minarea/synthesis_area_report_toll_minarea.rep

|
gui_show
```

The following is the constraint file for synthesizing under strict timing restrictions.

```
# Constraints File for Toll Managment System Controller (Toll Module)

# Clock Definition
create_clock -name clk -period 3.5 [get_ports "clk"]

# Clock Latency
set_clock_latency 0.01 -source -late [get_clocks "clk"]
set_clock_latency 0.01 -source -early [get_clocks "clk"]

# Clock Transition
set_clock_transition -rise 0.6 [get_clocks "clk"]
set_clock_transition -fall 0.6 [get_clocks "clk"]

# Optional: Uncomment to add uncertainty
# set_clock_uncertainty -setup 0.01 [get_clocks "clk"]
# set_clock_uncertainty -hold 0.01 [get_clocks "clk"]

# Input Delays
set_input_delay -min 0.5 -clock [get_clocks "clk"] [all_inputs]
```

The area report after the synthesis using the above constraint file is shown below.

| Generated by: Genus(TM) Synthesis Solution 19.13-s073_1 |                   |            |           |          |            |            |
|---------------------------------------------------------|-------------------|------------|-----------|----------|------------|------------|
| Generated on: Oct 11 2025 07:10:30 pm                   |                   |            |           |          |            |            |
| Module: toll_gate                                       |                   |            |           |          |            |            |
| Operating conditions: slow (balanced_tree)              |                   |            |           |          |            |            |
| Wireload mode: enclosed                                 |                   |            |           |          |            |            |
| Area mode: timing library                               |                   |            |           |          |            |            |
| Instance                                                | Module            | Cell Count | Cell Area | Net Area | Total Area | Wireload   |
| toll_gate                                               |                   | 706        | 7220.826  | 0.000    | 7220.826   | <none> (D) |
| u_account_balance                                       | account_balance   | 623        | 6715.974  | 0.000    | 6715.974   | <none> (D) |
| u_barrier_ctrl                                          | barrier_ctrl      | 30         | 183.170   | 0.000    | 183.170    | <none> (D) |
| u_vehicle_interface                                     | vehicle_interface | 14         | 120.347   | 0.000    | 120.347    | <none> (D) |
| (D) = wireload is default in technology library         |                   |            |           |          |            |            |

The power report for the synthesized netlist is shown in the figure below.

| Instance: /toll_gate        |              |              |              |              |         |
|-----------------------------|--------------|--------------|--------------|--------------|---------|
| Power Unit: W               |              |              |              |              |         |
| PDB Frames: /stim#0/frame#0 |              |              |              |              |         |
| Category                    | Leakage      | Internal     | Switching    | Total        | Row%    |
| memory                      | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.00%   |
| register                    | 3.23906e-05  | 1.44280e-03  | 1.86450e-05  | 1.49384e-03  | 84.06%  |
| latch                       | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.00%   |
| logic                       | 9.49587e-06  | 1.03257e-04  | 7.06531e-05  | 1.83406e-04  | 10.32%  |
| bbox                        | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.00%   |
| clock                       | 0.000000e+00 | 0.000000e+00 | 9.99077e-05  | 9.99077e-05  | 5.62%   |
| pad                         | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.00%   |
| pm                          | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.00%   |
| Subtotal                    | 4.18865e-05  | 1.54606e-03  | 1.89206e-04  | 1.77715e-03  | 100.00% |
| Percentage                  | 2.36%        | 87.00%       | 10.65%       | 100.00%      | 100.00% |

The timing report for the netlist is shown in the two figures below.



```

=====
Generated by:      Genus(TM) Synthesis Solution 19.13-s073_1
Generated on:      Oct 11 2025  07:10:29 pm
Module:           toll_gate
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

```

Path 1: MET (5 ps) Setup Check with Pin u\_account\_balance/balance\_db\_reg[9][15]/CK->SE

```

Group: clk
Startpoint: (R) u_vehicle_interface/id_to_check_reg[0]/CK
Clock: (R) clk
Endpoint: (R) u_account_balance/balance_db_reg[9][15]/SE
Clock: (R) clk

```

```

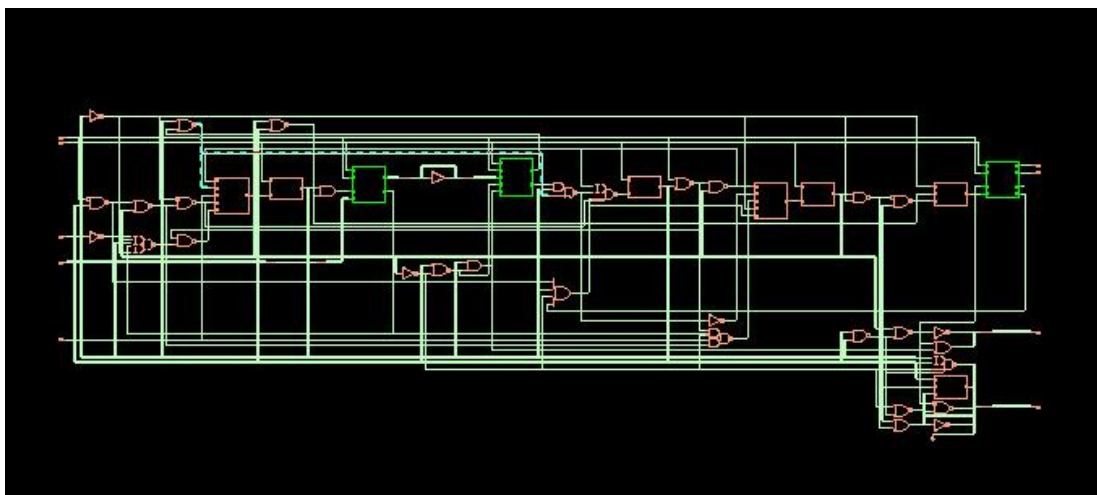
          Capture      Launch
Clock Edge:+ 3500      0
Src Latency:+ 10       10
Net Latency:+ 0 (I)    0 (I)
Arrival:=    3510      10

Setup:-      301
Required Time:= 3209
Launch Clock:- 10
Data Path:-  3194
Slack:=      5

```

| # | Timing Point                                   | Flags | Arc   | Edge | Cell      | Fanout | Load (fF) | Trans (ps) | Delay (ps) | Arrival (ps) | Instance Location |
|---|------------------------------------------------|-------|-------|------|-----------|--------|-----------|------------|------------|--------------|-------------------|
| # | u_vehicle_interface/id_to_check_reg[0]/CK      | -     | -     | R    | (arrival) | 254    | -         | 600        | -          | 10           | (-, -)            |
| # | u_vehicle_interface/id_to_check_reg[0]/Q       | -     | CK->Q | F    | DFFQX1    | 2      | 7.0       | 74         | 439        | 449          | (-, -)            |
| # | g484/Y                                         | -     | A->Y  | F    | BUFEX6    | 30     | 86.5      | 119        | 177        | 625          | (-, -)            |
| # | u_account_balance/g17373/Y                     | -     | A->Y  | R    | CLKINVX2  | 17     | 47.0      | 202        | 182        | 807          | (-, -)            |
| # | u_account_balance/g17369/Y                     | -     | A->Y  | F    | NAND2X1   | 6      | 11.0      | 179        | 169        | 976          | (-, -)            |
| # | u_account_balance/g17319/Y                     | -     | B->Y  | R    | NOR2XL    | 5      | 13.8      | 424        | 350        | 1326         | (-, -)            |
| # | u_account_balance/g17427/Y                     | -     | B1->Y | F    | A0I221X1  | 1      | 1.7       | 173        | 128        | 1453         | (-, -)            |
| # | u_account_balance/g17144/Y                     | -     | B->Y  | R    | NAND2XL   | 3      | 5.9       | 123        | 135        | 1589         | (-, -)            |
| # | u_account_balance/g17140/Y                     | -     | B->Y  | F    | NOR2XL    | 2      | 4.6       | 84         | 84         | 1673         | (-, -)            |
| # | u_account_balance/g17138/Y                     | -     | AN->Y | F    | NOR3BX1   | 2      | 4.5       | 77         | 128        | 1801         | (-, -)            |
| # | u_account_balance/g17136/Y                     | -     | D->Y  | R    | NAND4BXL  | 1      | 2.7       | 83         | 78         | 1879         | (-, -)            |
| # | u_account_balance/g17135/Y                     | -     | A1->Y | F    | OAI21X1   | 3      | 6.2       | 132        | 106        | 1986         | (-, -)            |
| # | u_account_balance/g17134/Y                     | -     | B->Y  | R    | NOR2BX1   | 4      | 8.0       | 158        | 150        | 2136         | (-, -)            |
| # | u_account_balance/g3655/Y                      | -     | B->Y  | R    | AND2X1    | 5      | 8.5       | 102        | 197        | 2332         | (-, -)            |
| # | u_account_balance/g3635/Y                      | -     | B->Y  | F    | NAND2XL   | 4      | 8.0       | 212        | 176        | 2509         | (-, -)            |
| # | u_account_balance/g3663/Y                      | -     | B->Y  | R    | NOR2BX1   | 15     | 57.0      | 918        | 696        | 3204         | (-, -)            |
| # | u_account_balance/balance_db_reg[9][15]/SE <<< | -     | -     | R    | SDFFQX1   | 15     | -         | -          | 0          | 3204         | (-, -)            |

The overall synthesized netlist schematic is shown below.



## Synthesis for Intermediate Constraints



The constraints for this synthesis are essentially a combination of the two syntheses we completed previously. The design offers a more relaxed area than the netlist synthesized for the situation of minimum area limits, as well as less precise timing than the prior netlist.

```
read_lib /home/prerana25172/Desktop/cmos65/VDF_Project/synthesis/intermediate/slow.lib
read_hdl /home/prerana25172/Desktop/cmos65/VDF_Project/synthesis/intermediate/toll.v

elaborate

read_sdc cons.sdc
synthesize -to_mapped -effort medium

write_sdf -timescale ns -nonegchecks -recrem split -edges check edge > syn_report_toll_minarea/delays_toll_minarea.sdf
check_timing_intent > syn_report_toll_minarea/timing_intent_toll_minarea.rep

write_hdl > syn_report_toll_minarea/netlist.v
write_sdc > syn_report_toll_minarea/dc_file_for_physical_design_toll_minarea.sdc
write_script > syn_report_toll_minarea/synthesis_script_sdc_toll_minarea.g

report_timing > syn_report_toll_minarea/synthesis_timing_report_toll_minarea.rep
report_power > syn_report_toll_minarea/synthesis_power_report_toll_minarea.rep
report_gates > syn_report_toll_minarea/synthesis_cell_report_toll_minarea.rep
report_area > syn_report_toll_minarea/synthesis_area_report_toll_minarea.rep

gui_show
```

The constraint file used for the synthesis is shown below.

---

```
# Constraints File for Toll Managment System Controller (Toll Module)

# Clock Definition
create_clock -name clk -period 8 [get_ports "clk"]

# Clock Latency
set_clock_latency 0.01 -source -late [get_clocks "clk"]
set_clock_latency 0.01 -source -early [get_clocks "clk"]

# Clock Transition
set_clock_transition -rise 0.6 [get_clocks "clk"]
set_clock_transition -fall 0.6 [get_clocks "clk"]

# Optional: Uncomment to add uncertainty
# set_clock_uncertainty -setup 0.01 [get_clocks "clk"]
# set_clock_uncertainty -hold 0.01 [get_clocks "clk"]

# Input Delays
set_input_delay -min 0.5 -clock [get_clocks "clk"] [all_inputs]
```

The area report for the synthesis is shown below.

|                                                 |                                           |            |           |          |            |            |
|-------------------------------------------------|-------------------------------------------|------------|-----------|----------|------------|------------|
| =====                                           |                                           |            |           |          |            |            |
| Generated by:                                   | Genus(TM) Synthesis Solution 19.13-s073_1 |            |           |          |            |            |
| Generated on:                                   | Oct 11 2025 07:18:41 pm                   |            |           |          |            |            |
| Module:                                         | toll_gate                                 |            |           |          |            |            |
| Operating conditions:                           | slow (balanced_tree)                      |            |           |          |            |            |
| Wireload mode:                                  | enclosed                                  |            |           |          |            |            |
| Area mode:                                      | timing library                            |            |           |          |            |            |
| =====                                           |                                           |            |           |          |            |            |
| Instance                                        | Module                                    | Cell Count | Cell Area | Net Area | Total Area | Wireload   |
| -----                                           |                                           |            |           |          |            |            |
| toll_gate                                       |                                           | 671        | 7105.777  | 0.000    | 7105.777   | <none> (D) |
| u_account_balance                               | account_balance                           | 589        | 6612.278  | 0.000    | 6612.278   | <none> (D) |
| u_barrier_ctrl                                  | barrier_ctrl                              | 30         | 183.170   | 0.000    | 183.170    | <none> (D) |
| u_vehicle_interface                             | vehicle_interface                         | 14         | 120.347   | 0.000    | 120.347    | <none> (D) |
| (D) = wireload is default in technology library |                                           |            |           |          |            |            |

The power report for the synthesis is shown below.

Instance: /toll\_gate

Power Unit: W

PDB Frames: /stim#0/frame#0

| Category   | Leakage     | Internal    | Switching   | Total       | Row%    |
|------------|-------------|-------------|-------------|-------------|---------|
| memory     | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| register   | 3.23906e-05 | 6.18952e-04 | 8.08945e-06 | 6.59432e-04 | 84.71%  |
| latch      | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| logic      | 8.81003e-06 | 3.97644e-05 | 2.67068e-05 | 7.52812e-05 | 9.67%   |
| bbox       | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| clock      | 0.00000e+00 | 0.00000e+00 | 4.37096e-05 | 4.37096e-05 | 5.62%   |
| pad        | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| pm         | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| Subtotal   | 4.12006e-05 | 6.58716e-04 | 7.85058e-05 | 7.78423e-04 | 100.00% |
| Percentage | 5.29%       | 84.62%      | 10.09%      | 100.00%     | 100.00% |

The timing report is shown below.

```

=====
Generated by:      Genus(TM) Synthesis Solution 19.13-s073_1
Generated on:     Oct 11 2025  07:18:40 pm
Module:          toll_gate
Operating conditions: slow (balanced_tree)
Wireload mode:   enclosed
Area mode:       timing library
=====

```

Path 1: MET (3161 ps) Setup Check with Pin u\_account\_balance/balance\_db\_reg[13][15]/CK->SE

```

Group: clk
Startpoint: (R) u_vehicle_interface/id_to_check_reg[3]/CK
Clock: (R) clk
Endpoint: (F) u_account_balance/balance_db_reg[13][15]/SE
Clock: (R) clk

```

```

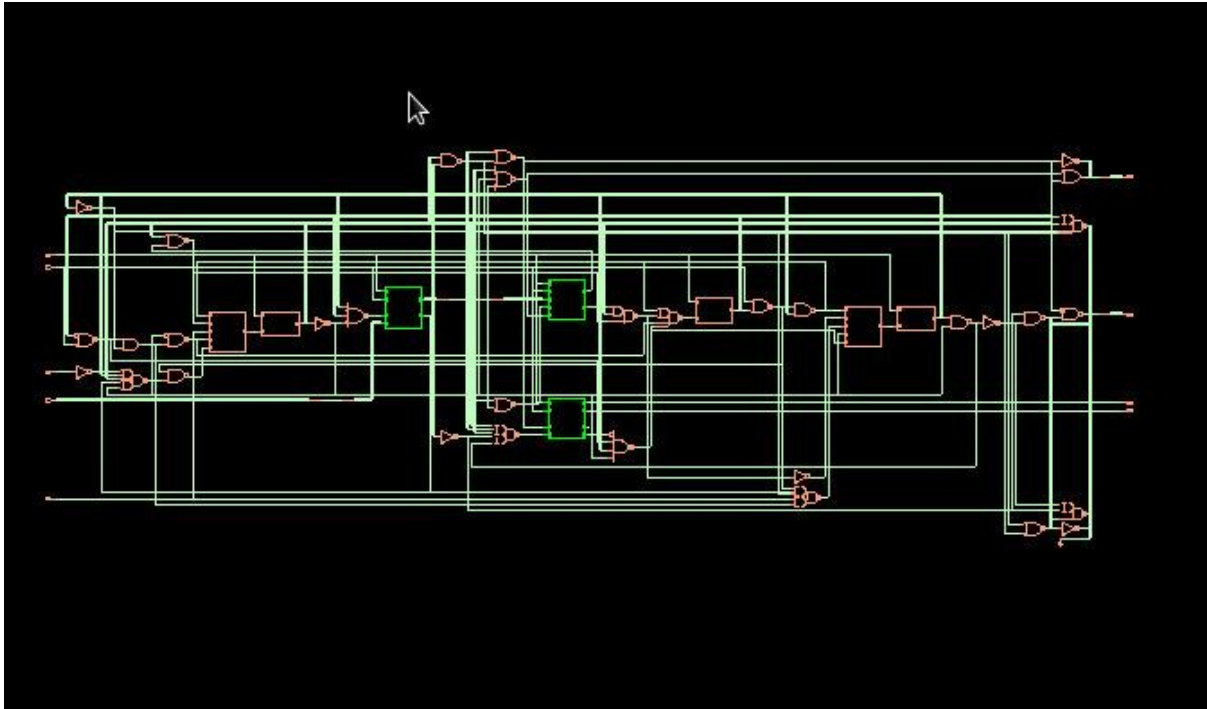
          Capture      Launch
Clock Edge:+ 8000      0
Src Latency:+ 10      10
Net Latency:+ 0 (I)    0 (I)
Arrival:= 8010      10

Setup:- 417
Required Time:= 7593
Launch Clock:- 10
Data Path:- 4422
Slack:= 3161

```

| # | Timing Point                                    | Flags | Arc   | Edge | Cell      | Fanout | Load (fF) | Trans (ps) | Delay (ps) | Arrival (ps) | Instance Location |
|---|-------------------------------------------------|-------|-------|------|-----------|--------|-----------|------------|------------|--------------|-------------------|
| # | u_vehicle_interface/id_to_check_reg[3]/CK       | -     | -     | R    | (arrival) | 254    | -         | 600        | -          | 10           | (-, -)            |
| # | u_vehicle_interface/id_to_check_reg[3]/Q        | -     | CK->Q | F    | DFFQX1    | 10     | 21.0      | 163        | 508        | 518          | (-, -)            |
| # | u_account_balance/g12166/Y                      | -     | A->Y  | R    | CLKINX1   | 23     | 79.9      | 662        | 512        | 1031         | (-, -)            |
| # | u_account_balance/g12164/Y                      | -     | B->Y  | R    | AND2X1    | 31     | 57.7      | 492        | 558        | 1589         | (-, -)            |
| # | u_account_balance/g12159/Y                      | -     | A->Y  | F    | CLKINX1   | 4      | 6.8       | 162        | 110        | 1699         | (-, -)            |
| # | u_account_balance/g12147/Y                      | -     | B->Y  | R    | NOR2XL    | 8      | 18.9      | 574        | 445        | 2144         | (-, -)            |
| # | u_account_balance/g12034/Y                      | -     | C0->Y | F    | A0I222XL  | 1      | 1.6       | 201        | 131        | 2275         | (-, -)            |
| # | u_account_balance/g11985/Y                      | -     | C->Y  | R    | NAND4XL   | 3      | 5.9       | 140        | 153        | 2428         | (-, -)            |
| # | u_account_balance/g11968/Y                      | -     | B->Y  | F    | NOR2XL    | 2      | 4.6       | 95         | 88         | 2516         | (-, -)            |
| # | u_account_balance/g11965/Y                      | -     | AN->Y | F    | NOR3BX1   | 2      | 4.5       | 90         | 132        | 2647         | (-, -)            |
| # | u_account_balance/g11962/Y                      | -     | D->Y  | R    | NAND4BXL  | 1      | 2.7       | 87         | 82         | 2729         | (-, -)            |
| # | u_account_balance/g11960/Y                      | -     | A1->Y | F    | OAI21X1   | 3      | 6.2       | 125        | 107        | 2837         | (-, -)            |
| # | u_account_balance/g11959/Y                      | -     | B->Y  | R    | NOR2BX1   | 4      | 9.2       | 174        | 161        | 2997         | (-, -)            |
| # | u_account_balance/g3648/Y                       | -     | B->Y  | F    | NAND2XL   | 5      | 12.9      | 329        | 270        | 3268         | (-, -)            |
| # | u_account_balance/g3631/Y                       | -     | B->Y  | R    | NOR2XL    | 4      | 9.6       | 323        | 312        | 3580         | (-, -)            |
| # | u_account_balance/g3623/Y                       | -     | A->Y  | F    | NAND3X1   | 15     | 57.0      | 1153       | 852        | 4432         | (-, -)            |
| # | u_account_balance/balance_db_reg[13][15]/SE <<< | -     | -     | F    | SdffQX1   | 15     | -         | -          | 0          | 4432         | (-, -)            |

The overall synthesized netlist schematic is shown below.



## Analysis of Results

| Constraint     | Cell Count | Area     | Slack(in ps) | Power(in Watts)          |
|----------------|------------|----------|--------------|--------------------------|
| Minimum Area   | 642        | 7082.313 | 6968         | $5.36491 \times 10^{-4}$ |
| Minimum Timing | 706        | 7220.826 | 5            | $1.77715 \times 10^{-3}$ |
| Intermediate   | 671        | 7105.777 | 3161         | $7.78423 \times 10^{-4}$ |

The results presented illustrate the outcomes of various synthesis methods where factors such as area, timing, and power constraints were considered. Each outcome highlights a trade-off between these factors.

In the Minimum Area scenario, the objective is to reduce the physical size of the design. The number of cells is 642, and the area is 7082.313, which is the smallest among the three cases. The slack is considerable, at 6968 ps, indicating that the design comfortably satisfies the timing requirements. This implies that the timing constraints are lenient, enabling the synthesis tool to primarily optimize for area, thereby resulting in minimal power consumption ( $5.36491 \times 10^{-4}$  Watts). This outcome is accurate and anticipated for a situation where area reduction is the main focus, leading to efficient resource utilization without the need for stringent timing optimizations.

In the Minimum Timing case, the synthesis concentrated on achieving strict timing requirements. The cell count of 706 and area of 7220.826 are both greater than in the previous scenario, signifying that additional cells and a larger area were required to meet the timing constraints. Further, power consumption is elevated to  $1.77715 \times 10^{-3}$  Watts. As minimum timing constraints necessitate increased switching activity and the insertion of extra buffers,

along with the utilization of more standard cells in the design, the increased power consumption is justified.

The Intermediate constraints results illustrate a compromise between area and timing. With 671 cells and an area of 7.78423, it is comparable to the minimum area design but possesses a slack of 3161 ps, indicating that it has more timing margin than required without being as aggressive as the minimum timing scenario. The power consumption is a moderate value at  $7.78423 \times 10^{-4}$  Watts situated between the two other scenarios.

## **Chapter 4**

# **FORMAL EQUIVALENCE CHECKING**

Formal Equivalence Checking (FEC) is an essential verification technique that confirms two representations of an identical design—usually a Register Transfer Level (RTL) representation and a synthesized netlist—demonstrate equivalent functionality.

The procedure starts with the creation of a one-to-one correspondence between the primary inputs and outputs of both the RTL and the netlist, ensuring accurate alignment. Following this, common flip-flops present in both representations are identified and mapped, which is vital for verifying consistency in state and transitions. The tool then analyzes the logic between the matched principal inputs and the inputs of the flip-flops, ensuring that the behavior of incoming signals is maintained. Moreover, the logic between the outputs of one flip-flop and the inputs of another is scrutinized to confirm that the data flow through the sequential elements remains intact. Lastly, the outputs of the flip-flops are compared to the primary outputs of the RTL, confirming that the synthesized netlist yields outputs that align with those defined in the RTL representation.

In summary, FEC offers a structured method to verify that the synthesized design preserves the intended functionality of the original RTL, aiding in the identification of any discrepancies that may impact the integrity of the design. The necessary inputs for the simulations include the RTL code, the synthesized netlist, and a do file that specifies the tasks required by the tool. The do file provided below pertains to one of the synthesis cases (intermediate constraints). For the other cases, only the corresponding netlist should be supplied in the revised netlist.

## Equivalence Checking for Minimum Area

The outputs of the equivalence checking for the minimum area constraint are shown below.

The screenshot displays the Cadence Conformal(R) Logic Equivalence Checking interface. The main window is titled "Conformal(R) Logic Equivalence Checking" and features a menu bar with "File", "Compare", "Tools", "Custom", "Preferences", "Window", and "Help". Below the menu bar is a toolbar with various icons and buttons, including "Setup" and "LEC".

The interface is divided into two main sections: "Golden" and "Revised".

**Golden Design:**

- toll\_gate
  - 15 library cells and 45 primitives
  - u\_account\_balance(account\_balance)
  - u\_barrier\_ctrl(barrier\_ctrl)
  - u\_vehicle\_interface(vehicle\_interface)

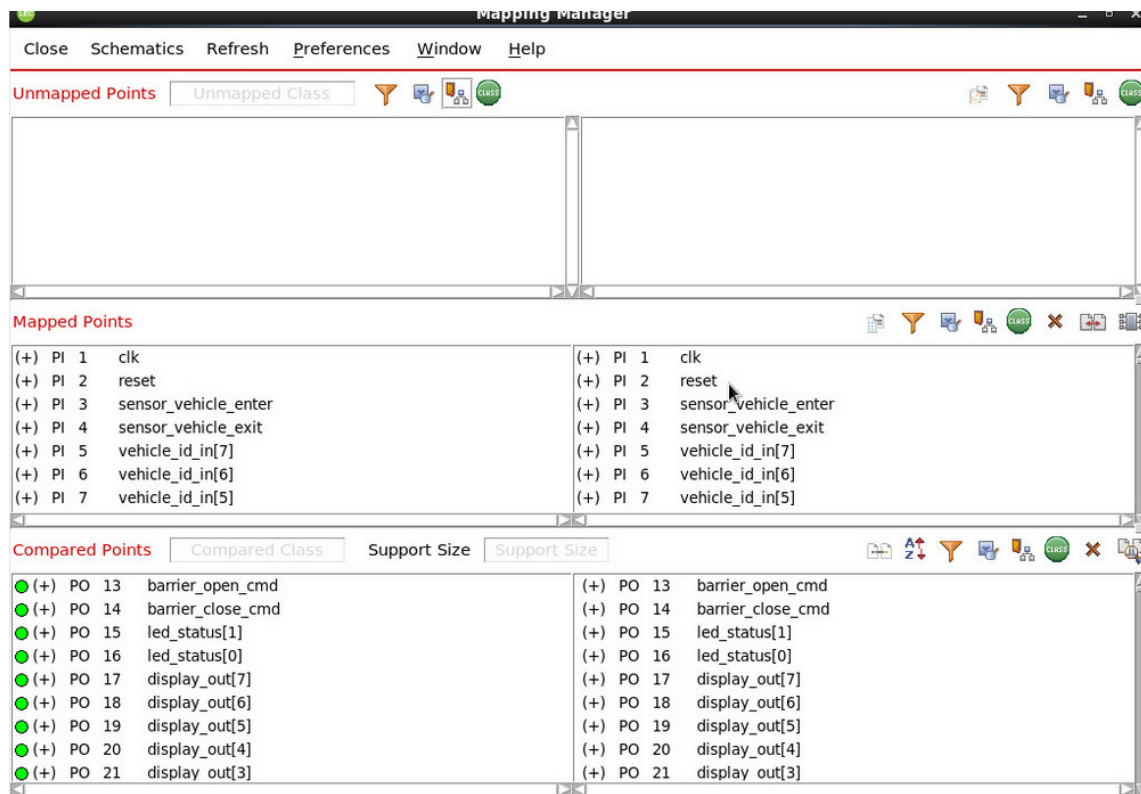
**Revised Design:**

- toll\_gate
  - 38 library cells
  - u\_account\_balance(account\_balance)
  - u\_barrier\_ctrl(barrier\_ctrl)
  - u\_vehicle\_interface(vehicle\_interface)

Below the design comparison, a "Verification Report" is displayed, showing the following results:

| Category                                                                   | Count |
|----------------------------------------------------------------------------|-------|
| 1. Non-standard modeling options used:                                     | 0     |
| 2. Incomplete verification:                                                | 0     |
| 3. User modification to design:                                            | 0     |
| 4. Conformal Constraint Designer clock domain crossing checks recommended: | 1     |
| Multiple clocks in the design:                                             | yes   |
| 5. Design ambiguity:                                                       | 0     |
| 6. Compare Results:                                                        | PASS  |

At the bottom of the window, a status bar indicates "Compare done!" and "100% completed". The taskbar at the very bottom shows several open applications, including "cmos65", "VDF\_Project", "CEC", "min\_area", "prerana25172...", and "Conformal(R) L...".



```
set log file logical_equivalence_checking_minarea.log -replace

read library /home/prerana25172/Desktop/cmos65/VDF_Project/CEC/min_area/slow.v -verilog -both
read design /home/prerana25172/Desktop/cmos65/VDF_Project/CEC/min_area/toll.v -golden
read design /home/prerana25172/Desktop/cmos65/VDF_Project/CEC/min_area/netlist.v -verilog -revised

set system mode lec
add compared points -all
compare
report messages -compare -verb
report compare data -noneq
report verification
write compared points -replace lec compared_points_minarea
write mapped points -replace lec mapped_points_minarea
set verification information logical_eq_check_minarea
write verification information
exit
```

Based on the results presented, it is evident that there are no discrepancies in the equivalence checking for the minimum area scenario. All points are correctly mapped, and every compare point is found to be equivalent in both the golden netlist (RTL code) and the modified netlist (synthesized netlist), confirming that equivalence is established for all comparison points. This confirms that our design is consistent between the RTL and the synthesized netlist.

## Equivalence Checking for Minimum Time

The outputs of the equivalence checking for the minimum timing constraint are shown below.

```
set log file logical_equivalence_checking_minarea.log -replace
|
read library /home/vanshika25155/Desktop/VDF Project/VDF Project/VDF Project/CEC/min_area/slow.v -verilog -both
read design /home/vanshika25155/Desktop/VDF Project/VDF Project/VDF Project/CEC/min_area/toll.v -golden
read design /home/vanshika25155/Desktop/VDF Project/VDF Project/VDF Project/CEC/min_area/netlist.v -verilog -revised

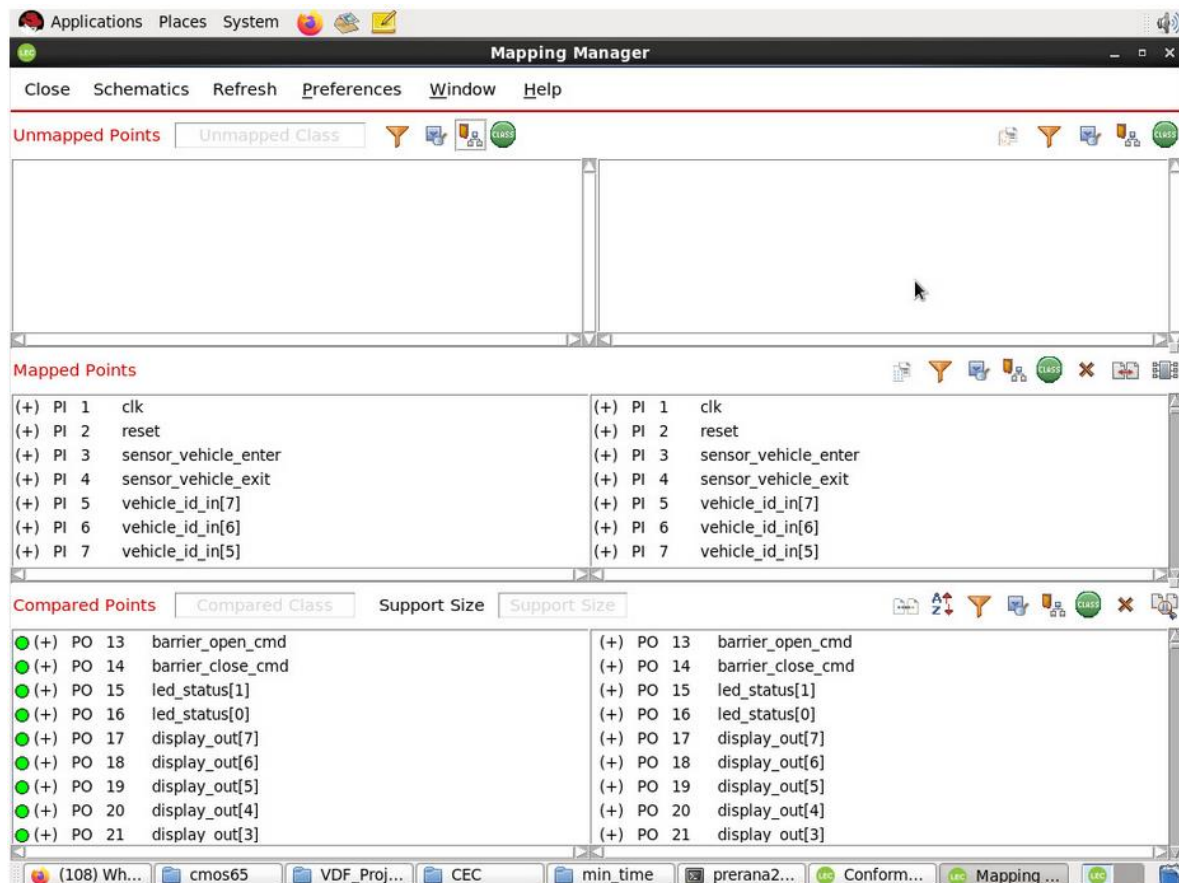
set system mode lec
add compared points -all
compare
report messages -compare -verb
report compare data -noneq
report verification
write compared points -replace lec_compared_points_minarea
write mapped points -replace lec_mapped_points_minarea
set verification information logical_eq_check_minarea
write verification information
exit
```

The screenshot displays the Cadence Equivalence Checking tool interface. The top menu bar includes File, Compare, Tools, Custom, Preferences, Window, and Help. Below the menu is a toolbar with icons for various functions, and a 'Setup' button. The main workspace is divided into two panes: 'Golden' and 'Revised'. Both panes show a hierarchical tree structure of the design, starting with 'toll\_gate' and listing components like 'u\_account\_balance', 'u\_barrier\_ctrl', and 'u\_vehicle\_interface'. Below the design panes is a 'Verification Report' section, which contains a table with the following data:

| Category                                                                  | Count |
|---------------------------------------------------------------------------|-------|
| . Non-standard modeling options used:                                     | 0     |
| . Incomplete verification:                                                | 0     |
| . User modification to design:                                            | 0     |
| . Conformal Constraint Designer clock domain crossing checks recommended: | 1     |
| Multiple clocks in the design:                                            | yes   |
| . Design ambiguity:                                                       | 0     |
| . Compare Results:                                                        | PASS  |

At the bottom of the interface, a status bar indicates 'Compare done!' and '100% completed'.





Based on the results presented, we observe that there are no discrepancies in the equivalence checking for the tight timing scenario. All compare points are matched, and there are no unmapped points, affirming that the golden netlist (RTL code) and the revised netlist (synthesized netlist) are equivalent. This confirms that our design is consistent between both the RTL and the synthesized netlist.

## Equivalence Checking for Intermediate Constraints

The outputs of the equivalence checking for the intermediate timing and area constraints are shown below.

```
set log file logical_equivalence_checking_minarea.log -replace

read library /home/prerana25172/Desktop/cmos65/VDF_Project/CEC/intermediate/slow.v -verilog -both
read design /home/prerana25172/Desktop/cmos65/VDF_Project/CEC/intermediate/toll.v -golden
read design /home/prerana25172/Desktop/cmos65/VDF_Project/CEC/intermediate/netlist.v -verilog -revised

set system mode lec
add compared points -all
compare
report messages -compare -verb
report compare data -noneq
report verification
write compared points -replace lec_compared_points_minarea
write mapped points -replace lec_mapped_points_minarea
set verification information logical_eq_check_minarea
write verification information
exit
```

Applications Places System

### Conformal(R) Logic Equivalence Checking

File Compare Tools Custom Preferences Window Help

Design Library Tool Setup Design Data Rule Checking

Golden

- toll\_gate
  - 15 library cells and 45 primitives
  - u\_account\_balance(account\_balance)
  - u\_barrier\_ctrl(barrier\_ctrl)
  - u\_vehicle\_interface(vehicle\_interface)

Revised

- toll\_gate
  - 38 library cells
  - u\_account\_balance(account\_balance)
  - u\_barrier\_ctrl(barrier\_ctrl)
  - u\_vehicle\_interface(vehicle\_interface)

```
// Command: report verification
```

Verification Report

| Category                                                                   | Count |
|----------------------------------------------------------------------------|-------|
| 1. Non-standard modeling options used:                                     | 0     |
| 2. Incomplete verification:                                                | 0     |
| 3. User modification to design:                                            | 0     |
| 4. Conformal Constraint Designer clock domain crossing checks recommended: | 1     |
| Multiple clocks in the design:                                             | yes   |
| 5. Design ambiguity:                                                       | 0     |
| 6. Compare Results:                                                        | PASS  |

Compare done! 100% completed

(108) What... cmos65 VDF\_Project CEC intermediate prerana251... Conformal(...)

Applications Places System

### Mapping Manager

Close Schematics Refresh Preferences Window Help

Unmapped Points Unmapped Class

Report Unmapped Points Summary

Mapped Points

|                               |                               |
|-------------------------------|-------------------------------|
| (+) PI 1 clk                  | (+) PI 1 clk                  |
| (+) PI 2 reset                | (+) PI 2 reset                |
| (+) PI 3 sensor_vehicle_enter | (+) PI 3 sensor_vehicle_enter |
| (+) PI 4 sensor_vehicle_exit  | (+) PI 4 sensor_vehicle_exit  |
| (+) PI 5 vehicle_id_in[7]     | (+) PI 5 vehicle_id_in[7]     |
| (+) PI 6 vehicle_id_in[6]     | (+) PI 6 vehicle_id_in[6]     |
| (+) PI 7 vehicle_id_in[5]     | (+) PI 7 vehicle_id_in[5]     |

Compared Points Compared Class Support Size Support Size

|                             |                             |
|-----------------------------|-----------------------------|
| (+) PO 13 barrier_open_cmd  | (+) PO 13 barrier_open_cmd  |
| (+) PO 14 barrier_close_cmd | (+) PO 14 barrier_close_cmd |
| (+) PO 15 led_status[1]     | (+) PO 15 led_status[1]     |
| (+) PO 16 led_status[0]     | (+) PO 16 led_status[0]     |
| (+) PO 17 display_out[7]    | (+) PO 17 display_out[7]    |
| (+) PO 18 display_out[6]    | (+) PO 18 display_out[6]    |
| (+) PO 19 display_out[5]    | (+) PO 19 display_out[5]    |
| (+) PO 20 display_out[4]    | (+) PO 20 display_out[4]    |
| (+) PO 21 display_out[3]    | (+) PO 21 display_out[3]    |

(108) Wh... cmos65 VDF\_Proj... CEC intermed... prerana2... Conform... Mapping ...

Based on the results presented, it is evident that there are no discrepancies in the equivalence checking for the intermediate case. There are no points that remain unmapped, and all comparison points show equivalence in both the golden netlist (RTL code) and the revised netlist (synthesized netlist). The equivalence has been confirmed for every comparison point, validating that our design is consistent in both the RTL and the synthesized netlist.

## BAD NETLIST:

### CEC For Bad Netlist -1

The original netlist:

```

wire n_309, n_310, n_311, n_312, n_313, n_314, n_315, n_316;
wire n_317, n_320, n_323, n_324, n_325;
INVXL g1597(.A (n_309), .Y (sufficient_balance));
DFFQX1 charge_done_reg(.CK (clk), .D (n_323), .Q (charge_done));
NOR2BX1 g16885(.AN (fsm_charge_req), .B (n_309), .Y (n_310));
OAI31X1 g16886(.A0 (n_307), .A1 (n_308), .A2 (n_293), .B0
(fsm_check_req), .Y (n_309));
NAND3BX1 g16887(.AN (n_313), .B (n_315), .C (n_312), .Y (n_293));
OR4X1 g16888(.A (n_314), .B (n_299), .C (n_292), .D (n_300), .Y
(n_313));
NOR3BX1 g16889(.AN (n_316), .B (n_303), .C (n_304), .Y (n_312));
INVX1 g16890(.A (n_317), .Y (n_292));
NOR2XL g16891(.A (n_297), .B (n_298), .Y (n_317));
NOR2XL g16892(.A (n_305), .B (n_306), .Y (n_315));
NAND2XL g16893(.A (n_226), .B (n_291), .Y (n_300));
NAND4XL g16894(.A (n_237), .B (n_236), .C (n_238), .D (n_290), .Y
(n_304));
AOI221X1 g16895(.A0 (\balance_db[0] [7]), .A1 (n_160), .B0
(\balance_db[1] [7]), .B1 (n_161), .C0 (n_288), .Y (n_291));
NAND2XL g16896(.A (n_235), .B (n_289), .Y (n_298));
NOR2XL g16897(.A (n_301), .B (n_302), .Y (n_316));
NAND3X1 g16898(.A (n_176), .B (n_198), .C (n_285), .Y (n_299));
NAND4XL g16899(.A (n_247), .B (n_229), .C (n_232), .D (n_283), .Y
(n_305));
NAND4XL g16900(.A (n_223), .B (n_225), .C (n_224), .D (n_287), .Y
(n_306));
AOI211X1 g16901(.A0 (\balance_db[6] [11]), .A1 (n_155), .B0 (n_234),
.C0 (n_278), .Y (n_290));
NAND3X1 g16902(.A (n_201), .B (n_202), .C (n_284), .Y (n_303));
OAI211X1 g16903(.A0 (n_139), .A1 (n_246), .B0 (n_211), .C0 (n_280),
.Y (n_301));
AOI211X1 g16904(.A0 (\balance_db[0] [5]), .A1 (n_160), .B0 (n_265),
.C0 (n_286), .Y (n_289));
NAND4X1 g16905(.A (n_217), .B (n_216), .C (n_218), .D (n_281), .Y

```

Bad Netlist: NAND gates replace with NOR gate and NOR gate replace with NAND gate

```

File Edit View Search Tools Documents Help
Open Save Undo
netlist.v
wire n_283, n_284, n_285, n_286, n_287, n_288, n_289, n_290;
wire n_291, n_292, n_293, n_296, n_297, n_298, n_299, n_300;
wire n_301, n_302, n_303, n_304, n_305, n_306, n_307, n_308;
wire n_309, n_310, n_311, n_312, n_313, n_314, n_315, n_316;
wire n_317, n_320, n_323, n_324, n_325;
INVXL g1597(.A (n_309), .Y (sufficient_balance));
DFFQX1 charge_done_reg(.CK (clk), .D (n_323), .Q (charge_done));
NOR2BX1 g16885(.AN (fsm_charge_req), .B (n_309), .Y (n_310));
OAI31X1 g16886(.A0 (n_307), .A1 (n_308), .A2 (n_293), .B0
(fsm_check_req), .Y (n_309));
NAND3BX1 g16887(.AN (n_313), .B (n_315), .C (n_312), .Y (n_293));
OR4X1 g16888(.A (n_314), .B (n_299), .C (n_292), .D (n_300), .Y
(n_313));
NOR3BX1 g16889(.AN (n_316), .B (n_303), .C (n_304), .Y (n_312));
INVX1 g16890(.A (n_317), .Y (n_292));
NOR2XL g16891(.A (n_297), .B (n_298), .Y (n_317));
NAND2XL g16892(.A (n_305), .B (n_306), .Y (n_315));
NOR2XL g16893(.A (n_226), .B (n_291), .Y (n_300));
NAND4XL g16894(.A (n_237), .B (n_236), .C (n_238), .D (n_290), .Y
(n_304));
AOI221X1 g16895(.A0 (\balance_db[0] [7]), .A1 (n_160), .B0
(\balance_db[1] [7]), .B1 (n_161), .C0 (n_288), .Y (n_291));
NAND2XL g16896(.A (n_235), .B (n_289), .Y (n_298));
NOR2XL g16897(.A (n_301), .B (n_302), .Y (n_316));
NAND3X1 g16898(.A (n_176), .B (n_198), .C (n_285), .Y (n_299));
NAND4XL g16899(.A (n_247), .B (n_229), .C (n_232), .D (n_283), .Y
(n_305));
NAND4XL g16900(.A (n_223), .B (n_225), .C (n_224), .D (n_287), .Y
(n_306));
AOI211X1 g16901(.A0 (\balance_db[6] [11]), .A1 (n_155), .B0 (n_234),
.C0 (n_278), .Y (n_290));
NAND3X1 g16902(.A (n_201), .B (n_202), .C (n_284), .Y (n_303));
OAI211X1 g16903(.A0 (n_139), .A1 (n_246), .B0 (n_211), .C0 (n_280),
.Y (n_301));

```

## Results: Equivalence Checking

Applications Places System

Conformal(R) Logic Equivalence Checking

File Compare Tools Custom Preferences Window Help

Setup LEC

Design Library Tool Setup Design Data Rule Checking

Golden

- toll\_gate
  - 15 library cells and 45 primitives
  - u\_account\_balance(account\_balance)
  - u\_barrier\_ctrl(barrier\_ctrl)
  - u\_vehicle\_interface(vehicle\_interface)

Revised

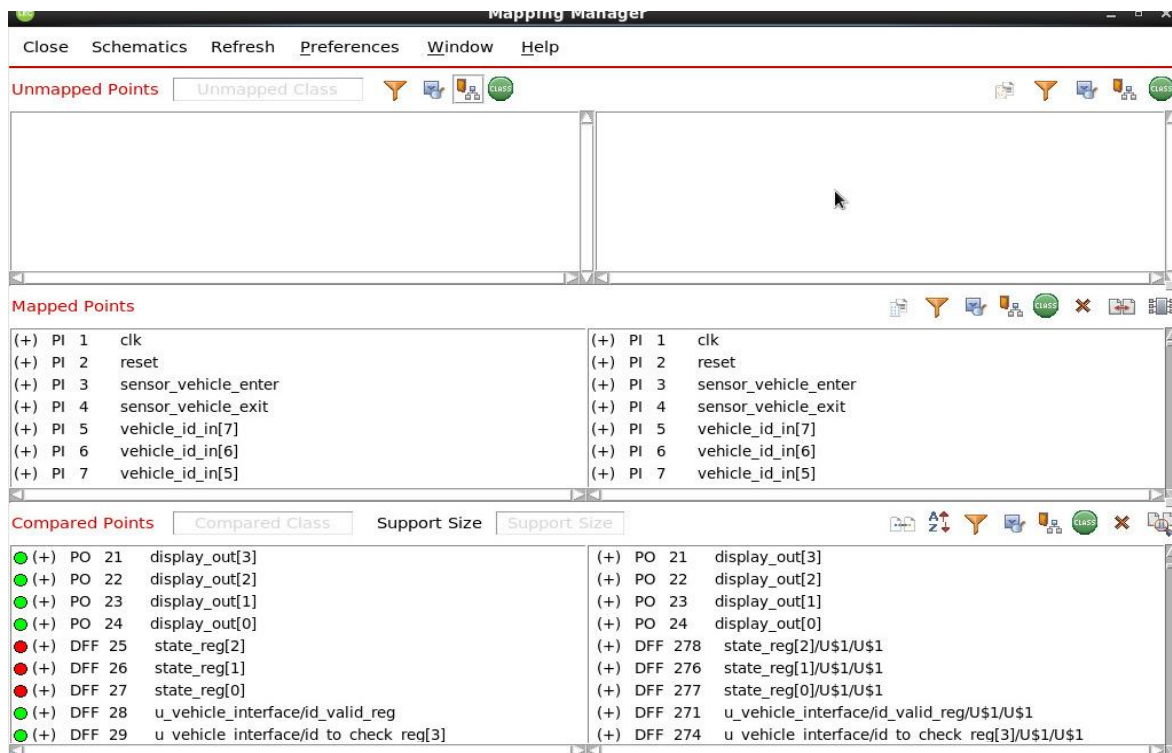
- toll\_gate
  - 38 library cells
  - u\_account\_balance(account\_balance)
  - u\_barrier\_ctrl(barrier\_ctrl)
  - u\_vehicle\_interface(vehicle\_interface)

| Category                                                                   | Count      |
|----------------------------------------------------------------------------|------------|
| 1. Non-standard modeling options used:                                     | 0          |
| 2. Incomplete verification:                                                | 0          |
| 3. User modification to design:                                            | 0          |
| 4. Conformal Constraint Designer clock domain crossing checks recommended: | 1          |
| Multiple clocks in the design:                                             | yes        |
| 5. Design ambiguity:                                                       | 0          |
| 6. Compare Results:                                                        | FAIL:NONEQ |

CPU time : 1.50 seconds  
Elapse time : 3 seconds

Compare done! 100% completed

(108) ... cmos65 VDF\_... CEC min\_... netlis... prera... Confo... Save ... Save ...



## CEC For Bad Netlist –2

The original netlist:

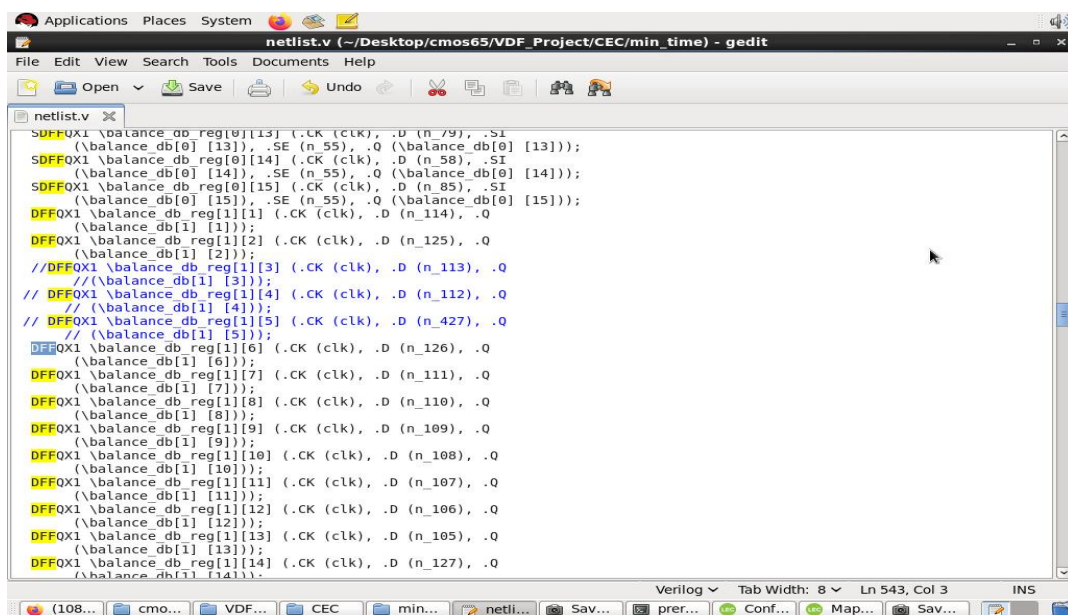


```

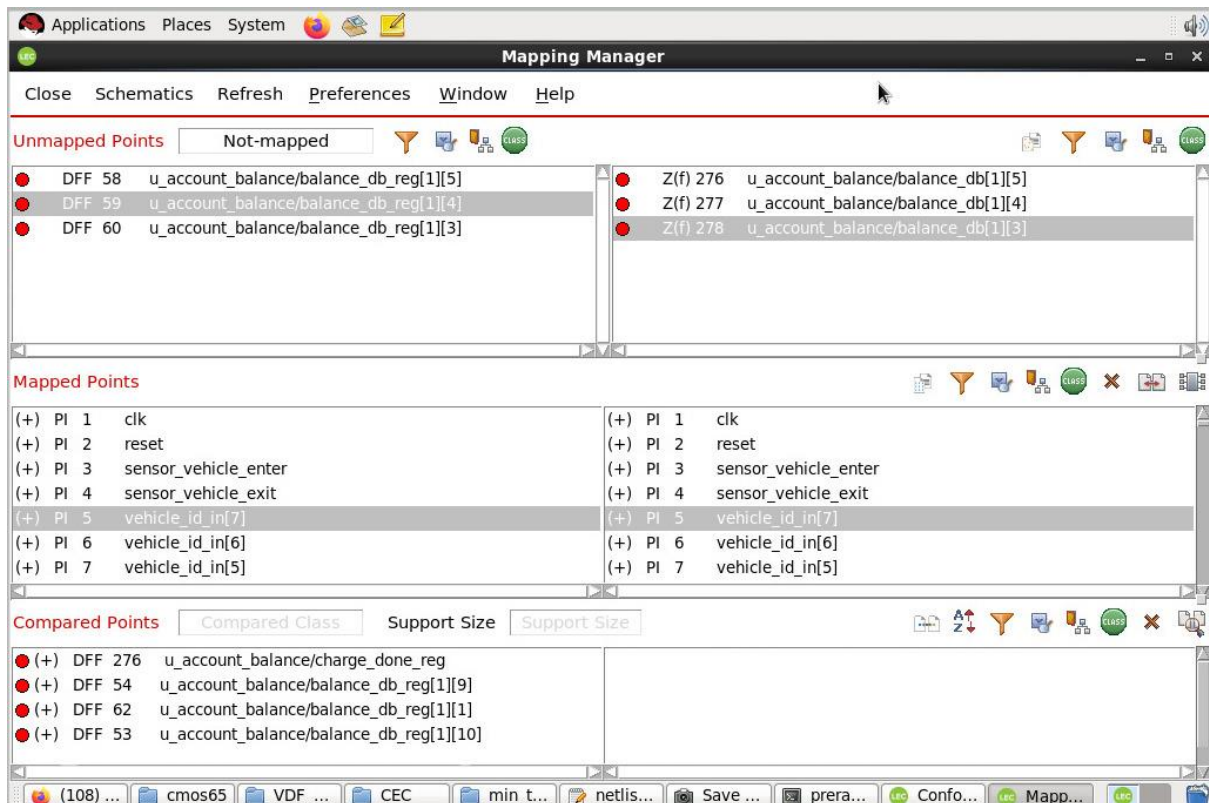
SDFFQX1 \balance_db_reg[0][13] (.CK (clk), .D (n_79), .SI
(\balance_db[0] [13]), .SE (n_55), .Q (\balance_db[0] [13]));
SDFFQX1 \balance_db_reg[0][14] (.CK (clk), .D (n_58), .SI
(\balance_db[0] [14]), .SE (n_55), .Q (\balance_db[0] [14]));
SDFFQX1 \balance_db_reg[0][15] (.CK (clk), .D (n_85), .SI
(\balance_db[0] [15]), .SE (n_55), .Q (\balance_db[0] [15]));
DFFQX1 \balance_db_reg[1][1] (.CK (clk), .D (n_114), .Q
(\balance_db[1] [1]));
DFFQX1 \balance_db_reg[1][2] (.CK (clk), .D (n_125), .Q
(\balance_db[1] [2]));
DFFQX1 \balance_db_reg[1][3] (.CK (clk), .D (n_113), .Q
(\balance_db[1] [3]));
DFFQX1 \balance_db_reg[1][4] (.CK (clk), .D (n_112), .Q
(\balance_db[1] [4]));
DFFQX1 \balance_db_reg[1][5] (.CK (clk), .D (n_427), .Q
(\balance_db[1] [5]));
DFFQX1 \balance_db_reg[1][6] (.CK (clk), .D (n_126), .Q
(\balance_db[1] [6]));
DFFQX1 \balance_db_reg[1][7] (.CK (clk), .D (n_111), .Q
(\balance_db[1] [7]));
DFFQX1 \balance_db_reg[1][8] (.CK (clk), .D (n_110), .Q
(\balance_db[1] [8]));
DFFQX1 \balance_db_reg[1][9] (.CK (clk), .D (n_109), .Q
(\balance_db[1] [9]));

```

In this case we have commented out 3 flip flops in the synthesized netlist as shown below.



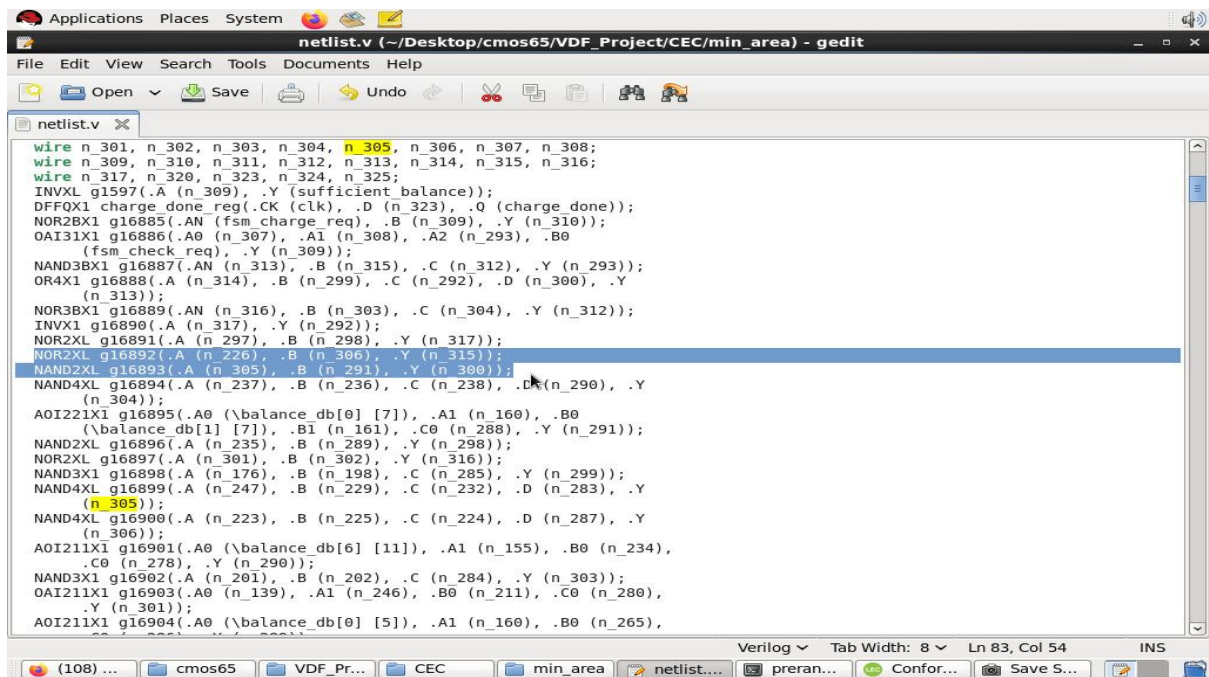
Results: Equivalence Checking



Because some of the flip flops were missing from the synthesised netlist, there are several places where the RTL and netlist do not match. The motion of the flip-flops may also cause some of the compare points to fail. Because there are both unmapped and failed compare points, the equivalence check is unsuccessful.

### CEC For Bad Netlist –3

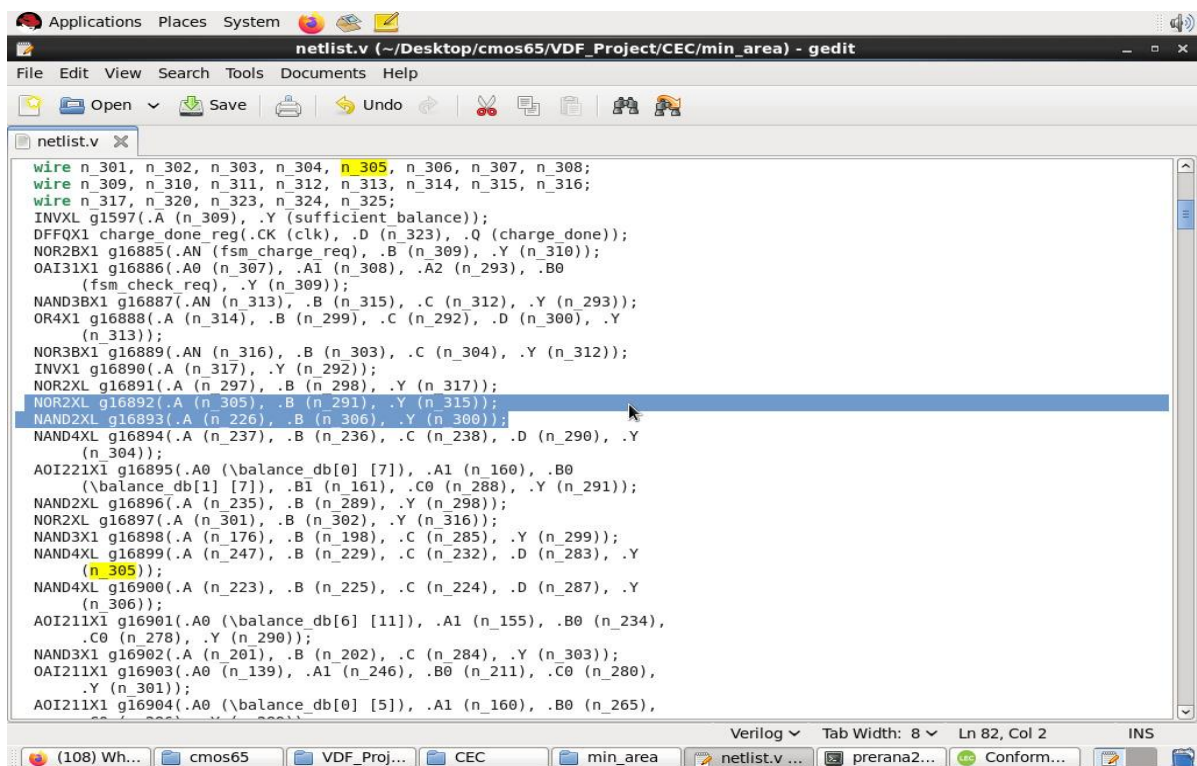
The original netlist:



```
netlist.v
File Edit View Search Tools Documents Help
(108) ... cmos65 VDF_Pr... CEC min_area netlist... preran... Confor... Save S...

wire n_301, n_302, n_303, n_304, n_305, n_306, n_307, n_308;
wire n_309, n_310, n_311, n_312, n_313, n_314, n_315, n_316;
wire n_317, n_320, n_323, n_324, n_325;
INVXL g1597(.A (n_309), .Y (sufficient_balance));
DFFQX1 charge_done_reg(.CK (clk), .D (n_323), .Q (charge_done));
NOR2BX1 g16885(.AN (fsm_charge_req), .B (n_309), .Y (n_310));
OAI31X1 g16886(.A0 (n_307), .A1 (n_308), .A2 (n_293), .B0
(fsm_check_req), .Y (n_309));
NAND3BX1 g16887(.AN (n_313), .B (n_315), .C (n_312), .Y (n_293));
OR4X1 g16888(.A (n_314), .B (n_299), .C (n_292), .D (n_300), .Y
(n_313));
NOR3BX1 g16889(.AN (n_316), .B (n_303), .C (n_304), .Y (n_312));
INVX1 g16890(.A (n_317), .Y (n_292));
NOR2XL g16891(.A (n_297), .B (n_298), .Y (n_317));
NOR2XL g16892(.A (n_226), .B (n_306), .Y (n_315));
NAND2XL g16893(.A (n_305), .B (n_291), .Y (n_300));
NAND4XL g16894(.A (n_237), .B (n_236), .C (n_238), .D (n_290), .Y
(n_304));
AOI221X1 g16895(.A0 (\balance_db[0] [7]), .A1 (n_160), .B0
(\balance_db[1] [7]), .B1 (n_161), .C0 (n_288), .Y (n_291));
NAND2XL g16896(.A (n_235), .B (n_289), .Y (n_298));
NOR2XL g16897(.A (n_301), .B (n_302), .Y (n_316));
NAND3X1 g16898(.A (n_176), .B (n_198), .C (n_285), .Y (n_299));
NAND4XL g16899(.A (n_247), .B (n_229), .C (n_232), .D (n_283), .Y
(n_305));
NAND4XL g16900(.A (n_223), .B (n_225), .C (n_224), .D (n_287), .Y
(n_306));
AOI211X1 g16901(.A0 (\balance_db[6] [11]), .A1 (n_155), .B0 (n_234),
.C0 (n_278), .Y (n_290));
NAND3X1 g16902(.A (n_201), .B (n_202), .C (n_284), .Y (n_303));
OAI211X1 g16903(.A0 (n_139), .A1 (n_246), .B0 (n_211), .C0 (n_280),
.Y (n_301));
AOI211X1 g16904(.A0 (\balance_db[0] [5]), .A1 (n_160), .B0 (n_265),
```

Interchange the port of 2 cells:

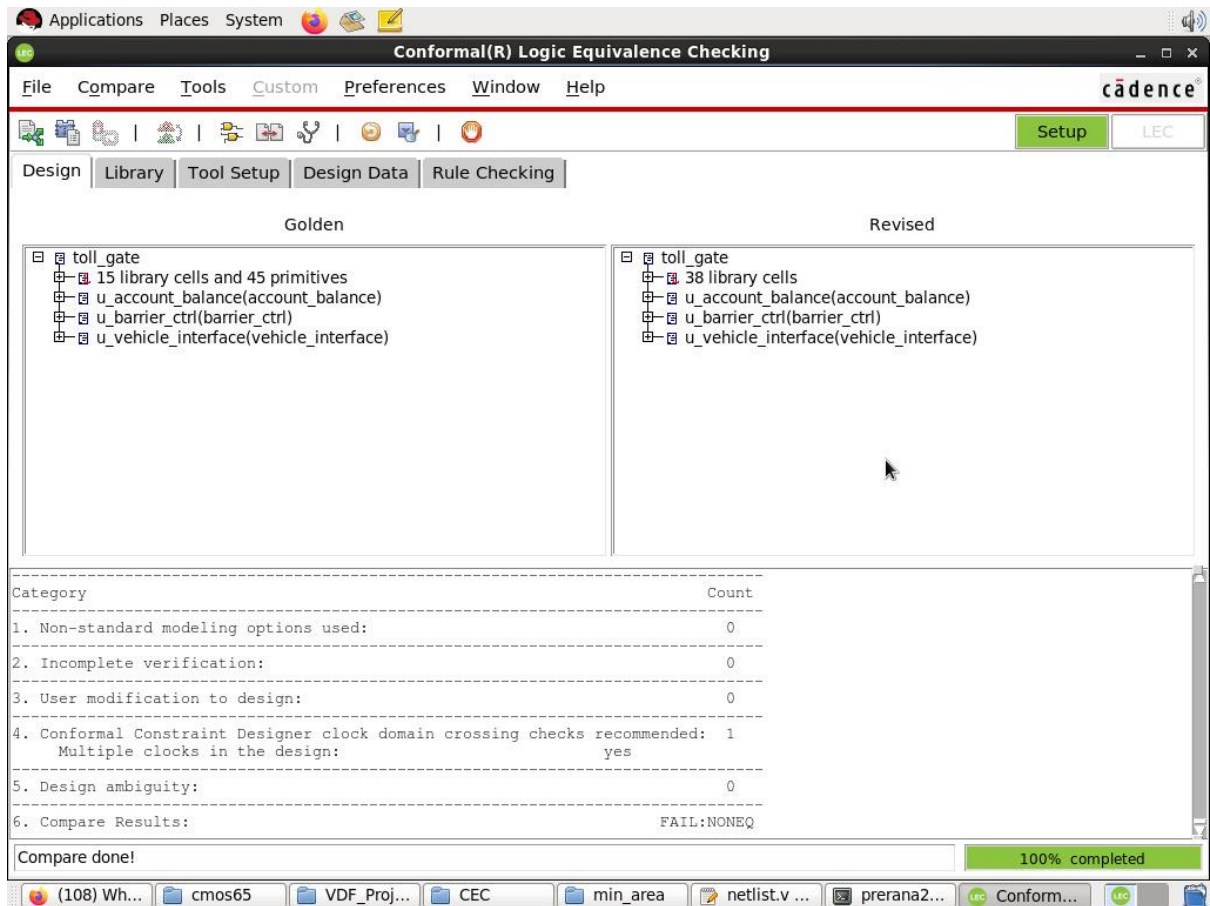


```
netlist.v
File Edit View Search Tools Documents Help
(108) Wh... cmos65 VDF_Proj... CEC min_area netlist.v ... prerana2... Conform...

wire n_301, n_302, n_303, n_304, n_305, n_306, n_307, n_308;
wire n_309, n_310, n_311, n_312, n_313, n_314, n_315, n_316;
wire n_317, n_320, n_323, n_324, n_325;
INVXL g1597(.A (n_309), .Y (sufficient_balance));
DFFQX1 charge_done_reg(.CK (clk), .D (n_323), .Q (charge_done));
NOR2BX1 g16885(.AN (fsm_charge_req), .B (n_309), .Y (n_310));
OAI31X1 g16886(.A0 (n_307), .A1 (n_308), .A2 (n_293), .B0
(fsm_check_req), .Y (n_309));
NAND3BX1 g16887(.AN (n_313), .B (n_315), .C (n_312), .Y (n_293));
OR4X1 g16888(.A (n_314), .B (n_299), .C (n_292), .D (n_300), .Y
(n_313));
NOR3BX1 g16889(.AN (n_316), .B (n_303), .C (n_304), .Y (n_312));
INVX1 g16890(.A (n_317), .Y (n_292));
NOR2XL g16891(.A (n_297), .B (n_298), .Y (n_317));
NOR2XL g16892(.A (n_305), .B (n_291), .Y (n_315));
NAND2XL g16893(.A (n_226), .B (n_306), .Y (n_300));
NAND4XL g16894(.A (n_237), .B (n_236), .C (n_238), .D (n_290), .Y
(n_304));
AOI221X1 g16895(.A0 (\balance_db[0] [7]), .A1 (n_160), .B0
(\balance_db[1] [7]), .B1 (n_161), .C0 (n_288), .Y (n_291));
NAND2XL g16896(.A (n_235), .B (n_289), .Y (n_298));
NOR2XL g16897(.A (n_301), .B (n_302), .Y (n_316));
NAND3X1 g16898(.A (n_176), .B (n_198), .C (n_285), .Y (n_299));
NAND4XL g16899(.A (n_247), .B (n_229), .C (n_232), .D (n_283), .Y
(n_305));
NAND4XL g16900(.A (n_223), .B (n_225), .C (n_224), .D (n_287), .Y
(n_306));
AOI211X1 g16901(.A0 (\balance_db[6] [11]), .A1 (n_155), .B0 (n_234),
.C0 (n_278), .Y (n_290));
NAND3X1 g16902(.A (n_201), .B (n_202), .C (n_284), .Y (n_303));
OAI211X1 g16903(.A0 (n_139), .A1 (n_246), .B0 (n_211), .C0 (n_280),
.Y (n_301));
AOI211X1 g16904(.A0 (\balance_db[0] [5]), .A1 (n_160), .B0 (n_265),
```

Results: Equivalence Checking



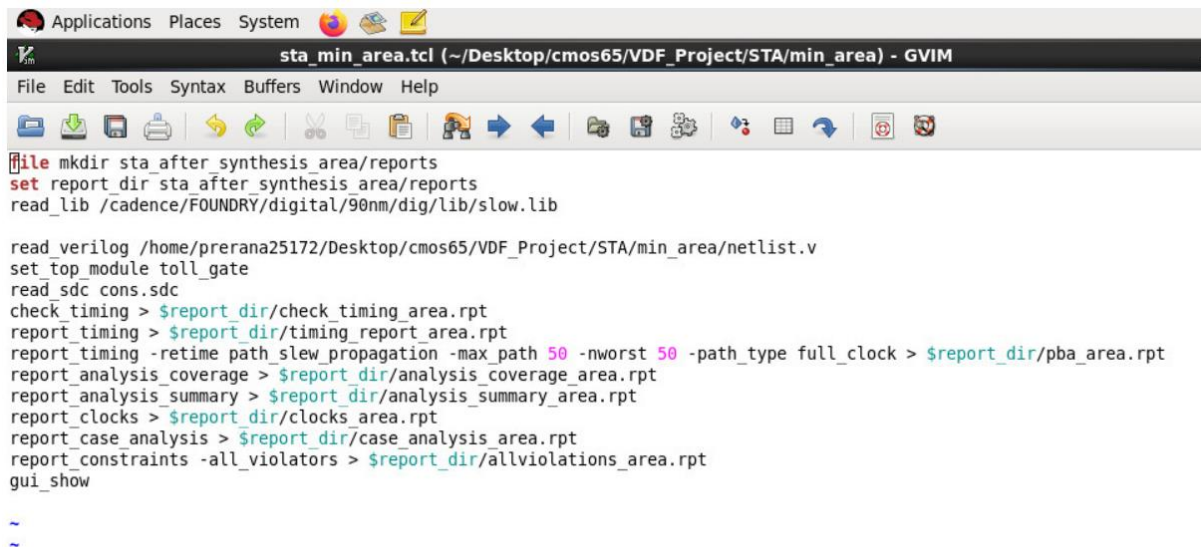


## Chapter 5

# STATIC TIMING ANALYSIS

Static Timing Analysis (STA) is a method utilized to verify the timing performance of a digital circuit without requiring simulation. It assesses the timing paths in the design by calculating the arrival times of signals at various timing locations and comparing these to the required times, ensuring that the circuit meets its timing constraints. STA is a crucial tool for maintaining timing precision in complex digital designs, as it identifies issues such as setup and hold time violations and uncovers potential delays that could impact the circuit's functionality. The inputs required for performing static timing analysis include the synthesized netlist, the technology library, and the constraints under which we want the tool to operate. The output of the tool is a timing report that details any violations, including both setup and hold types. Using the constraints file for the intermediate step, we will analyze the static timing of each of the three netlists synthesized previously.

Below is the tcl script used to execute the simulation for one of the netlists. Additionally, the tcl script remains unchanged for the other netlists, except that they will be passed accordingly. For each case, we have performed both path-based and graph-based analyses.

A screenshot of a GVIM editor window. The title bar shows the application name and the file path: 'sta\_min\_area.tcl (~/Desktop/cmos65/VDF\_Project/STA/min\_area) - GVIM'. The menu bar includes File, Edit, Tools, Syntax, Buffers, Window, and Help. The toolbar contains various icons for file operations and editing. The main text area displays a TCL script for performing Static Timing Analysis. The script starts with creating a directory for reports, setting the report directory, and reading a library file. It then reads a Verilog netlist, sets the top module, and reads an SDC constraints file. A series of commands are used to generate various timing reports, such as check\_timing, report\_timing, report\_analysis\_coverage, report\_analysis\_summary, report\_clocks, report\_case\_analysis, and report\_constraints. The script ends with a 'gui\_show' command.

```
file mkdir sta_after_synthesis_area/reports
set report_dir sta_after_synthesis_area/reports
read_lib /cadence/FOUNDRY/digital/90nm/dig/lib/slow.lib

read_verilog /home/prerana25172/Desktop/cmos65/VDF_Project/STA/min_area/netlist.v
set_top_module toll_gate
read_sdc cons.sdc
check_timing > $report_dir/check_timing_area.rpt
report_timing > $report_dir/timing_report_area.rpt
report_timing -retime path_slew_propagation -max_path 50 -nworst 50 -path_type full_clock > $report_dir/pba_area.rpt
report_analysis_coverage > $report_dir/analysis_coverage_area.rpt
report_analysis_summary > $report_dir/analysis_summary_area.rpt
report_clocks > $report_dir/clocks_area.rpt
report_case_analysis > $report_dir/case_analysis_area.rpt
report_constraints -all_violators > $report_dir/allviolations_area.rpt
gui_show

~
~
```

## Static Timing Analysis for Minimum Area Netlist

The violation reports for the static timing analysis for minimum area netlist are shown below in the two figures.

```
#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sat Oct 11 19:12:38 2025
# Design: toll_gate
# Command: report_constraints -all_violators > sta_after_synthesis_area/reports/allviolations_area.rpt
#####
# format : frame 0 : split 1

max_delay/setup
-----
No paths found

min_delay/hold
-----
No paths found

Check type : clock_period
-----
No paths found

Check type : skew
-----
No paths found

Check type : pulse_width
-----
No violating Checks with given description found

Check type : max_transition
-----
No Violations found

Check type : min_transition
-----
No Violations found

Check type : max_capacitance
-----
No Violations found

Check type : min_capacitance
-----
No Violations found

Check type : max_fanout
-----
No Violations found

Check type : min_fanout
-----
No Violations found
```

The coverage report is shown below.

```
#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sat Oct 11 19:12:38 2025
# Design: toll_gate
# Command: report_analysis_coverage > $report_dir/analysis_coverage_area.rpt
#####

-----
                        TIMING CHECK COVERAGE SUMMARY
-----
Check Type              No. of  Met      Violated    Untested
                        Checks
-----
Hold                    644    644 (100%)  0 (0%)      0 (0%)
PulseWidth              508    508 (100%)  0 (0%)      0 (0%)
Setup                   644    644 (100%)  0 (0%)      0 (0%)
-----
```

The Graph Based Analysis (GBA) report is shown below.

```

#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sat Oct 11 19:12:38 2025
# Design: toll_gate
# Command: report_timing > $report_dir/timing_report_area.rpt
#####
Path 1: MET Setup Check with Pin u_account_balance/balance_db_reg[6][1]/CK
Endpoint: u_account_balance/balance_db_reg[6][1]/SE (v) checked with leading edge of 'clk'
Beginpoint: u_vehicle_interface/id_to_check_reg[3]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time 0.010
- Setup 0.418
+ Phase Shift 5.600
= Required Time 5.192
- Arrival Time 4.650
= Slack Time 0.542
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.010
= Beginpoint Arrival Time 0.010
#####

```

| Instance                               | Arc         | Cell     | Delay | Arrival Time | Required Time |
|----------------------------------------|-------------|----------|-------|--------------|---------------|
| u_vehicle_interface/id_to_check_reg[3] | CK ^        | -        | -     | 0.010        | 0.552         |
| u_vehicle_interface/id_to_check_reg[3] | CK ^ -> Q ^ | DFFQX1   | 0.531 | 0.542        | 1.084         |
| u_account_balance/g17076               | A ^ -> Y v  | CLKINXV1 | 0.445 | 0.986        | 1.529         |
| u_account_balance/g17074               | A v -> Y ^  | NOR2XL   | 0.751 | 1.738        | 2.280         |
| u_account_balance/g17070               | A ^ -> Y v  | CLKINXV1 | 0.136 | 1.874        | 2.416         |
| u_account_balance/g17044               | B v -> Y ^  | NOR2XL   | 0.542 | 2.416        | 2.959         |
| u_account_balance/g16923               | B1 ^ -> Y v | A0I221X1 | 0.166 | 2.582        | 3.125         |
| u_account_balance/g16910               | B0 v -> Y ^ | OAI21X1  | 0.095 | 2.677        | 3.219         |
| u_account_balance/g16904               | C0 ^ -> Y v | A0I211X1 | 0.052 | 2.729        | 3.272         |
| u_account_balance/g16896               | B v -> Y ^  | NAND2XL  | 0.115 | 2.845        | 3.387         |
| u_account_balance/g16891               | B ^ -> Y v  | NOR2XL   | 0.093 | 2.938        | 3.480         |
| u_account_balance/g16890               | A v -> Y ^  | INVX1    | 0.048 | 2.986        | 3.528         |
| u_account_balance/g16888               | C ^ -> Y ^  | OR4X1    | 0.139 | 3.126        | 3.668         |
| u_account_balance/g16887               | AN ^ -> Y ^ | NAND3BX1 | 0.117 | 3.243        | 3.785         |
| u_account_balance/g16886               | A2 ^ -> Y v | OAI31X1  | 0.114 | 3.357        | 3.899         |
| u_account_balance/g16885               | B v -> Y ^  | NOR2BX1  | 0.126 | 3.482        | 4.025         |
| u_account_balance/g3679                | AN ^ -> Y ^ | NOR2BX1  | 0.146 | 3.629        | 4.171         |
| u_account_balance/g3668                | B ^ -> Y ^  | AND2X1   | 0.199 | 3.828        | 4.370         |
| u_account_balance/g3643                | C ^ -> Y v  | NAND3X1  | 0.822 | 4.650        | 5.192         |
| u_account_balance/balance_db_reg[6][1] | SE v        | SDFQX1   | 0.000 | 4.650        | 5.192         |

The Path Based Analysis(PBA) report is shown below

```

Beginpoint: u_vehicle_interface/id_to_check_reg[3]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Retime Analysis { Data Path-Slew }
Other End Arrival Time 0.010
- Setup 0.418
+ Phase Shift 5.600
= Required Time 5.192
- Arrival Time 4.554
= Slack Time 0.638
= Slack Time(original) 0.542
Clock Rise Edge 0.000
+ Source Insertion Delay 0.010
= Beginpoint Arrival Time 0.010
Timing Path:

```

| Instance                               | Arc         | Cell     | Retime Delay | Arrival Time | Required Time |
|----------------------------------------|-------------|----------|--------------|--------------|---------------|
| -                                      | clk ^       | -        | -            | 0.010        | 0.648         |
| u_vehicle_interface/id_to_check_reg[3] | -           | DFFQX1   | 0.000        | 0.010        | 0.648         |
| u_vehicle_interface/id_to_check_reg[3] | CK ^ -> Q ^ | DFFQX1   | 0.531        | 0.542        | 1.180         |
| u_account_balance/g17076               | -           | CLKINXV1 | 0.000        | 0.542        | 1.180         |
| u_account_balance/g17076               | A ^ -> Y v  | CLKINXV1 | 0.445        | 0.986        | 1.625         |
| u_account_balance/g17074               | -           | NOR2XL   | 0.000        | 0.986        | 1.625         |
| u_account_balance/g17074               | A v -> Y ^  | NOR2XL   | 0.752        | 1.738        | 2.377         |
| u_account_balance/g17070               | -           | CLKINXV1 | 0.000        | 1.738        | 2.377         |
| u_account_balance/g17070               | A ^ -> Y v  | CLKINXV1 | 0.136        | 1.874        | 2.513         |
| u_account_balance/g17044               | -           | NOR2XL   | 0.000        | 1.874        | 2.513         |
| u_account_balance/g17044               | B v -> Y ^  | NOR2XL   | 0.542        | 2.417        | 3.055         |
| u_account_balance/g16923               | -           | A0I221X1 | 0.000        | 2.417        | 3.055         |
| u_account_balance/g16923               | B1 ^ -> Y v | A0I221X1 | 0.166        | 2.582        | 3.221         |
| u_account_balance/g16910               | -           | OAI21X1  | 0.000        | 2.582        | 3.221         |
| u_account_balance/g16910               | B0 v -> Y ^ | OAI21X1  | 0.090        | 2.673        | 3.311         |

**Static Timing Analysis for Minimum Timing Constraint Netlist**

The violation report is shown below in the two figures.

```
#####
# Generated by:      Cadence Tempus 20.10-p003_1
# OS:               Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on:      Sat Oct 11 18:42:10 2025
# Design:           toll_gate
# Command:          report constraints -all violators > sta_after_synthesis_area/reports/allviolations_area.rpt
#####
# format : frame 0 : split 1

max_delay/setup
-----
No paths found

min_delay/hold
-----
No paths found

Check type : clock_period
-----
No paths found

Check type : skew
-----
No paths found

Check type : pulse_width
-----
No violating Checks with given description found

Check type : max_transition
-----
No Violations found

Check type : min_transition
-----
No Violations found

Check type : max_capacitance
-----
No Violations found

Check type : min_capacitance
-----
No Violations found

Check type : max_fanout
-----
No Violations found

Check type : min_fanout
-----
No Violations found
```

The coverage report is shown below.

```
#####
# Generated by:      Cadence Tempus 20.10-p003_1
# OS:               Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on:      Sat Oct 11 18:42:10 2025
# Design:           toll_gate
# Command:          report analysis_coverage > $report_dir/analysis_coverage_area.rpt
#####

-----
                        TIMING CHECK COVERAGE SUMMARY
-----
Check Type              No. of  Met          Violated      Untested
                        Checks
-----
Hold                    644     644 (100%)    0 (0%)        0 (0%)
PulseWidth              508     508 (100%)    0 (0%)        0 (0%)
Setup                   644     644 (100%)    0 (0%)        0 (0%)
-----|
```

The Graph Based Analysis (GBA) report is shown below.

```
#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sat Oct 11 18:42:10 2025
# Design: toll_gate
# Command: report_timing > $report_dir/timing_report_area.rpt
#####
Path 1: MET Setup Check with Pin u_account_balance/balance_db_reg[6][1]/CK
Endpoint: u_account_balance/balance_db_reg[6][1]/SE (v) checked with leading edge of 'clk'
Beginpoint: u_vehicle_interface/id_to_check_reg[3]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time 0.010
- Setup 0.418
+ Phase Shift 12.000
= Required Time 11.592
- Arrival Time 4.650
= Slack Time 6.942
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.010
= Beginpoint Arrival Time 0.010
```

| Instance                               | Arc         | Cell      | Delay | Arrival Time | Required Time |
|----------------------------------------|-------------|-----------|-------|--------------|---------------|
| u_vehicle_interface/id_to_check_reg[3] | CK ^        | -         | -     | 0.010        | 6.952         |
| u_vehicle_interface/id_to_check_reg[3] | CK ^ -> Q ^ | DFFQX1    | 0.531 | 0.542        | 7.484         |
| u_account_balance/g17076               | A ^ -> Y v  | CLKIN VX1 | 0.445 | 0.986        | 7.929         |
| u_account_balance/g17074               | A v -> Y ^  | NOR2XL    | 0.751 | 1.738        | 8.680         |
| u_account_balance/g17070               | A ^ -> Y v  | CLKIN VX1 | 0.136 | 1.874        | 8.816         |
| u_account_balance/g17044               | B v -> Y ^  | NOR2XL    | 0.542 | 2.416        | 9.359         |
| u_account_balance/g16923               | B1 ^ -> Y v | A0I221X1  | 0.166 | 2.582        | 9.525         |
| u_account_balance/g16910               | B0 v -> Y ^ | OAI21X1   | 0.095 | 2.677        | 9.619         |
| u_account_balance/g16904               | C0 ^ -> Y v | A0I211X1  | 0.052 | 2.729        | 9.672         |
| u_account_balance/g16896               | B v -> Y ^  | NAND2XL   | 0.115 | 2.845        | 9.787         |
| u_account_balance/g16891               | B ^ -> Y v  | NOR2XL    | 0.093 | 2.938        | 9.880         |
| u_account_balance/g16890               | A v -> Y ^  | INVX1     | 0.048 | 2.986        | 9.928         |
| u_account_balance/g16888               | C ^ -> Y ^  | OR4X1     | 0.139 | 3.126        | 10.068        |
| u_account_balance/g16887               | AN ^ -> Y ^ | NAND3BX1  | 0.117 | 3.243        | 10.185        |
| u_account_balance/g16886               | A2 ^ -> Y v | OAI31X1   | 0.114 | 3.357        | 10.299        |
| u_account_balance/g16885               | B v -> Y ^  | NOR2BX1   | 0.126 | 3.482        | 10.425        |
| u_account_balance/g3679                | AN ^ -> Y ^ | NOR2BX1   | 0.146 | 3.629        | 10.571        |
| u_account_balance/g3668                | B ^ -> Y ^  | AND2X1    | 0.199 | 3.828        | 10.770        |
| u_account_balance/g3643                | C ^ -> Y v  | NAND3X1   | 0.822 | 4.650        | 11.592        |
| u_account_balance/balance_db_reg[6][1] | SE v        | SdffQX1   | 0.000 | 4.650        | 11.592        |

The Path Based Analysis (PBA) report is shown below.

```
Beginpoint: u_vehicle_interface/id_to_check_reg[3]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Retime Analysis { Data Path-Slew }
Other End Arrival Time 0.010
- Setup 0.418
+ Phase Shift 12.000
= Required Time 11.592
- Arrival Time 4.554
= Slack Time 7.038
= Slack Time(original) 6.942
Clock Rise Edge 0.000
+ Source Insertion Delay 0.010
= Beginpoint Arrival Time 0.010
Timing Path:
```

| Instance                               | Arc         | Cell      | Retime Delay | Arrival Time | Required Time |
|----------------------------------------|-------------|-----------|--------------|--------------|---------------|
| -                                      | clk ^       | -         | -            | 0.010        | 7.048         |
| u_vehicle_interface/id_to_check_reg[3] | -           | DFFQX1    | 0.000        | 0.010        | 7.048         |
| u_vehicle_interface/id_to_check_reg[3] | CK ^ -> Q ^ | DFFQX1    | 0.531        | 0.542        | 7.580         |
| u_account_balance/g17076               | -           | CLKIN VX1 | 0.000        | 0.542        | 7.580         |
| u_account_balance/g17076               | A ^ -> Y v  | CLKIN VX1 | 0.445        | 0.986        | 8.025         |
| u_account_balance/g17074               | -           | NOR2XL    | 0.000        | 0.986        | 8.025         |
| u_account_balance/g17074               | A v -> Y ^  | NOR2XL    | 0.752        | 1.738        | 8.777         |
| u_account_balance/g17070               | -           | CLKIN VX1 | 0.000        | 1.738        | 8.777         |
| u_account_balance/g17070               | A ^ -> Y v  | CLKIN VX1 | 0.136        | 1.874        | 8.913         |
| u_account_balance/g17044               | -           | NOR2XL    | 0.000        | 1.874        | 8.913         |
| u_account_balance/g17044               | B v -> Y ^  | NOR2XL    | 0.542        | 2.417        | 9.455         |
| u_account_balance/g16923               | -           | A0I221X1  | 0.000        | 2.417        | 9.455         |
| u_account_balance/g16923               | B1 ^ -> Y v | A0I221X1  | 0.166        | 2.583        | 9.621         |
| u_account_balance/g16910               | -           | OAI21X1   | 0.000        | 2.583        | 9.621         |
| u_account_balance/g16910               | B0 v -> Y ^ | OAI21X1   | 0.090        | 2.673        | 9.711         |

**Static Timing Analysis for Intermediate Timing and Area Constraint Netlist**

The violation report is shown below in the two figures.

```
#####
# Generated by:      Cadence Tempus 20.10-p003_1
# OS:               Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on:      Sat Oct 11 19:29:49 2025
# Design:           toll_gate
# Command:          report_constraints -all violators > sta_after_synthesis_area/reports/allviolations_area.rpt
#####
# format : frame 0 : split 1

max_delay/setup
-----
No paths found

min_delay/hold
-----
No paths found

Check type : clock_period
-----
No paths found

Check type : skew
-----
No paths found

Check type : pulse_width
-----
No violating Checks with given description found

Check type : max_transition
-----
No Violations found

Check type : min_transition
-----
No Violations found

Check type : max_capacitance
-----
No Violations found

Check type : min_capacitance
-----
No Violations found

Check type : max_fanout
-----
No Violations found

Check type : min_fanout
-----
No Violations found
```

The coverage report is shown below.

```
#####
# Generated by:      Cadence Tempus 20.10-p003_1
# OS:               Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on:      Sat Oct 11 19:29:49 2025
# Design:           toll_gate
# Command:          report_analysis_coverage > $report_dir/analysis_coverage_area.rpt
#####

-----
                        TIMING CHECK COVERAGE SUMMARY
-----
Check Type              No. of Met          Violated          Untested
                        Checks
-----
Hold                    644      644 (100%)    0 (0%)            0 (0%)
PulseWidth              508      508 (100%)    0 (0%)            0 (0%)
Setup                   644      644 (100%)    0 (0%)            0 (0%)
-----
```

The Graph Based Analysis (GBA) report is shown below.



```
#####
# Generated by:      Cadence Tempus 20.10-p003_1
# OS:               Linux x86 64(Host ID edatools-server2.iiitd.edu.in)
# Generated on:      Sat Oct 11 19:29:49 2025
# Design:           toll_gate
# Command:          report_timing > $report_dir/timing_report_area.rpt
#####
Path 1: MET Setup Check with Pin u_account_balance/balance_db_reg[13][1]/CK
Endpoint:  u_account_balance/balance_db_reg[13][1]/SE (v) checked with leading edge of 'clk'
Beginpoint: u_vehicle_interface/id_to_check_reg[3]/Q (v) triggered by leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time      0.010
- Setup                    0.418
+ Phase Shift              8.000
= Required Time            7.592
- Arrival Time            4.456
= Slack Time              3.136
  Clock Rise Edge          0.000
  + Clock Network Latency (Ideal) 0.010
  = Beginpoint Arrival Time 0.010
-----
-----
```

| Instance                                | Arc         | Cell     | Delay | Arrival Time | Required Time |
|-----------------------------------------|-------------|----------|-------|--------------|---------------|
| u_vehicle_interface/id_to_check_reg[3]  | CK ^        | -        | -     | 0.010        | 3.146         |
| u_vehicle_interface/id_to_check_reg[3]  | CK ^ -> Q v | DFFQX1   | 0.511 | 0.521        | 3.657         |
| u_account_balance/g12166                | A v -> Y ^  | CLKINVX1 | 0.515 | 1.036        | 4.172         |
| u_account_balance/g12164                | B ^ -> Y ^  | AND2X1   | 0.560 | 1.596        | 4.732         |
| u_account_balance/g12159                | A ^ -> Y v  | CLKINVX1 | 0.115 | 1.711        | 4.847         |
| u_account_balance/g12147                | B v -> Y ^  | NOR2XL   | 0.444 | 2.155        | 5.290         |
| u_account_balance/g12034                | C0 ^ -> Y v | A0I222XL | 0.134 | 2.289        | 5.424         |
| u_account_balance/g11985                | C v -> Y ^  | NAND4XL  | 0.156 | 2.444        | 5.580         |
| u_account_balance/g11968                | B ^ -> Y v  | NOR2XL   | 0.090 | 2.534        | 5.670         |
| u_account_balance/g11965                | AN v -> Y v | NOR3BX1  | 0.132 | 2.667        | 5.803         |
| u_account_balance/g11962                | D v -> Y ^  | NAND4BX1 | 0.083 | 2.750        | 5.886         |
| u_account_balance/g11960                | A1 ^ -> Y v | OAI21X1  | 0.108 | 2.858        | 5.993         |
| u_account_balance/g11959                | B v -> Y ^  | NOR2BX1  | 0.160 | 3.018        | 6.154         |
| u_account_balance/g3648                 | B ^ -> Y v  | NAND2XL  | 0.269 | 3.287        | 6.423         |
| u_account_balance/g3631                 | B v -> Y ^  | NOR2XL   | 0.315 | 3.602        | 6.738         |
| u_account_balance/g3623                 | A ^ -> Y v  | NAND3X1  | 0.854 | 4.456        | 7.592         |
| u_account_balance/balance_db_reg[13][1] | SE v        | SDFQX1   | 0.000 | 4.456        | 7.592         |

The Path Based Analysis (PBA) report is shown below.

```
Beginpoint: u_vehicle_interface/id_to_check_reg[3]/Q (v) triggered by leading edge of 'clk'
Path Groups: {clk}
Retime Analysis { Data Path-Slew }
Other End Arrival Time      0.010
- Setup                    0.417
+ Phase Shift              8.000
= Required Time            7.593
- Arrival Time            4.435
= Slack Time              3.157
= Slack Time(original)    3.136
  Clock Rise Edge          0.000
  + Source Insertion Delay 0.010
  = Beginpoint Arrival Time 0.010
Timing Path:
-----
-----
```

| Instance                               | Arc         | Cell     | Retime Delay | Arrival Time | Required Time |
|----------------------------------------|-------------|----------|--------------|--------------|---------------|
| -                                      | clk ^       | -        | -            | 0.010        | 3.167         |
| u_vehicle_interface/id_to_check_reg[3] | -           | DFFQX1   | 0.000        | 0.010        | 3.167         |
| u_vehicle_interface/id_to_check_reg[3] | CK ^ -> Q v | DFFQX1   | 0.511        | 0.521        | 3.678         |
| u_account_balance/g12166               | -           | CLKINVX1 | 0.000        | 0.521        | 3.678         |
| u_account_balance/g12166               | A v -> Y ^  | CLKINVX1 | 0.515        | 1.036        | 4.193         |
| u_account_balance/g12164               | -           | AND2X1   | 0.000        | 1.036        | 4.193         |
| u_account_balance/g12164               | B ^ -> Y ^  | AND2X1   | 0.560        | 1.596        | 4.753         |
| u_account_balance/g12159               | -           | CLKINVX1 | 0.000        | 1.596        | 4.753         |
| u_account_balance/g12159               | A ^ -> Y v  | CLKINVX1 | 0.115        | 1.711        | 4.868         |
| u_account_balance/g12147               | -           | NOR2XL   | 0.000        | 1.711        | 4.868         |
| u_account_balance/g12147               | B v -> Y ^  | NOR2XL   | 0.444        | 2.155        | 5.312         |
| u_account_balance/g12034               | -           | A0I222XL | 0.000        | 2.155        | 5.312         |
| u_account_balance/g12034               | C0 ^ -> Y v | A0I222XL | 0.134        | 2.289        | 5.446         |
| u_account_balance/g11985               | -           | NAND4XL  | 0.000        | 2.289        | 5.446         |
| u_account_balance/g11985               | C v -> Y ^  | NAND4XL  | 0.156        | 2.445        | 5.602         |

## Analysis of the Results

The coverage reports for the three netlists—minimum area, strict timing, and intermediate constraints—show that every point is checked and there are no set-up or hold violations. This demonstrates that, within the given limits, the Intelligent Tool Booth Management System's



design functions flawlessly. The table below provides a summary of the differences in slack time between the graph-based and path-based analyses for each of the three netlists.

| Constraints    | SlackforGBA(inns) | SlackforPBA(inns) |
|----------------|-------------------|-------------------|
| Minimum Area   | 0.542             | 0.638             |
| Minimum Timing | 6.942             | 7.038             |
| Intermediate   | 3.136             | 3.157             |

## **Chapter 6**

### **DESIGN FOR TESTABILITY (Scan Chain Insertion)**

Scan chain insertion is an essential process in the design and testing of Very Large-Scale Integration (VLSI) circuits. Its primary purpose is to improve the testability of digital circuits by making internal states more observable and manageable. During the design stage, the complexity of a circuit often makes it impractical to conduct exhaustive testing with standard functional methods, as there can be millions or even billions of transistors and logic gates involved. Scan chains help overcome this issue by transforming sequential elements, like flip-flops, into a series of shift registers that can be monitored and controlled more efficiently.

The fundamental concept of scan chain insertion is to introduce a specific operational mode, termed "scan mode," in addition to the regular functional operation of the circuit. When in scan mode, the flip-flops (which are typically integrated within the design) are reconfigured into a sequential chain, allowing for test patterns to be inputted into the circuit and the output responses to be extracted for validation. This involves connecting the output of one flip-flop to the input of the subsequent one, creating a lengthy shift register or chain. These chains allow external testing equipment to access the internal states of the design, which would otherwise be challenging to observe during standard functional operation.

The scan insertion procedure necessitates adjustments at both design and physical implementation stages. During the design process, scanable flip-flops replace regular flip-flops, and additional control logic is integrated to manage the transition between normal and scan modes. Though this often leads to a slight increase in area and power consumption due to the added hardware, the gains in testability and fault coverage far exceed these downsides. In the physical implementation phase, it is crucial to ensure that the scan chain does not create excessive routing complexity or cause timing violations.

Moreover, scan chain insertion supports methods like Automatic Test Pattern Generation (ATPG), which autonomously produces test patterns to optimize fault coverage. The scan chains enable ATPG to directly control internal nodes, greatly enhancing the detection of manufacturing defects, such as stuck-at faults, bridging faults, or transition faults. This approach is particularly effective in uncovering defects that may not be identified through functional testing alone.

To perform scan chain insertion, we provide the synthesized netlist, the associated constraint file, the library file, and the tcl script to the tool. The tool then generates the netlist with the incorporated scan chains.

Below is the tcl script used for simulation based on one of the constraints.

## **Scan Chains Insertion for Minimum Area**

We provide the constraint file for minimum area synthesis and the netlist synthesized in the logic synthesis step.

The area report for minimum area is shown below.

| Generated by: Genus(TM) Synthesis Solution 19.13-s073_1 |                   |            |           |          |            |            |
|---------------------------------------------------------|-------------------|------------|-----------|----------|------------|------------|
| Generated on: Oct 13 2025 02:00:04 am                   |                   |            |           |          |            |            |
| Module: toll_gate                                       |                   |            |           |          |            |            |
| Technology library: slow                                |                   |            |           |          |            |            |
| Operating conditions: slow (balanced_tree)              |                   |            |           |          |            |            |
| Wireload mode: enclosed                                 |                   |            |           |          |            |            |
| Area mode: timing library                               |                   |            |           |          |            |            |
| Instance                                                | Module            | Cell Count | Cell Area | Net Area | Total Area | Wireload   |
| toll_gate                                               |                   | 648        | 7136.053  | 0.000    | 7136.053   | <none> (D) |
| u_account_balance                                       | account_balance   | 571        | 6581.245  | 0.000    | 6581.245   | <none> (D) |
| u_barrier_ctrl                                          | barrier_ctrl      | 25         | 201.335   | 0.000    | 201.335    | <none> (D) |
| u_vehicle_interface                                     | vehicle_interface | 11         | 140.026   | 0.000    | 140.026    | <none> (D) |
| (D) = wireload is default in technology library         |                   |            |           |          |            |            |

The power report for minimum area is shown below.

| Instance: /toll_gate        |              |              |              |              |         |
|-----------------------------|--------------|--------------|--------------|--------------|---------|
| Power Unit: W               |              |              |              |              |         |
| PDB Frames: /stim#0/frame#0 |              |              |              |              |         |
| Category                    | Leakage      | Internal     | Switching    | Total        | Row%    |
| memory                      | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.00%   |
| register                    | 3.42620e-05  | 4.21123e-04  | 6.11318e-06  | 4.61498e-04  | 85.46%  |
| latch                       | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.00%   |
| logic                       | 7.61116e-06  | 2.41513e-05  | 1.79888e-05  | 4.97513e-05  | 9.21%   |
| bbox                        | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.00%   |
| clock                       | 0.000000e+00 | 0.000000e+00 | 2.87955e-05  | 2.87955e-05  | 5.33%   |
| pad                         | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.00%   |
| pm                          | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.00%   |
| Subtotal                    | 4.18732e-05  | 4.45274e-04  | 5.28975e-05  | 5.40045e-04  | 100.00% |
| Percentage                  | 7.75%        | 82.45%       | 9.80%        | 100.00%      | 100.00% |

The timing report for minimum area is shown below.

| Generated by: Genus(TM) Synthesis Solution 19.13-s073_1 |          |        |           |           |            |              |
|---------------------------------------------------------|----------|--------|-----------|-----------|------------|--------------|
| Generated on: Oct 13 2025 02:00:03 am                   |          |        |           |           |            |              |
| Module: toll_gate                                       |          |        |           |           |            |              |
| Technology library: slow                                |          |        |           |           |            |              |
| Operating conditions: slow (balanced_tree)              |          |        |           |           |            |              |
| Wireload mode: enclosed                                 |          |        |           |           |            |              |
| Area mode: timing library                               |          |        |           |           |            |              |
| Pin                                                     | Type     | Fanout | Load (fF) | Slew (ps) | Delay (ps) | Arrival (ps) |
| (clock clk)                                             | launch   |        |           |           |            | 0 R          |
|                                                         | latency  |        |           |           | +10        | 10 R         |
| u_vehicle_interface                                     |          |        |           |           |            |              |
| id_to_check_reg[0]/CK                                   |          |        |           | 600       |            | 10 R         |
| id_to_check_reg[0]/Q                                    | SdffQX1  | 17     | 36.7      | 268       | +564       | 574 F        |
| u_vehicle_interface/id_to_check[0]                      |          |        |           |           |            |              |
| u_account_balance/id_to_check[0]                        |          |        |           |           |            |              |
| g12137/B                                                |          |        |           |           | +0         | 574          |
| g12137/Y                                                | NOR2XL   | 10     | 23.1      | 689       | +555       | 1129 R       |
| g12131/A                                                |          |        |           |           | +0         | 1129         |
| g12131/Y                                                | CLKINVX1 | 8      | 15.8      | 273       | +212       | 1341 F       |
| g12100/B                                                |          |        |           |           | +0         | 1341         |
| g12100/Y                                                | NOR2XL   | 10     | 20.5      | 614       | +505       | 1846 R       |
| g11978/A1                                               |          |        |           |           | +0         | 1846         |
| g11978/Y                                                | A0I221X1 | 1      | 2.8       | 279       | +186       | 2032 F       |
| g11964/C0                                               |          |        |           |           | +0         | 2032         |
| g11964/Y                                                | OAI221X1 | 6      | 12.9      | 342       | +190       | 2222 R       |
| g11941/A                                                |          |        |           |           | +0         | 2222         |
| g11941/Y                                                | NOR2XL   | 3      | 7.1       | 176       | +153       | 2374 F       |
| g11936/B                                                |          |        |           |           | +0         | 2374         |
| g11934/Y                                                | NOR2XL   | 3      | 6.2       | 111       | +109       | 2624 F       |
| g11932/B                                                |          |        |           |           | +0         | 2624         |
| g11932/Y                                                | NAND2XL  | 3      | 6.1       | 113       | +117       | 2742 R       |
| g11930/B                                                |          |        |           |           | +0         | 2742         |
| g11930/Y                                                | NOR2XL   | 3      | 6.2       | 103       | +96        | 2838 F       |
| g11928/B                                                |          |        |           |           | +0         | 2838         |
| g11928/Y                                                | NAND2XL  | 5      | 10.7      | 179       | +161       | 2999 R       |
| g12145/D                                                |          |        |           |           | +0         | 2999         |
| g12145/Y                                                | NOR4BBX1 | 2      | 5.6       | 89        | +84        | 3083 F       |
| g11921/C                                                |          |        |           |           | +0         | 3083         |
| g11921/Y                                                | NAND3BX1 | 3      | 7.3       | 155       | +91        | 3174 R       |
| g11917/C                                                |          |        |           |           | +0         | 3174         |
| g11917/Y                                                | NAND3X1  | 3      | 7.2       | 197       | +174       | 3348 F       |
| g11915/A                                                |          |        |           |           | +0         | 3348         |
| g11915/Y                                                | INVX1    | 2      | 4.5       | 85        | +90        | 3437 R       |
| g11912/B                                                |          |        |           |           | +0         | 3437         |
| g11912/Y                                                | NAND2XL  | 2      | 6.9       | 211       | +155       | 3592 F       |
| g11909/B                                                |          |        |           |           | +0         | 3592         |
| g11909/Y                                                | NOR2XL   | 3      | 8.3       | 274       | +252       | 3844 R       |
| g11906/A                                                |          |        |           |           | +0         | 3844         |
| g11906/Y                                                | INVX1    | 2      | 5.6       | 107       | +85        | 3930 F       |
| g11898/B                                                |          |        |           |           | +0         | 3930         |
| g11898/Y                                                | NOR2X1   | 15     | 57.0      | 925       | +669       | 4598 R       |
| balance_db_reg[14][15]/SE <<<                           | SdffQX1  |        |           |           | +0         | 4598         |
| balance_db_reg[14][15]/CK                               | setup    |        |           | 600       | +302       | 4900 R       |
| (clock clk)                                             | capture  |        |           |           |            | 12000 R      |
|                                                         | latency  |        |           |           | +10        | 12010 R      |
| Cost Group : 'clk' (path_group 'clk')                   |          |        |           |           |            |              |
| Timing slack : 7110ps                                   |          |        |           |           |            |              |
| Start-point : u_vehicle_interface/id_to_check_reg[0]/CK |          |        |           |           |            |              |
| End-point : u_account_balance/balance_db_reg[14][15]/SE |          |        |           |           |            |              |

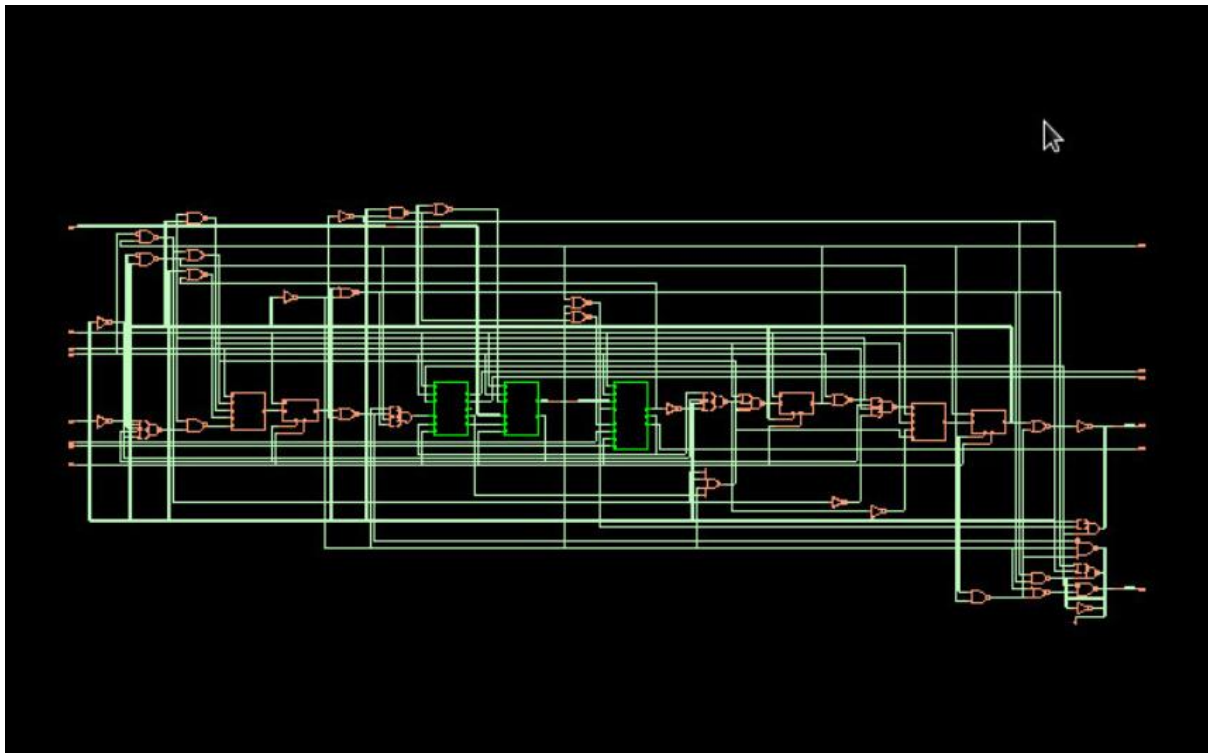
A portion of the netlist showing the scan chains inserted is shown below.

```

AOI211X1 g16901(.A0 (\balance_db[6] [11]), .A1 (n_155), .B0 (n_234),
.C0 (n_278), .Y (n_290));
NAND3X1 g16902(.A (n_201), .B (n_202), .C (n_284), .Y (n_303));
OAI211X1 g16903(.A0 (n_139), .A1 (n_246), .B0 (n_211), .C0 (n_280),
.Y (n_301));
AOI211X1 g16904(.A0 (\balance_db[0] [5]), .A1 (n_160), .B0 (n_265),
.C0 (n_286), .Y (n_289));
NAND4XL g16905(.A (n_217), .B (n_216), .C (n_218), .D (n_281), .Y
(n_307));
OAI211X1 g16906(.A0 (n_139), .A1 (n_244), .B0 (n_203), .C0 (n_277),
.Y (n_302));
NAND4XL g16907(.A (n_209), .B (n_210), .C (n_208), .D (n_282), .Y
(n_308));
NAND3X1 g16908(.A (n_221), .B (n_222), .C (n_279), .Y (n_288));
AOI221X1 g16909(.A0 (\balance_db[13] [13]), .A1 (n_164), .B0
(\balance_db[5] [13]), .B1 (n_165), .C0 (n_273), .Y (n_287));
OAI21X1 g16910(.A0 (n_136), .A1 (n_239), .B0 (n_276), .Y (n_286));
AOI211X1 g16911(.A0 (\balance_db[0] [6]), .A1 (n_160), .B0 (n_274),
.C0 (n_267), .Y (n_285));
AOI211X1 g16912(.A0 (\balance_db[2] [10]), .A1 (n_156), .B0 (n_231),
.C0 (n_270), .Y (n_284));
AOI211X1 g16913(.A0 (\balance_db[1] [12]), .A1 (n_161), .B0 (n_255),
.C0 (n_275), .Y (n_283));
AOI28B1X1 g16914(.A0N (n_311), .A1N (n_296), .B0 (n_320), .Y (n_314));
AOI221X1 g16915(.A0 (\balance_db[0] [15]), .A1 (n_160), .B0
(\balance_db[1] [15]), .B1 (n_161), .C0 (n_261), .Y (n_282));
AOI221X1 g16916(.A0 (\balance_db[3] [14]), .A1 (n_153), .B0
(\balance_db[11] [14]), .B1 (n_157), .C0 (n_260), .Y (n_281));
AOI211X1 g16917(.A0 (n_137), .A1 (n_245), .B0 (n_256), .C0 (n_272),
.Y (n_320));
NAND3X1 g16918(.A (n_227), .B (n_264), .C (n_259), .Y (n_297));

```

The schematic for the synthesis is shown below.



**Scan Chain Insertion for Minimum Timing Constraint**

The scan chains insertion synthesis area report for minimum timing constraint is shown below.

```
=====
Generated by:      Genus(TM) Synthesis Solution 19.13-s073_1
Generated on:      Oct 13 2025 02:06:35 am
Module:           toll_gate
Technology library: slow
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

| Instance            | Module            | Cell Count | Cell Area | Net Area | Total Area | Wireload   |
|---------------------|-------------------|------------|-----------|----------|------------|------------|
| toll_gate           |                   | 712        | 7369.178  | 0.000    | 7369.178   | <none> (D) |
| u_account_balance   | account_balance   | 633        | 6805.288  | 0.000    | 6805.288   | <none> (D) |
| u_barrier_ctrl      | barrier_ctrl      | 27         | 199.822   | 0.000    | 199.822    | <none> (D) |
| u_vehicle_interface | vehicle_interface | 11         | 145.325   | 0.000    | 145.325    | <none> (D) |

(D) = wireload is default in technology library

The power report for minimum time is shown below.

|                             |             |             |             |             |         |
|-----------------------------|-------------|-------------|-------------|-------------|---------|
| Instance: /toll_gate        |             |             |             |             |         |
| Power Unit: W               |             |             |             |             |         |
| PDB Frames: /stim#0/frame#0 |             |             |             |             |         |
| -----                       |             |             |             |             |         |
| Category                    | Leakage     | Internal    | Switching   | Total       | Row%    |
| -----                       |             |             |             |             |         |
| memory                      | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| register                    | 3.37916e-05 | 1.52068e-03 | 2.25574e-05 | 1.57703e-03 | 84.58%  |
| latch                       | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| logic                       | 9.16576e-06 | 1.02126e-04 | 7.14476e-05 | 1.82739e-04 | 9.80%   |
| bbox                        | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| clock                       | 0.00000e+00 | 0.00000e+00 | 1.04735e-04 | 1.04735e-04 | 5.62%   |
| pad                         | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| pm                          | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| -----                       |             |             |             |             |         |
| Subtotal                    | 4.29574e-05 | 1.62281e-03 | 1.98740e-04 | 1.86450e-03 | 100.00% |
| Percentage                  | 2.30%       | 87.04%      | 10.66%      | 100.00%     | 100.00% |
| -----                       |             |             |             |             |         |

The timing report for minimum area is shown below.

Generated by:

Genus(TM) Synthesis Solution 19.13-s073\_1

Generated on:

Oct 13 2025 02:06:34 am

Module:

toll\_gate

Technology library:

slow

Operating conditions:

slow (balanced\_tree)

Wireload mode:

enclosed

Area mode:

timing library

=====

| Pin                                | Type      | Fanout | Load (fF) | Slew (ps) | Delay (ps) | Arrival (ps) |
|------------------------------------|-----------|--------|-----------|-----------|------------|--------------|
| -----                              |           |        |           |           |            |              |
| (clock clk)                        | launch    |        |           |           |            | 0 R          |
|                                    | latency   |        |           |           | +10        | 10 R         |
| u_vehicle_interface                |           |        |           |           |            |              |
| id_to_check_reg[3]/CK              |           |        |           | 600       |            | 10 R         |
| id_to_check_reg[3]/Q               | SDFFQX1   | 3      | 8.7       | 94        | +390       | 400 R        |
| u_vehicle_interface/id_to_check[3] |           |        |           |           |            |              |
| g222/A                             |           |        |           |           | +0         | 400          |
| g222/Y                             | BUFX6     | 23     | 75.5      | 121       | +176       | 575 R        |
| u_account_balance/id_to_check[3]   |           |        |           |           |            |              |
| g13463/A                           |           |        |           |           | +0         | 575          |
| g13463/Y                           | INVX8     | 17     | 41.0      | 64        | +64        | 639 F        |
| g13456/B                           |           |        |           |           | +0         | 639          |
| g13456/Y                           | CLKAND2X3 | 10     | 21.4      | 75        | +148       | 786 F        |
| g13449/A                           |           |        |           |           | +0         | 786          |
| g13449/Y                           | CLKINVX2  | 4      | 9.8       | 55        | +63        | 850 R        |
| g13398/B                           |           |        |           |           | +0         | 850          |
| g13398/Y                           | NOR2BXL   | 1      | 2.8       | 58        | +53        | 903 F        |
| g13286/C0                          |           |        |           |           | +0         | 903          |
| g13286/Y                           | A0I221X1  | 1      | 1.7       | 133       | +111       | 1014 R       |
| g13252/B                           |           |        |           |           | +0         | 1014         |

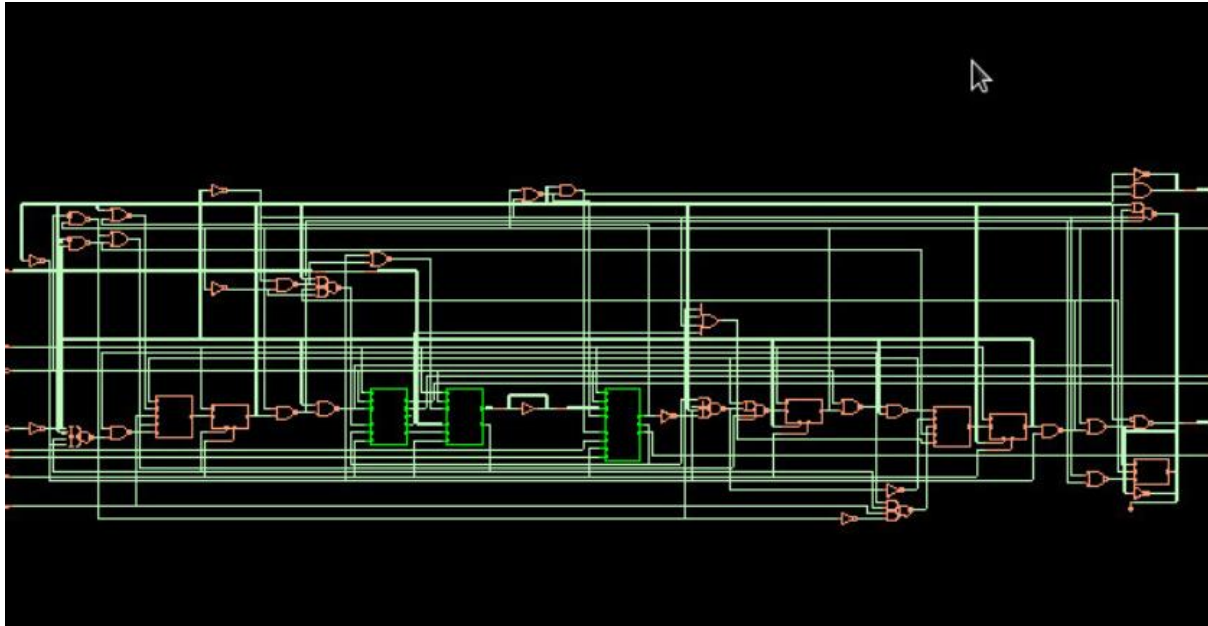
|                                                         |         |    |      |     |      |        |
|---------------------------------------------------------|---------|----|------|-----|------|--------|
| g13194/B0                                               |         |    |      |     | +0   | 1348   |
| g13194/Y                                                | OAI21X1 | 3  | 7.2  | 146 | +128 | 1476 F |
| g13192/B                                                |         |    |      |     | +0   | 1476   |
| g13192/Y                                                | NOR2X1  | 3  | 9.1  | 174 | +166 | 1642 R |
| g13187/B                                                |         |    |      |     | +0   | 1642   |
| g13187/Y                                                | NAND2X1 | 4  | 7.7  | 142 | +135 | 1778 F |
| g13184/D                                                |         |    |      |     | +0   | 1778   |
| g13184/Y                                                | OR4X1   | 4  | 7.9  | 127 | +295 | 2073 F |
| g13179/B                                                |         |    |      |     | +0   | 2073   |
| g13179/Y                                                | OR2X1   | 3  | 8.5  | 93  | +195 | 2267 F |
| g13170/B                                                |         |    |      |     | +0   | 2267   |
| g13170/Y                                                | NAND2X1 | 3  | 5.9  | 81  | +80  | 2347 R |
| g4361/B                                                 |         |    |      |     | +0   | 2347   |
| g4361/Y                                                 | OR2X1   | 4  | 7.8  | 82  | +130 | 2477 R |
| g4357/B                                                 |         |    |      |     | +0   | 2477   |
| g4357/Y                                                 | NAND2XL | 2  | 4.4  | 134 | +115 | 2592 F |
| g4343/A                                                 |         |    |      |     | +0   | 2592   |
| g4343/Y                                                 | NAND2XL | 13 | 20.8 | 331 | +276 | 2868 R |
| g4332/A                                                 |         |    |      |     | +0   | 2868   |
| g4332/Y                                                 | NAND2XL | 1  | 1.7  | 113 | +98  | 2966 F |
| g4291/B0                                                |         |    |      |     | +0   | 2966   |
| g4291/Y                                                 | OAI21XL | 1  | 1.8  | 171 | +72  | 3038 R |
| balance_db_reg[2][12]/D <<<                             | SDFFQX1 |    |      |     | +0   | 3038   |
| balance_db_reg[2][12]/CK                                | setup   |    |      | 600 | +272 | 3310 R |
| -----                                                   |         |    |      |     |      |        |
| (clock clk)                                             | capture |    |      |     |      | 3300 R |
|                                                         | latency |    |      |     | +10  | 3310 R |
| -----                                                   |         |    |      |     |      |        |
| Cost Group : 'clk' (path_group 'clk')                   |         |    |      |     |      |        |
| Timing slack : 0ps                                      |         |    |      |     |      |        |
| Start-point : u_vehicle_interface/id_to_check_reg[3]/CK |         |    |      |     |      |        |
| End-point : u_account_balance/balance_db_reg[2][12]/D   |         |    |      |     |      |        |

A portion of the netlist showing the scan chains inserted is shown below.

```
OAI222XL g17166(.A0 (n_166), .A1 (n_271), .B0 (n_165), .B1 (n_301),
.C0 (n_263), .C1 (n_168), .Y (n_356));
OA22X1 g17167(.A0 (n_166), .A1 (n_312), .B0 (n_165), .B1 (n_314), .Y
(n_355));
OAI21X1 g17168(.A0 (n_165), .A1 (n_300), .B0 (n_255), .Y (n_354));
OA22X1 g17169(.A0 (n_166), .A1 (n_304), .B0 (n_165), .B1 (n_305), .Y
(n_353));
NAND2XL g17170(.A (n_337), .B (n_339), .Y (n_371));
NAND3X1 g17171(.A (n_336), .B (n_331), .C (n_341), .Y (n_386));
AOI21X1 g17172(.A0 (\balance_db[5] [9]), .A1 (n_210), .B0 (n_319), .Y
(n_352));
NOR2XL g17173(.A (n_166), .B (n_299), .Y (n_351));
OAI211X1 g17174(.A0 (n_171), .A1 (n_240), .B0 (n_287), .C0 (n_328),
.Y (n_350));
OAI2BB1X1 g17175(.A0N (\balance_db[5] [8]), .A1N (n_210), .B0
(n_321), .Y (n_349));
NOR2XL g17176(.A (n_166), .B (n_302), .Y (n_348));
AOI21X1 g17177(.A0 (\balance_db[5] [5]), .A1 (n_210), .B0 (n_320), .Y
(n_347));
AOI21X1 g17178(.A0 (n_167), .A1 (n_294), .B0 (n_318), .Y (n_346));
AOI221X1 g17179(.A0 (\balance_db[4] [13]), .A1 (n_209), .B0
(\balance_db[6] [13]), .B1 (n_215), .C0 (n_335), .Y (n_345));
AOI22X1 g17180(.A0 (n_167), .A1 (n_264), .B0 (n_170), .B1 (n_267), .Y
(n_344));
OAI211X1 g17181(.A0 (n_165), .A1 (n_243), .B0 (n_257), .C0 (n_281),
.Y (n_343));
AOI221X1 g17182(.A0 (\balance_db[12] [11]), .A1 (n_216), .B0
(\balance_db[11] [11]), .B1 (n_218), .C0 (n_327), .Y (n_342));
AOI22X1 g17183(.A0 (n_167), .A1 (n_259), .B0 (n_170), .B1 (n_266), .Y
(n_341));
AOI221X1 g17184(.A0 (\balance_db[10] [6]), .A1 (n_213), .B0
(\balance_db[11] [6]), .B1 (n_218), .C0 (n_313), .Y (n_340));
AOI22X1 g17185(.A0 (n_167), .A1 (n_260), .B0 (n_170), .B1 (n_262), .Y
```

The schematic for the synthesis is shown below.





## Scan Chain Insertion for Intermediate Time and Area Constraint

The area report for the synthesized netlist is shown below.

| <pre> ===== Generated by:      Genus(TM) Synthesis Solution 19.13-s073_1 Generated on:      Oct 11 2025  08:27:03 pm Module:           toll_gate Technology library: slow Operating conditions: slow (balanced_tree) Wireload mode:    enclosed Area mode:        timing library ===== </pre> |                   |            |           |          |            |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------|-----------|----------|------------|------------|
| Instance                                                                                                                                                                                                                                                                                      | Module            | Cell Count | Cell Area | Net Area | Total Area | Wireload   |
| toll_gate                                                                                                                                                                                                                                                                                     |                   | 666        | 7142.865  | 0.000    | 7142.865   | <none> (D) |
| u_account_balance                                                                                                                                                                                                                                                                             | account_balance   | 588        | 6588.815  | 0.000    | 6588.815   | <none> (D) |
| u_barrier_ctrl                                                                                                                                                                                                                                                                                | barrier_ctrl      | 26         | 198.308   | 0.000    | 198.308    | <none> (D) |
| u_vehicle_interface                                                                                                                                                                                                                                                                           | vehicle_interface | 11         | 140.026   | 0.000    | 140.026    | <none> (D) |
| (D) = wireload is default in technology library                                                                                                                                                                                                                                               |                   |            |           |          |            |            |

The power report for minimum time is shown below.

| <pre> Instance: /toll_gate Power Unit: W PDB Frames: /stim#0/frame#0 </pre> |             |             |             |             |         |
|-----------------------------------------------------------------------------|-------------|-------------|-------------|-------------|---------|
| Category                                                                    | Leakage     | Internal    | Switching   | Total       | Row%    |
| memory                                                                      | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| register                                                                    | 3.42620e-05 | 6.64850e-04 | 9.01628e-06 | 7.08128e-04 | 86.09%  |
| latch                                                                       | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| logic                                                                       | 7.20475e-06 | 3.39656e-05 | 2.71739e-05 | 6.83442e-05 | 8.31%   |
| bbox                                                                        | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| clock                                                                       | 0.00000e+00 | 0.00000e+00 | 4.60728e-05 | 4.60728e-05 | 5.60%   |
| pad                                                                         | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| pm                                                                          | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00000e+00 | 0.00%   |
| Subtotal                                                                    | 4.14667e-05 | 6.98816e-04 | 8.22630e-05 | 8.22545e-04 | 100.00% |
| Percentage                                                                  | 5.04%       | 84.96%      | 10.00%      | 100.00%     | 100.00% |



The timing report for minimum area is shown below.

|                                    |                                              |        |           |           |            |              |
|------------------------------------|----------------------------------------------|--------|-----------|-----------|------------|--------------|
| =====                              |                                              |        |           |           |            |              |
| Generated by:                      | Genus(TM) Synthesis Solution 19.13-s073_1    |        |           |           |            |              |
| Generated on:                      | Oct 11 2025 08:27:02 pm                      |        |           |           |            |              |
| Module:                            | toll_gate                                    |        |           |           |            |              |
| Technology library:                | slow                                         |        |           |           |            |              |
| Operating conditions:              | slow (balanced_tree)                         |        |           |           |            |              |
| Wireload mode:                     | enclosed                                     |        |           |           |            |              |
| Area mode:                         | timing library                               |        |           |           |            |              |
| =====                              |                                              |        |           |           |            |              |
| Pin                                | Type                                         | Fanout | Load (fF) | Slew (ps) | Delay (ps) | Arrival (ps) |
| -----                              |                                              |        |           |           |            |              |
| (clock clk)                        | launch                                       |        |           |           |            | 0 R          |
|                                    | latency                                      |        |           |           | +10        | 10 R         |
| u_vehicle_interface                |                                              |        |           |           |            |              |
| id_to_check_reg[3]/CK              |                                              |        |           | 600       |            | 10 R         |
| id_to_check_reg[3]/Q               | SDFQX1                                       | 9      | 19.0      | 149       | +483       | 493 F        |
| u_vehicle_interface/id_to_check[3] |                                              |        |           |           |            |              |
| u_account_balance/id_to_check[3]   |                                              |        |           |           |            |              |
| g11418/A                           |                                              |        |           |           | +0         | 493          |
| g11418/Y                           | CLKINVX1                                     | 7      | 15.7      | 148       | +150       | 644 R        |
| g11413/B                           |                                              |        |           |           | +0         | 644          |
| g11413/Y                           | NAND2XL                                      | 8      | 16.8      | 421       | +324       | 967 F        |
| g11377/B                           |                                              |        |           |           | +0         | 967          |
| g11377/Y                           | NOR2XL                                       | 11     | 24.9      | 747       | +635       | 1602 R       |
| g11313/B0                          |                                              |        |           |           | +0         | 1602         |
| g11313/Y                           | A0I22XL                                      | 1      | 1.6       | 207       | +121       | 1723 F       |
| g11257/A                           |                                              |        |           |           | +0         | 1723         |
| g11257/Y                           | NAND2XL                                      | 1      | 2.8       | 100       | +108       | 1831 R       |
| g11219/C0                          |                                              |        |           |           | +0         | 1831         |
| g11219/Y                           | A0I211X1                                     | 1      | 2.9       | 233       | +54        | 1886 F       |
| g11213/C                           |                                              |        |           |           | +0         | 1886         |
|                                    |                                              |        |           |           |            |              |
| g11207/Y                           | NOR2XL                                       | 4      | 10.2      | 320       | +262       | 2434 R       |
| g11205/D                           |                                              |        |           |           | +0         | 2434         |
| g11205/Y                           | NOR4BX1                                      | 3      | 6.2       | 132       | +100       | 2535 F       |
| g11201/B                           |                                              |        |           |           | +0         | 2535         |
| g11201/Y                           | NAND2XL                                      | 3      | 6.1       | 116       | +124       | 2659 R       |
| g11197/B                           |                                              |        |           |           | +0         | 2659         |
| g11197/Y                           | NOR2XL                                       | 4      | 9.1       | 137       | +121       | 2780 F       |
| g11195/C                           |                                              |        |           |           | +0         | 2780         |
| g11195/Y                           | NAND3X1                                      | 4      | 8.9       | 108       | +116       | 2896 R       |
| g11189/B                           |                                              |        |           |           | +0         | 2896         |
| g11189/Y                           | OR2X1                                        | 2      | 4.5       | 57        | +117       | 3013 R       |
| g11186/B                           |                                              |        |           |           | +0         | 3013         |
| g11186/Y                           | NAND2XL                                      | 4      | 7.9       | 210       | +165       | 3177 F       |
| g11185/AN                          |                                              |        |           |           | +0         | 3177         |
| g11185/Y                           | NAND2BX1                                     | 2      | 4.4       | 127       | +180       | 3357 F       |
| g11178/B                           |                                              |        |           |           | +0         | 3357         |
| g11178/Y                           | NOR2XL                                       | 5      | 8.5       | 273       | +232       | 3589 R       |
| g11171/B                           |                                              |        |           |           | +0         | 3589         |
| g11171/Y                           | NAND2XL                                      | 4      | 7.6       | 233       | +212       | 3800 F       |
| g11164/B                           |                                              |        |           |           | +0         | 3800         |
| g11164/Y                           | NOR2X1                                       | 15     | 57.0      | 918       | +704       | 4504 R       |
| balance_db_reg[9][15]/SE <<<       | SDFQX1                                       |        |           |           | +0         | 4504         |
| balance_db_reg[9][15]/CK           | setup                                        |        |           | 600       | +301       | 4805 R       |
| -----                              |                                              |        |           |           |            |              |
| (clock clk)                        | capture                                      |        |           |           |            | 7500 R       |
|                                    | latency                                      |        |           |           | +10        | 7510 R       |
| -----                              |                                              |        |           |           |            |              |
| Cost Group                         | : 'clk' (path_group 'clk')                   |        |           |           |            |              |
| Timing slack                       | : 2705ps                                     |        |           |           |            |              |
| Start-point                        | : u_vehicle_interface/id_to_check_reg[3]/CK  |        |           |           |            |              |
| End-point                          | : u_account_balance/balance_db_reg[9][15]/SE |        |           |           |            |              |

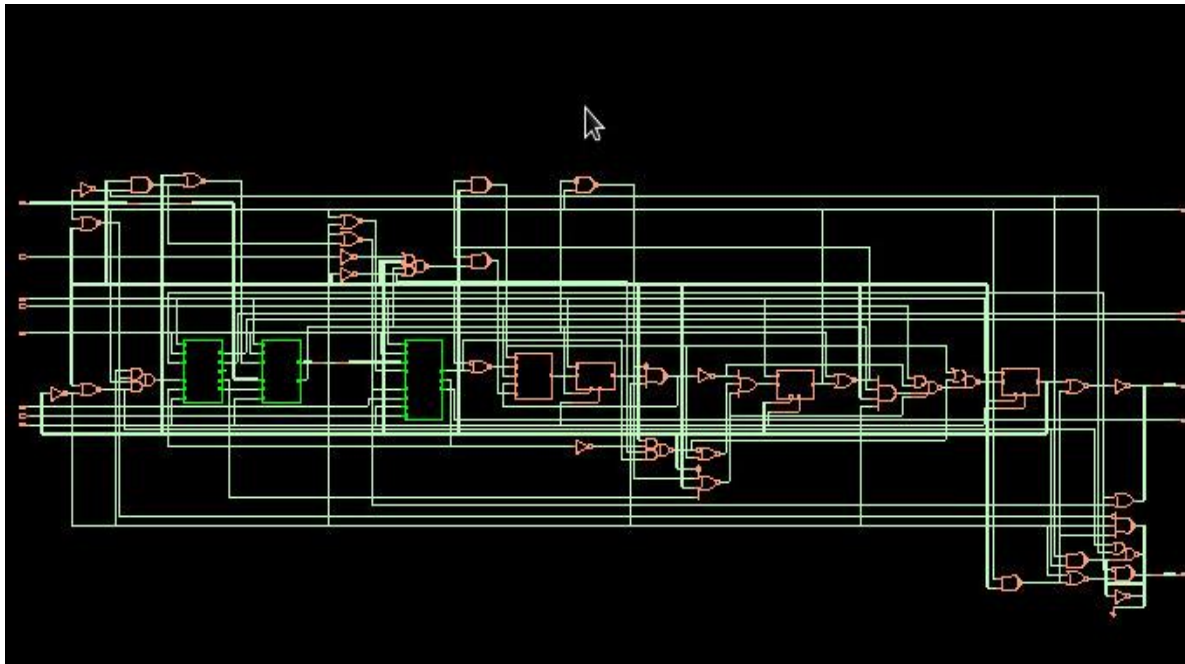
A portion of the netlist showing the scan chains inserted is shown below.

```

OAI221X1 g11972(.A0 (n_153), .A1 (n_265), .B0 (n_155), .B1 (n_266),
.C0 (n_320), .Y (n_332));
OAI221X1 g11973(.A0 (n_153), .A1 (n_267), .B0 (n_155), .B1 (n_279),
.C0 (n_321), .Y (n_331));
AOI221X1 g11974(.A0 (n_158), .A1 (n_220), .B0 (n_157), .B1 (n_244),
.C0 (n_319), .Y (n_352));
AOI221X1 g11976(.A0 (\balance db[10] [6]), .A1 (n_179), .B0
(\balance db[3] [6]), .B1 (n_172), .C0 (n_312), .Y (n_321));
AOI221X1 g11977(.A0 (\balance db[10] [7]), .A1 (n_179), .B0
(\balance db[3] [7]), .B1 (n_172), .C0 (n_317), .Y (n_320));
OAI221X1 g11978(.A0 (n_159), .A1 (n_222), .B0 (n_156), .B1 (n_251),
.C0 (n_305), .Y (n_328));
OAI221X1 g11979(.A0 (n_156), .A1 (n_259), .B0 (n_298), .C0 (n_314),
.Y (n_334));
NAND4XL g11980(.A (n_225), .B (n_226), .C (n_286), .D (n_304), .Y
(n_335));
NAND4XL g11981(.A (n_236), .B (n_282), .C (n_235), .D (n_302), .Y
(n_339));
NOR2XL g11982(.A (n_310), .B (n_295), .Y (n_322));
OAI221X1 g11983(.A0 (n_155), .A1 (n_243), .B0 (n_214), .C0 (n_316),
.Y (n_319));
AOI221X1 g11984(.A0 (\balance db[15] [4]), .A1 (n_180), .B0 (n_291),
.C0 (n_311), .Y (n_318));
NAND4XL g11985(.A (n_228), .B (n_272), .C (n_273), .D (n_306), .Y
(n_333));
OAI221X1 g11986(.A0 (n_156), .A1 (n_260), .B0 (n_159), .B1 (n_258),
.C0 (n_308), .Y (n_336));
OAI221X1 g11987(.A0 (n_159), .A1 (n_221), .B0 (n_156), .B1 (n_253),
.C0 (n_313), .Y (n_343));
NAND4XL g11988(.A (n_238), .B (n_277), .C (n_278), .D (n_307), .Y
(n_337));
NAND4XL g11989(.A (n_237), .B (n_275), .C (n_276), .D (n_303), .Y
(n_338));

```

The schematic for the synthesis is shown below.



## Analysis of Results

The quality of results before the scan chain insertion is summarized in the two tables below.

# Chapter 7

## POST STA VS DFT RESULTS

### for Minimum Time

The violation reports for the static timing analysis for minimum area netlist are shown below in the two figures.

```
#####
# Generated by:      Cadence Tempus 20.10-p003_1
# OS:               Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on:      Sat Oct 11 20:38:07 2025
# Design:           toll_gate
# Command:          report constraints -all_violators > sta_after_dft_area/reports/allviolations_area.rpt
#####
# format : frame 0 : split 1

max_delay/setup
-----
No paths found

min_delay/hold
-----
No paths found

Check type : clock_period
-----
No paths found

Check type : skew
-----
No paths found

Check type : pulse_width
-----
No violating Checks with given description found

Check type : max_transition
-----
No Violations found

Check type : min_transition
-----
No Violations found

Check type : max_capacitance
-----
No Violations found

Check type : min_capacitance
-----
No Violations found

Check type : max_fanout
-----
No Violations found

Check type : min_fanout
-----
No Violations found
```

The coverage report is shown below.

| TIMING CHECK COVERAGE SUMMARY |               |            |          |          |
|-------------------------------|---------------|------------|----------|----------|
| Check Type                    | No. of Checks | Met        | Violated | Untested |
| Hold                          | 755           | 755 (100%) | 0 (0%)   | 0 (0%)   |
| PulseWidth                    | 502           | 502 (100%) | 0 (0%)   | 0 (0%)   |
| Setup                         | 755           | 755 (100%) | 0 (0%)   | 0 (0%)   |

Applications Places System

timing\_report\_area.rpt (~/Desktop/cmos65/VDF\_Project/STA\_DFT/min\_time/sta\_after\_dft\_area/reports) - gedit

File Edit View Search Tools Documents Help

Open Save Undo

timing\_report\_area.rpt

```
#####
# Generated by: Cadence Tempus 20.10-p003 1
# OS: Linux x86 64(Host ID edatools-server2.iitd.edu.in)
# Generated on: Sat Oct 11 20:38:07 2025
# Design: toll_gate
# Command: report_timing > $report_dir/timing_report_area.rpt
#####
Path 1: MET Setup Check with Pin u_account_balance/balance_db_reg[14][1]/CK
Endpoint: u_account_balance/balance_db_reg[14][1]/SE (^) checked with leading edge of 'clk'
Beginpoint: u_vehicle_interface/id_to_check_reg[0]/Q (v) triggered by leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time 0.010
- Setup 0.301
+ Phase Shift 5.600
= Required Time 5.309
- Arrival Time 4.624
= Slack Time 0.685
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.010
= Beginpoint Arrival Time 0.010
-----
Instance Arc Cell Delay Arrival Time Required Time
-----
u_vehicle_interface/id_to_check_reg[0] CK ^ - - 0.010 0.695
u_vehicle_interface/id_to_check_reg[0] CK ^ -> Y v SdffQX1 0.567 0.577 1.262
u_account_balance/g12137 B v -> Y ^ NOR2XL 0.555 1.132 1.817
u_account_balance/g12131 A ^ -> Y v CLKINVX1 0.216 1.348 2.033
u_account_balance/g12100 B v -> Y ^ NOR2XL 0.501 1.849 2.534
u_account_balance/g11978 A1 ^ -> Y v AOI221X1 0.187 2.036 2.721
u_account_balance/g11964 C0 v -> Y ^ OAI221X1 0.193 2.229 2.914
u_account_balance/g11941 A ^ -> Y v NOR2XL 0.157 2.385 3.070
u_account_balance/g11936 B v -> Y ^ NOR2XL 0.142 2.528 3.213
-----
```

Plain Text Tab Width: 8 v Ln 1, Col 1 INS

(107) Wh... cmos65 VDF Proj... STA DFT min\_time sta after... reports timing re...

The Path Based Analysis (PBA) report is shown below.

```
#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sat Oct 11 20:38:07 2025
# Design: toll_gate
# Command: report_timing -retime path_slew_propagation -max_path 50 -nworst 50 -path_type full_clock >
$report_dir/pba_area.rpt
#####
Path 1: MET Setup Check with Pin u_account_balance/balance_db_reg[14][15]/CK
Endpoint: u_account_balance/balance_db_reg[14][15]/SE (^) checked with leading edge of 'clk'
Beginpoint: u_vehicle_interface/id_to_check_reg[0]/Q (v) triggered by leading edge of 'clk'
Path Groups: {clk}
Retime Analysis { Data Path-Slew }
Other End Arrival Time 0.010
- Setup 0.301
+ Phase Shift 5.600
= Required Time 5.309
- Arrival Time 4.534
= Slack Time 0.775
= Slack Time(original) 0.685
Clock Rise Edge 0.000
+ Source Insertion Delay 0.010
= Beginpoint Arrival Time 0.010
Timing Path:
-----
Instance Arc Cell Retime Delay Arrival Time Required Time
-----
- clk ^ - - 0.010 0.785
u_vehicle_interface/id_to_check_reg[0] - SdffQX1 0.000 0.010 0.785
u_vehicle_interface/id_to_check_reg[0] CK ^ -> Q v SdffQX1 0.567 0.577 1.352
u_account_balance/g12137 - NOR2XL 0.000 0.577 1.352
u_account_balance/g12137 B v -> Y ^ NOR2XL 0.555 1.132 1.907
```

## For Minimum Area

The violation reports for the static timing analysis for minimum area netlist are shown below in the two figures.

```
#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sat Oct 11 20:34:11 2025
# Design: toll_gate
# Command: report_constraints -all_violators > sta_after_dft_area/reports/allviolations_area.rpt
#####
# format : frame 0 : split 1

max_delay/setup
-----
No paths found

min_delay/hold
-----
No paths found

Check type : clock_period
-----
No paths found

Check type : skew
-----
No paths found

Check type : pulse_width
-----
No violating Checks with given description found

Check type : max_transition
-----
No Violations found
```

```

Check type : min_transition
-----
No Violations found

Check type : max_capacitance
-----
No Violations found

Check type : min_capacitance
-----
No Violations found

Check type : max_fanout
-----
No Violations found

Check type : min_fanout
-----
No Violations found

```

The coverage report is shown below.

```

#####
# Generated by:      Cadence Tempus 20.10-p003_1
# OS:               Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on:      Sat Oct 11 20:34:11 2025
# Design:           toll_gate
# Command:           report_analysis_coverage > $report_dir/analysis_coverage_area.rpt
#####

```

| TIMING CHECK COVERAGE SUMMARY |               |            |          |          |
|-------------------------------|---------------|------------|----------|----------|
| Check Type                    | No. of Checks | Met        | Violated | Untested |
| Hold                          | 755           | 755 (100%) | 0 (0%)   | 0 (0%)   |
| PulseWidth                    | 502           | 502 (100%) | 0 (0%)   | 0 (0%)   |
| Setup                         | 755           | 755 (100%) | 0 (0%)   | 0 (0%)   |

The Graph Based Analysis (GBA) report is shown below.

```

#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sat Oct 11 20:34:11 2025
# Design: toll_gate
# Command: report_timing > $report_dir/timing_report_area.rpt
#####
Path 1: MET Setup Check with Pin u_account_balance/balance_db_reg[14][1]/CK
Endpoint: u_account_balance/balance_db_reg[14][1]/SE (^) checked with leading edge of 'clk'
Beginpoint: u_vehicle_interface/id_to_check_reg[0]/Q (v) triggered by leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time 0.010
- Setup 0.301
+ Phase Shift 12.000
= Required Time 11.709
- Arrival Time 4.624
= Slack Time 7.085
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.010
= Beginpoint Arrival Time 0.010
-----
Instance Arc Cell Delay Arrival Time Required Time
-----
u_vehicle_interface/id_to_check_reg[0] CK ^ - - 0.010 7.095
u_vehicle_interface/id_to_check_reg[0] CK ^ -> Q v SdffQX1 0.567 0.577 7.662
u_account_balance/g12137 B v -> Y ^ NOR2XL 0.555 1.132 8.217
u_account_balance/g12131 A ^ -> Y v CLKINVX1 0.216 1.348 8.433
u_account_balance/g12100 B v -> Y ^ NOR2XL 0.501 1.849 8.934
u_account_balance/g11978 A1 ^ -> Y v AOI221X1 0.187 2.036 9.121
u_account_balance/g11964 C0 v -> Y ^ OAI221X1 0.193 2.229 9.314
u_account_balance/g11941 A ^ -> Y v NOR2XL 0.157 2.385 9.470
u_account_balance/g11936 B v -> Y ^ NOR2XL 0.142 2.527 9.613

u_account_balance/g11936 B v -> Y ^ NOR2XL 0.142 2.527 9.613
u_account_balance/g11934 A ^ -> Y v NOR2XL 0.109 2.637 9.722
u_account_balance/g11932 B v -> Y ^ NAND2XL 0.117 2.754 9.839
u_account_balance/g11930 B ^ -> Y v NOR2XL 0.096 2.851 9.936
u_account_balance/g11928 B v -> Y ^ NAND2XL 0.161 3.011 10.096
u_account_balance/g12145 D ^ -> Y v NOR4BBX1 0.086 3.097 10.182
u_account_balance/g11921 C v -> Y ^ NAND3BX1 0.092 3.189 10.274
u_account_balance/g11917 C ^ -> Y v NAND3X1 0.175 3.364 10.449
u_account_balance/g11915 A v -> Y ^ INVX1 0.091 3.455 10.540
u_account_balance/g11912 B ^ -> Y v NAND2XL 0.156 3.611 10.696
u_account_balance/g11909 B v -> Y ^ NOR2XL 0.254 3.866 10.951
u_account_balance/g11906 A ^ -> Y v INVX1 0.087 3.953 11.038
u_account_balance/g11898 B v -> Y ^ NOR2X1 0.671 4.624 11.709
u_account_balance/balance_db_reg[14][1] SE ^ SdffQX1 0.000 4.624 11.709
-----

```

The Path Based Analysis (PBA) report is shown below.

```

Beginpoint: u_vehicle_interface/id_to_check_reg[0]/Q (v) triggered by leading edge of 'clk'
Path Groups: {clk}
Retime Analysis { Data Path-Slew }
Other End Arrival Time 0.010
- Setup 0.301
+ Phase Shift 12.000
= Required Time 11.709
- Arrival Time 4.534
= Slack Time 7.175
= Slack Time(original) 7.085
Clock Rise Edge 0.000
+ Source Insertion Delay 0.010
= Beginpoint Arrival Time 0.010
Timing Path:
-----
Instance Arc Cell Retime Delay Arrival Time Required Time
-----
- clk ^ - - 0.010 7.185
u_vehicle_interface/id_to_check_reg[0] - SDFFX1 0.000 0.010 7.185
u_vehicle_interface/id_to_check_reg[0] CK ^ -> Q v SDFFX1 0.567 0.577 7.752
u_account_balance/g12137 - NOR2XL 0.000 0.577 7.752
u_account_balance/g12137 B v -> Y ^ NOR2XL 0.555 1.132 8.307
u_account_balance/g12131 - CLKINX1 0.000 1.132 8.307
u_account_balance/g12131 A ^ -> Y v CLKINX1 0.215 1.348 8.523
u_account_balance/g12100 - NOR2XL 0.000 1.348 8.523
u_account_balance/g12100 B v -> Y ^ NOR2XL 0.502 1.849 9.024
u_account_balance/g11978 - AOI221X1 0.000 1.849 9.024
u_account_balance/g11978 A1 ^ -> Y v AOI221X1 0.186 2.036 9.211
u_account_balance/g11964 - OAI221X1 0.000 2.036 9.211
u_account_balance/g11964 C0 v -> Y ^ OAI221X1 0.184 2.219 9.394
u_account_balance/g11941 - NOR2XL 0.000 2.219 9.394
u_account_balance/g11941 A ^ -> Y v NOR2XL 0.127 2.346 9.521

```

## For Intermediate Area and Timing Constraints

The violation reports for the static timing analysis for minimum area netlist are shown below in the two figures.

```

#####
# Generated by: Cadence Tempus 20.10-p003.1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sat Oct 11 20:42:07 2025
# Design: toll_gate
# Command: report_constraints -all_violators > sta_after_dft_area/reports/allviolations_area.rpt
#####
# format : frame 0 : split 1

max_delay/setup
-----
No paths found

min_delay/hold
-----
No paths found

Check type : clock_period
-----
No paths found

Check type : skew
-----
No paths found

Check type : pulse_width
-----
No violating Checks with given description found

Check type : max_transition
-----
No Violations found

```



```

Check type : min_transition
-----
No Violations found

Check type : max_capacitance
-----
No Violations found

Check type : min_capacitance
-----
No Violations found

Check type : max_fanout
-----
No Violations found

Check type : min_fanout
-----
No Violations found

```

The coverage report is shown below.

```

#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sat Oct 11 20:42:07 2025
# Design: toll_gate
# Command: report_analysis_coverage > $report_dir/analysis_coverage_area.rpt
#####

```

| TIMING CHECK COVERAGE SUMMARY |               |            |          |          |
|-------------------------------|---------------|------------|----------|----------|
| Check Type                    | No. of Checks | Met        | Violated | Untested |
| Hold                          | 755           | 755 (100%) | 0 (0%)   | 0 (0%)   |
| PulseWidth                    | 502           | 502 (100%) | 0 (0%)   | 0 (0%)   |
| Setup                         | 755           | 755 (100%) | 0 (0%)   | 0 (0%)   |

The Graph Based Analysis (GBA) report is shown below.

```

#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64(Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sat Oct 11 20:42:07 2025
# Design: toll_gate
# Command: report_timing > $report_dir/timing_report_area.rpt
#####
Path 1: MET Setup Check with Pin u_account_balance/balance_db_reg[9][1]/CK
Endpoint: u_account_balance/balance_db_reg[9][1]/SE (^) checked with leading edge of 'clk'
Beginpoint: u_vehicle_interface/id_to_check_reg[3]/Q (v) triggered by leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time      0.010
- Setup                    0.301
+ Phase Shift              8.000
= Required Time            7.709
- Arrival Time             4.526
= Slack Time               3.183
  Clock Rise Edge          0.000
+ Clock Network Latency (Ideal) 0.010
= Beginpoint Arrival Time  0.010

```

| Instance                               | Arc         | Cell     | Delay | Arrival Time | Required Time |
|----------------------------------------|-------------|----------|-------|--------------|---------------|
| u_vehicle_interface/id_to_check_reg[3] | CK ^        | -        | -     | 0.010        | 3.193         |
| u_vehicle_interface/id_to_check_reg[3] | CK ^ -> Q v | SDFFXQX1 | 0.486 | 0.496        | 3.679         |
| u_account_balance/g11418               | A v -> Y ^  | CLKINVX1 | 0.151 | 0.647        | 3.830         |
| u_account_balance/g11413               | B ^ -> Y v  | NAND2XL  | 0.324 | 0.971        | 4.154         |
| u_account_balance/g11377               | B v -> Y ^  | NOR2XL   | 0.634 | 1.605        | 4.788         |
| u_account_balance/g11313               | B0 ^ -> Y v | A0I22XL  | 0.121 | 1.726        | 4.909         |
| u_account_balance/g11257               | A v -> Y ^  | NAND2XL  | 0.110 | 1.836        | 5.019         |
| u_account_balance/g11219               | C0 ^ -> Y v | A0I211X1 | 0.055 | 1.891        | 5.074         |
| u_account_balance/g11213               | C v -> Y ^  | NAND3X1  | 0.178 | 2.069        | 5.253         |

|                                        |             |          |       |       |       |
|----------------------------------------|-------------|----------|-------|-------|-------|
| u_account_balance/g11209               | B ^ -> Y v  | NOR2XL   | 0.113 | 2.183 | 5.366 |
| u_account_balance/g11207               | B v -> Y ^  | NOR2XL   | 0.261 | 2.443 | 5.627 |
| u_account_balance/g11205               | D ^ -> Y v  | NOR4BX1  | 0.102 | 2.546 | 5.729 |
| u_account_balance/g11201               | B v -> Y ^  | NAND2XL  | 0.126 | 2.671 | 5.855 |
| u_account_balance/g11197               | B ^ -> Y v  | NOR2XL   | 0.121 | 2.793 | 5.976 |
| u_account_balance/g11195               | C v -> Y ^  | NAND3X1  | 0.116 | 2.909 | 6.092 |
| u_account_balance/g11189               | B ^ -> Y ^  | OR2X1    | 0.117 | 3.026 | 6.209 |
| u_account_balance/g11186               | B ^ -> Y v  | NAND2XL  | 0.165 | 3.191 | 6.374 |
| u_account_balance/g11185               | AN v -> Y v | NAND2BX1 | 0.181 | 3.372 | 6.555 |
| u_account_balance/g11178               | B v -> Y ^  | NOR2XL   | 0.231 | 3.603 | 6.787 |
| u_account_balance/g11171               | B ^ -> Y v  | NAND2XL  | 0.216 | 3.819 | 7.003 |
| u_account_balance/g11164               | B v -> Y ^  | NOR2X1   | 0.706 | 4.526 | 7.709 |
| u_account_balance/balance_db_reg[9][1] | SE ^        | SDFFOX1  | 0.000 | 4.526 | 7.709 |

The Path Based Analysis (PBA) report is shown below.

| Beginpoint: u_vehicle_interface/id_to_check_reg[3]/Q (v) triggered by leading edge of 'clk' |             |          |              |              |               |
|---------------------------------------------------------------------------------------------|-------------|----------|--------------|--------------|---------------|
| Path Groups: {clk}                                                                          |             |          |              |              |               |
| Retime Analysis { Data Path-Slew }                                                          |             |          |              |              |               |
| Other End Arrival Time 0.010                                                                |             |          |              |              |               |
| - Setup 0.301                                                                               |             |          |              |              |               |
| + Phase Shift 8.000                                                                         |             |          |              |              |               |
| = Required Time 7.709                                                                       |             |          |              |              |               |
| - Arrival Time 4.437                                                                        |             |          |              |              |               |
| = Slack Time 3.272                                                                          |             |          |              |              |               |
| = Slack Time(original) 3.183                                                                |             |          |              |              |               |
| Clock Rise Edge 0.000                                                                       |             |          |              |              |               |
| + Source Insertion Delay 0.010                                                              |             |          |              |              |               |
| = Beginpoint Arrival Time 0.010                                                             |             |          |              |              |               |
| Timing Path:                                                                                |             |          |              |              |               |
| Instance                                                                                    | Arc         | Cell     | Retime Delay | Arrival Time | Required Time |
| -                                                                                           | clk ^       | -        | -            | 0.010        | 3.282         |
| u_vehicle_interface/id_to_check_reg[3]                                                      | -           | SDFFOX1  | 0.000        | 0.010        | 3.282         |
| u_vehicle_interface/id_to_check_reg[3]                                                      | CK ^ -> Q v | SDFFOX1  | 0.486        | 0.496        | 3.767         |
| u_account_balance/g11418                                                                    | -           | CLKINX1  | 0.000        | 0.496        | 3.767         |
| u_account_balance/g11418                                                                    | A v -> Y ^  | CLKINX1  | 0.151        | 0.647        | 3.919         |
| u_account_balance/g11413                                                                    | -           | NAND2XL  | 0.000        | 0.647        | 3.919         |
| u_account_balance/g11413                                                                    | B ^ -> Y v  | NAND2XL  | 0.324        | 0.971        | 4.243         |
| u_account_balance/g11377                                                                    | -           | NOR2XL   | 0.000        | 0.971        | 4.243         |
| u_account_balance/g11377                                                                    | B v -> Y ^  | NOR2XL   | 0.632        | 1.603        | 4.875         |
| u_account_balance/g11313                                                                    | -           | A0I22XL  | 0.000        | 1.603        | 4.875         |
| u_account_balance/g11313                                                                    | B0 ^ -> Y v | A0I22XL  | 0.121        | 1.724        | 4.996         |
| u_account_balance/g11257                                                                    | -           | NAND2XL  | 0.000        | 1.724        | 4.996         |
| u_account_balance/g11257                                                                    | A v -> Y ^  | NAND2XL  | 0.110        | 1.834        | 5.106         |
| u_account_balance/g11219                                                                    | -           | A0I211X1 | 0.000        | 1.834        | 5.106         |
| u_account_balance/g11219                                                                    | C0 ^ -> Y v | A0I211X1 | 0.055        | 1.889        | 5.161         |

**RESULT :**

**STA:**

| Constraint     | Cell Count | Area     | Slack(in ps) | Power(in Watts)          |
|----------------|------------|----------|--------------|--------------------------|
| Minimum Area   | 642        | 7082.313 | 6968         | $5.36491 \times 10^{-4}$ |
| Minimum Timing | 706        | 7220.826 | 5            | $1.77715 \times 10^{-3}$ |
| Intermediate   | 671        | 7105.777 | 3161         | $7.78423 \times 10^{-4}$ |

| Constraints    | SlackforGBA(inns) | SlackforPBA(inns) |
|----------------|-------------------|-------------------|
| Minimum Area   | 0.542             | 0.638             |
| Minimum Timing | 6.942             | 7.038             |
| Intermediate   | 3.136             | 3.157             |

### DFT:

| Constraint   | CellCount | Area     | Slack(in ps) | Power(in Watts)          |
|--------------|-----------|----------|--------------|--------------------------|
| MinimumArea  | 648       | 7136.053 | 7110         | $5.40045 \times 10^{-4}$ |
| Tight Timing | 712       | 7369.178 | 0            | $1.8645 \times 10^{-3}$  |
| Intermediate | 666       | 7142.865 | 2705         | $8.22545 \times 10^{-4}$ |

| Constraints    | Slack for GBA(inns) | Slack for PBA(inns) |
|----------------|---------------------|---------------------|
| Minimum Area   | 7.085               | 7.175               |
| Minimum Timing | 8.685               | 8.775               |
| Intermediate   | 3.183               | 3.272               |